When Can an Agent Attribute Properties to Itself? A Cognitive Algebra for Identity, Integrity, and Freedom

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Abstract

An agent's cognitive architecture may possess every regional carrier required for a capability and still emit a wrong verdict, because the content those carriers hold is corrupted. Over the MCK observer-relative classification of Bragetti, Asaro and Italiano [12], this paper formalises the boundary between what an architecture can *structurally realise* and what it can *semantically get right*.

The central object is a two-layer state $\kappa_a(t) = (\Sigma_a(t), \Delta_a(t))$: Σ_a records regional carriers (the M/C/K predicate profile of each artefact in $\mathrm{Sto}(a)$), Δ_a records denotational and evidence content. A capability q decomposes into $\mathrm{Req}^{\Sigma}_a(q)$ (structural carriers) and $\mathrm{Req}^{\Delta}_a(q)$ (content correctness), realisability being their conjunction. This yields a three-way status — *structurally untypable*, *typable-but-incorrect*, *realised* — and proves that structural extension is monotone in carrier presence but not in semantic calibration: adding a carrier with wrong content can worsen a previously realised capability.

Six constructions follow: (i) an attribute certificate, the licence-class supplying content to Foundations' step-relation licence slot; (ii) a self-symbol $s^a_{\mathrm{self}} \in \Theta_a$ with $\mathrm{denotes}_a(s^a_{\mathrm{self}}) = a$, supplying the object of Foundations' epistemogenic-self-representation conjecture (which remains open) and anchoring de se certification and its reflexive companion MetaCert; (iii) an other-model triad $(\widehat{M}_a(b), \widehat{O}_a(b), \widehat{\mathrm{Mon}}^{\,b}_a(\Phi_b))$, the actual observation-relative monitor $\mathrm{Mon}^{O_b}_b(\Phi_b, u)$ generalising the setting of [13]; (iv) an operator-composition typing separating *structurally* from *semantically* composable: the pipeline $\widehat{M} \to \widehat{\mathrm{Effect}} \to \widehat{O} \to \widehat{\mathrm{Mon}}$ can execute end-to-end and still emit a miscalibrated verdict — the MCK-native root of downstream model-error phenomena; (v) a bounded transition system T_{MCK} over Foundations' GEN / MOV / EXIT steps plus two declared extension families, making reachability of Cert, MetaCert and W_B configurations decidable at finite scale and separating reachability from semantic realisability; (vi) a bounded integrity-consistency procedure deciding internal contradiction in $O(n^2 \cdot c)$ time.

Three worked self-properties — continuity, viability, owned branching, glossed *identity*, *integrity*, *freedom* — receive native $(\mathrm{Req}^{\Sigma}, \mathrm{Req}^{\Delta})$ signatures, and structural cuts and calibration cuts of the same capability have incomparable downstream blast radii. An attention layer completes the architecture: attention is an explicit third coordinate, focus shifts are a declared transition family, and a same-state / different-focus theorem shows that which critical properties are effectively attended conditions the admissible actions — and, through an attention-gated feedback channel, which temporal-implication regularities the agent can ever confirm. Of the thirty-two activation signatures combining structural realisability, semantic adequacy, attention, reflexive availability, and behavioural

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expression, exactly eighteen are type-consistent, and no link of the naive activation chain holds. Applied to four cognitive worlds (W_A first-order non-social, W_B calibrated help, W_C monitor-blind assistance, W_D monitored safety failure), the two-layer state proves same-Σ_a / different-Δ_a non-injectivity: W_B and $W_{D\text{-}F_{\mathrm{MODEL}}}$ share regional structure exactly and differ only on $\widehat{O}_a(b)$ content.

The answer to "*sapere vs. sapere di sapere*" is architectural: the ladder from Bel through Cert, Know, MetaCert, RefAware to operational consciousness is a strictly increasing tower of Req^{Σ} carriers plus Req^{Δ} content conditions — a difference in required carriers, not a modal-notation gloss. Prior work on positive introspection, awareness and unawareness logics, de se attitudes, epistemic planning, I-POMDPs, CIRL, and multi-agent runtime verification supplies formal antecedents for individual coordinates (Section 27); the construction here is the two-layer realisability decomposition and the typed pipeline it delivers inside a single agent's architecture, with its criticality analysis. Version 2 supersedes v1 [15] as an extension; v1 remains available at its version DOI 10.5281/zenodo.22050704. Every v1 finite model remains exhaustively validated by its original checker; the v2 results are validated by an extended regional checker and a mechanised dependency-graph extractor distributed with the article, whose gates the results ledgers cite individually.

1 Introduction

A monitoring system reports that a component is intact. A controller reports that it still has a choice. A process reports that it is the same process that started the job. An assistant reports that it has helped its user. Each report is about the reporting system itself — or about another system it acts upon — and each can fail in ordinary ways: the report can be wrong, or the system may not yet have seen enough to say. There are further failures. The system may lack the internal structure that would make the report mean anything at all — no symbol denoting itself, no representation of the other agent, no model of what that other agent observes. Such a system is not mistaken about its integrity, or about its user's welfare. It has nothing to be mistaken about.

The question we ask is not what an agent knows about itself and others, but the prior one:

> What must an agent internally represent in order to *certify* a property about itself, or about another agent it acts upon, at all — and how is that internal structure itself *formed*, through interaction with the world and through computation upon what the agent already holds?

Why formation, and not only possession. It would be simpler to treat the agent as a tuple that possesses, once and for all, its beliefs, self-model, incentives and picture of the other. We refuse that simplification for a definite reason: every failure we analyse — monitor-blind assistance, model-verdict mismatch, self-model error induced by another's feedback — is a failure of content laid down earlier in the agent's trajectory and never corrected. The paper therefore treats the agent as a trajectory $\kappa_a(0) \to \cdots \to \kappa_a(t)$ of two-layer states, each step either consuming new external content (*open*) or recombining what the agent holds (*closed*), and shows that belief, identity, incentive and social models co-evolve under precise, checkable dependency conditions — while none reduces to any other.

The path. We answer along the following ladder. First the MCK Framework Foundations [12] classification supplies a partial semantic geometry: an artifact x may or may not lie in an agent's classified domain $\mathcal{A}_a^+$, may or may not be a symbol ($x \in \Theta_a = M_a \cap C_a$), and if it is a symbol may or may not carry a denotation binding $\mathrm{denotes}_a(x) \in \mathrm{Ref}$. On top of that substrate the paper builds a *certificate calculus*: a claim φ about a referent r is certified

by a licence $z \in \mathcal{A}_a^+$ carrying (claim, referent, evidence, scope, verdict). Reflexive certification $\mathrm{MetaCert}_a$ is a certificate about that certificate; its architectural realisability requires a self-symbol $s^a_{\mathrm{self}} \in \Theta_a$ with $\mathrm{denotes}_a(s^a_{\mathrm{self}}) = a$ and a symbol denoting the first-order certificate itself. Neither symbol is supplied by default. *First-order certification does not, in this architecture, imply reflexive certification* — and this is not a weakness of classical epistemic logic but the honest architectural condition under which its introspection axioms hold. Above the carrier substrate the paper then adds an *attention* coordinate: which represented artefacts are effectively salient at a step, how a feedback loop over temporal implication (which antecedent–consequent instances the agent's focus lets it confirm) reshapes content, and why possibility, salience, and behavioural expression are three independently varying layers.

Cognition of another. An agent that certifies things about itself may or may not certify things about another agent, and the two capacities are independent (Proposition 4.6). Certification about another agent is a triad: a model of the agent $\widehat{M}_a(b)$, a model of what that agent observes $\widehat{O}_a(b)$, and a model of the monitor that agent runs on what it observes $\widehat{\mathrm{Mon}}^{\,b}_a(\Phi_b)$. Each may be absent, incomplete, mis-denoted, ill-typed, outdated, or false, and the six failure modes have different downstream consequences. The actual receiver monitor $\mathrm{Mon}_b^{O_b}(\Phi_b, u)$ generalises the observation-relative setting of Bragetti et al. [13], developed there for bounded temporal implication.

Four cognitive worlds. Interpretation over the certificate calculus and the other-model triad partitions actor stances into four cognitive worlds (Definition 21.1): W_A (first-order non-social: no represented predicted effect on the other), W_B (calibrated help: adequate triad, predicted verdict agrees with the actual one), W_C (monitor-blind assistance, with sub-cases C_1 / C_2 by whether the receiver cannot observe or the actor fails to model that observation), and W_D (monitored safety failure, with sub-cases D-F_{INT} / D-F_{MODEL} by coercive intent vs. miscalibrated other-model).

A crucial consequence, and the paper's bridge to trace-level analyses of transactional interaction, is that the projection π : (v2 cognitive state of a) $\to$ (observable trace) is *non-injective*: distinct cognitive worlds can produce identical observable traces (Theorem 21.5). Receiver-side observable failure therefore does not entail malicious intent, and receiver-side observable non-failure does not entail well-calibrated help.

Three properties, three different answers, one algebra. The three narrow computational self-properties of version 1 [15] — *continuity*, *viability*, *owned branching*, glossed *identity / integrity / freedom* — carry over into v2. They receive native MCK region signatures and act as *upstream certificates* on which downstream de se attributions depend. This upstream-dependency structure supplies "critical" with a formal criterion: a property is critical when the failure of its certificate invalidates the well-formedness of at least one downstream certificate over the same self-referent (Definition 23.1 and theorem 23.3). Version 1's minimality theorems *Theorems* 17.7*and* 18.8 and heterogeneity theorem *Theorem* 19.4 are preserved unchanged; v2 gives their content a native MCK reading and locates it within the certificate calculus.

What we do not claim. No new epistemic logic, de se semantics, awareness logic, free logic, or theory of intentionality. Positive introspection is prior art [24]; awareness and its introspection constraints are prior art [23, 40, 31, 21, 6, 29]; de se attitudes are prior art [43, 36, 34]; denotation failure is the setting of free logic [35, 5]; observation-relative monitorability is prior work of Bragetti et al. [13]. The contribution is the *integration* of these through MCK's region / symbol / denotation / licence substrate and the cognitive worlds it exposes

— an architectural, not a semantic, finding. No claim is made that any current AI system instantiates the full algebra.

Operational, not phenomenal. "Conscious" / "unconscious" are used operationally throughout (Definition 3.11); Remark 3.12 states the non-claims once: no phenomenal consciousness, no sentience, no psychoanalytic unconscious, no human-equivalent introspection.

Structure. The v2 core sits between carrier machinery and worked properties: Section 8 audits the regional realisation of every cognitive object; Section 9 splits the state into carriers and content; Section 10 decomposes capability requirements and types operator composition; Section 11 fixes the export contracts; Section 12 delivers integrity consistency, semantic repair, and bounded reachability; Sections 13 and 14 develop formation and co-evolution; Section 15 adds the attention coordinate. Upstream of the core, Sections 2 to 7 fix the four objects and status domain, the belief-to-meta-certificate ladder, the other-model triad, v1's three requirements, action and intention, and the common carrier (from v1, unchanged). Downstream, Sections 16 to 18 treat the three worked properties; Sections 19 to 25 carry incomparability, property-family reuse, the four worlds and trace non-injectivity, lifting, criticality, representational extension, and reflection; Sections 26 to 29 close, and Appendix A collects the finite witnesses.

Reproducibility. Every finite model, witness and countermodel in v1 remains exhaustively validated by the checker distributed with v1; v2 adds new finite witnesses in the same style. We call this exhaustive validation of finite models, not machine-checked proof: the proofs are in the text.

2 Attribution semantics

We fix the objects that a self-attribution involves, and keep apart four things that ordinary usage runs together: what is the case, what the agent can refer to, what the agent has seen, and what the agent returns.

Convention 2.1 (Traces). Σ is a finite alphabet of events, Σ^* the finite words over it, h a *history*, ε the empty history, and $h \cdot e$ the extension of h by $e \in \Sigma$. Where a property is stated over infinite behaviours we write Σ^ω and $u \sqsubseteq w$ for the prefix relation.

Definition 2.2 (The four objects)**.** Fix an agent a, a subject x and a property p.

1. $\mathrm{Truth}(p, x, h) \in \{\textsc{true}, \textsc{false}\}$ is the model-side truth of p about x after h. It is defined independently of a.
2. $\mathrm{Ref}_a(x)$ is a's referent for x, when it has one.
3. $\mathrm{Eval}_a(p)$ is a's recogniser for p: the structure by which a evaluates p, when it has one.
4. $\mathrm{Info}_a(h)$ is the information a retains after h.

The agent returns $\mathrm{Verdict}_a(p, x, h)$, which is not $\mathrm{Truth}(p, x, h)$ and is not asserted to track it.

Each failure in the opening paragraph of Section 1 is the failure of a different object: a wrong verdict mismatches $\mathrm{Verdict}_a$ and Truth; an uncertain verdict is a deficiency of Info_a; the third failure is the absence of $\mathrm{Ref}_a(x)$ or $\mathrm{Eval}_a(p)$ — a deficiency of neither.

2.1 The status domain

Definition 2.3 (Attribution status)**.** $\mathrm{Verdict}_a(p, x, h)$ takes values in

$$\{\,\textsc{true},\ \textsc{false},\ \textsc{undetermined},\ \textsc{untyped}\,\},$$

where

- UNTYPED holds when a semantic prerequisite for evaluating p about x is absent for a — $\mathrm{Ref}_a(x)$ or $\mathrm{Eval}_a(p)$ is undefined — so that no verdict function on a's information exists;
- UNDETERMINED holds when p is typed for a but $\mathrm{Info}_a(h)$ does not determine the truth value;
- TRUE and FALSE hold when p is typed and determined.

Remark 2.4 (The status domain is imported, not introduced)*.* Neither the fourth value nor its motivation is new here. That an agent may fail to entertain a proposition at all, as opposed to failing to know it, is the subject of the awareness and unawareness literatures [23, 40, 31], which also establish the introspection constraints such a notion must respect [21]. Board and Chung, who separate being unaware from not knowing, make the corresponding point about models: "if we ask a question that does not belong to the associated language, we cannot expect to find an answer" [6, p. 3]. We take that observation as given. What this paper adds is not the value but its *grounding*: in Definition 5.2 UNTYPED is determined by which structures an architecture holds, and Section 24 shows that structure can be added without any change in the world or the evidence.

Convention 2.5 (Scope)*.* Everything below is stated for *self*-attribution: x is a itself or a component a counts as its own. We do not claim the account transfers to attributions about the environment, and Definition 5.1 is scoped accordingly. Version 2 extends the scope to attribution about *other* agents ($x = b \neq a$); Section 4 supplies the additional primitives needed there, and the four-valued status domain above transfers verbatim.

2.2 Attribute certificates

The MCK Framework Foundations step relation [12] requires every MOV or EXIT step of agent a to carry a *licence* $z \in \mathcal{A}_a^+$ produced by a GEN step of a, whose content is "a certificate of understanding for $x \ldots$, a refutation of one, or an execution witness for x", x being the moved artefact. Foundations explicitly disclaims a certificate calculus for licences: "the framework carries no certificate calculus", and "the licensing clause is a stipulation of the step relation, not a derived condition". Version 2 supplies exactly that calculus, as a specific sub-class of the licence slot.

Definition 2.6 (Attribute certificate)**.** An *attribute certificate* held by agent a at time t for property φ about referent r is a licence $z_{\varphi,r} \in \mathcal{A}_a^+$, produced by a GEN step of a's architecture, whose content is a tuple

$$\begin{aligned}(\ &\text{claim } \varphi; \quad \text{referent } r \in \mathrm{Ref}; \quad \text{evidence } E \subseteq \mathrm{Lnk}(a);\\ &\text{scope } \Sigma_\varphi; \quad \text{verdict } v \in \{\textsc{true}, \textsc{false}, \textsc{undetermined}, \textsc{untyped}\}\ ).\end{aligned}$$

Well-formedness requires:

1. *property representation:* there exists $s_\varphi \in \Theta_a$ with $\mathrm{denotes}_a(s_\varphi) = \varphi$;
2. *referent typing:* $r \in \mathcal{A}_a^+$ within the active regions and $\mathrm{TypeReq}(\varphi) \subseteq \mathrm{Str}(\mathcal{A}_a)$;

3. *evidence support:* E is a subset of $\mathrm{Lnk}(a)$'s derivation-record entries whose composition determines the verdict.

When any of (1)–(3) fails, the certificate returns UNTYPED at (2), UNDETERMINED at (3), or is malformed at (1). We write $\mathrm{Cert}_a(\varphi, r, t)$ for the existence of such a licence, and $\mathrm{Cert}_a(\varphi, r, t) = v$ for its verdict.

Remark 2.7 (This is a licence-class, not a new region). An attribute certificate is a content-schema for the z-slot of Foundations' step relation, not a new primitive: conservativity in the sense of H2 is recorded once in the upstream contract (Section 11). What is new is the calculus over the content: (claim, referent, evidence, scope, verdict) as a typed structure.

2.3 The self-symbol

Foundations' Remark 6.7 [12] records the reflexive-self-model conjecture but introduces no primitive and is used by no result there. Version 2 supplies the primitive as a specific Θ_a-member.

Definition 2.8 (Self-symbol)**.** A *self-symbol for* a is an artefact $s^a_{\mathrm{self}} \in \Theta_a$ with $\mathrm{denotes}_a(s^a_{\mathrm{self}}) = a$, i.e. a symbol of a whose denotation target is a itself, viewed as an element of $\mathcal{A}^+_a$. Existence is architecture-conditional: an agent may not hold a self-symbol. When held, typically $s^a_{\mathrm{self}} \in R^a_7$ (full triple), since self-representation carries an operator character; the paper's results require only $s^a_{\mathrm{self}} \in \Theta_a = R^a_4 \cup R^a_7$.

Remark 2.9 (The object of Foundations' Remark 6.7). Definition 2.8 supplies the object about which Foundations conjectures; the conjecture itself, that representing one's own configuration places the self-artifact in R_7, is not proved here, since the results below require only $s^a_{\mathrm{self}} \in \Theta_a$. It introduces no new region and no primitive outside those of Foundations; s^a_{self} is a distinguished element of Θ_a, no more and no less. What Foundations leaves as "a formal treatment of the reflexive act and its cost is left to downstream work" is delivered as a symbol whose target is the agent-as-artifact. Note that s^a_{self}'s *content* — what a says about a via s^a_{self} — is a separate object; that content lives in Section 3's $\widehat{M}_a(a)$ (the self-model), and s^a_{self} is only its denotation anchor.

Remark 2.10 (Reference is not truth). $\mathrm{denotes}_a(s^a_{\mathrm{self}}) = a$ says only that the symbol *points at* a — not that a holds any true belief about itself, an accurate self-model, or consistent self-reference. Those are downstream questions about Cert_a, Bel_a, and $\widehat{M}_a(a)$.

3 Cognitive algebra: belief, evidence, certificate, knowledge, meta-certificate

The four objects of Definition 2.2 are the semantic anchors for a bounded agent's cognition; on them we build a layered algebra of belief, evidence, certificate, knowledge, and reflexive certificate. Every layer is defined so that its *failure* is a distinct architectural condition, not a shading of another layer.

3.1 Belief, evidence, knowledge

Definition 3.1 (Belief)**.** Agent a *believes* φ about r at time t, written $\mathrm{Bel}_a(\varphi, r, t)$, when $\mathrm{Sto}(a)$ holds an *assertion record* for $\varphi(r)$ at t: an artefact whose $\mathrm{Lnk}(a)$-registered content asserts $\varphi(r)$ under some internal representation. Belief is thus Δ_a-resident and a function of $\kappa_a(t)$. Belief may be false: $\mathrm{Bel}_a(\varphi, r, t)$ is compatible with $\mathrm{Truth}(\varphi, r, t) = \text{FALSE}$.

Definition 3.2 (Evidence)**.** The *evidence* $\mathrm{Ev}_a(\varphi, r, t) \subseteq \mathrm{Lnk}(a)$ is the set of derivation-record entries in a's linkage memory whose composition, under a's architecture, computes a verdict on $\varphi(r)$. $\mathrm{Ev}_a(\varphi, r, t) = \emptyset$ means no such entry exists and $\varphi(r)$ is unsupported at t; the entry may or may not compose to a definite verdict.

Definition 3.3 (Knowledge)**.** $\mathrm{Know}_a(\varphi, r, t)$ holds when: $\mathrm{Bel}_a(\varphi, r, t)$; $\mathrm{Cert}_a(\varphi, r, t) \in \{\textsc{true}\}$; and $\mathrm{Truth}(\varphi, r, t) = \textsc{true}$. Equivalently: a believes $\varphi(r)$, holds a well-formed attribute certificate (Definition 2.6) with verdict TRUE, and the world agrees.

Remark 3.4 (No new epistemology is claimed)*.* Definition 3.3 is a JTB-style skeleton with an MCK-native evidential clause. It is not offered as a new theory of knowledge. It is offered as the object that lets us formally state *when the introspection axioms of positive epistemic logic hold, and when they do not, in a bounded MCK agent.* Compare classical $K_a\varphi \to K_aK_a\varphi$ [24]: that axiom is about Know; below, MetaCert separates a strictly weaker structural condition, and $\mathrm{Know}_a\mathrm{Know}_a$ is recoverable only under further architectural clauses.

3.2 Meta-certificate

Definition 3.5 (Meta-certificate)**.** Assume $s^a_{\mathrm{self}} \in \Theta_a$ with $\mathrm{denotes}_a(s^a_{\mathrm{self}}) = a$ (Definition 2.8). Let $z_{\varphi,r}$ be the licence realising $\mathrm{Cert}_a(\varphi, r, t)$. The *meta-certificate* $\mathrm{MetaCert}_a(\varphi, r, t)$ holds when the following further architectural clauses are satisfied:

1. there exists $s_z \in \Theta_a$ with $\mathrm{denotes}_a(s_z) = z_{\varphi,r}$ (a symbol denoting the first-order certificate);
2. there exists an attribute certificate $\mathrm{Cert}_a(\eta, s_z, t')$ whose claim η is "the licence $z_{\varphi,r}$ is held by a at t", with verdict TRUE, and whose referent-typing clause is $s_z, s^a_{\mathrm{self}} \in \mathrm{Str}(\mathcal{A}_a)$.

If either clause fails, $\mathrm{MetaCert}_a(\varphi, r, t) = \textsc{untyped}$.

Theorem 3.6 (First-order does not imply meta)**.** *There exists a bounded MCK agent a with a valid first-order certificate* $\mathrm{Cert}_a(\varphi, r, t) = \textsc{true}$ *and* $\mathrm{MetaCert}_a(\varphi, r, t) = \textsc{untyped}$*, because clause (1) of Definition 3.5 fails: a holds no $s_z \in \Theta_a$ denoting the licence $z_{\varphi,r}$.*

Proof. Witness: take $\mathcal{A}_a = \{r, z, s^a_{\mathrm{self}}\}$ with $M_a = C_a = \{r, z, s^a_{\mathrm{self}}\}$, $K_a = \{z\}$. Then $s^a_{\mathrm{self}}, r \in R^a_4$ (understood, opaque residue, non-operator) and $z \in R^a_7$ (understood, opaque residue, operator). By Foundations' $\Theta_a = M_a \cap C_a = R^a_4 \cup R^a_7$, all three lie in Θ_a. Let $\mathrm{denotes}_a(s^a_{\mathrm{self}}) = a$, $\mathrm{denotes}_a(r) = r$, and *no other denotation.* In particular no artefact denotes z. Posit z as *held*: the licence $z_{\varphi,r} = z$ with content $(\varphi, r, E, \Sigma, \textsc{true})$ is part of the supplied configuration, for a well-typed φ and evidence $E \subseteq \mathrm{Lnk}(a)$ satisfying $E \vdash \varphi(r)$ — we make no claim about which step produced it. The first-order certificate is held with verdict TRUE. But no $s_z \in \Theta_a$ satisfies $\mathrm{denotes}_a(s_z) = z$ (the denotation function is what it is), so clause (1) of MetaCert fails and $\mathrm{MetaCert}_a(\varphi, r, t) = \textsc{untyped}$. Foundations' `cor:linkage-irreducible` confirms this is genuine: $\mathrm{Lnk}(a)$ is not a function of $\mathrm{Sto}(a)$, so a can hold the licence z without holding any symbol that denotes it. □

Remark 3.7 (Not a refutation of classical S4)*.* Theorem 3.6 says nothing about $K_a\varphi \to K_aK_a\varphi$ as an epistemic-logic axiom. We work with licences (Cert_a) and licences-about-licences ($\mathrm{MetaCert}_a$): an MCK agent holding a licence without a symbol for it cannot mount a licence about it. If $K_a\psi$ is instantiated as $\mathrm{Cert}_a(\psi, s^a_{\mathrm{self}}) = \textsc{true}$, the axiom fails precisely on such agents — v2 supplies the architectural realisability substrate for classical epistemic logic rather than competing with it.

3.3 Reflexive awareness and operational consciousness

Definition 3.8 (Reflexive awareness)**.** Agent a is *reflexively aware* of the certificate $\mathrm{Cert}_a(\varphi, r)$ at t, written $\mathrm{RefAware}_a(\varphi, r, t)$, when $\mathrm{MetaCert}_a(\varphi, r, t) = \textsc{true}$ *and* the meta-symbol s_z of Definition 3.5(1) lies in the effective attention support of a at t (in the sense of Definition 3.9 below).

Definition 3.9 (Effective attention support)**.** Fix an attention weighting $w_a(t) : \mathrm{Sto}(a) \to [0, 1]$ (an internal function of a's architecture, not further modelled here) and a threshold $\theta \in [0, 1)$. The *effective structure* at t is

$$\mathrm{Str}_{\mathrm{eff}}(a, t) \;=\; \{\, x \in \mathrm{Str}(\mathcal{A}_a) \;:\; w_a(t)(x) > \theta \,\}.$$

An artefact x is *effectively attended* at t when $x \in \mathrm{Str}_{\mathrm{eff}}(a, t)$.

Remark 3.10 (Attention is a modulator, not a new region)*.* $w_a(t)$ is not a further Foundations primitive. It is a bounded function whose domain and codomain are already Foundations-owned ($\mathrm{Sto}(a)$ and $[0, 1]$); its role is to modulate which artefacts participate in evaluation at t. Foundations does not discuss attention, and we do not fix a specific weighting mechanism here. The definition is intentionally minimal: it is what Remark 23.2 below requires to state the companion w-criticality criterion.

Definition 3.11 (Operational consciousness)**.** Artefact x is *operationally conscious* for a at t, written $\mathrm{Consc}_a(x, t)$, when:

1. $x \in \mathrm{Sto}(a)$ (represented);
2. $\exists\, s \in \Theta_a$ with $\mathrm{denotes}_a(s) = x$ (well-denoted);
3. $s \in \mathrm{Str}_{\mathrm{eff}}(a, t)$ (effectively attended);
4. s or x participates in some enabled MOV / EXIT step licence at t (action-available in the sense of Definition 3.13).

The negation *operationally unconscious* means $x \in \mathrm{Sto}(a)$ and at least one of (2)–(4) fails.

Remark 3.12 (Explicit non-claims)*.* Definition 3.11 is a predicate over MCK memory and attention. It does not claim phenomenal consciousness, sentience, or a psychoanalytic unconscious. It does not claim that current AI systems instantiate it. It does not claim a general theory of consciousness. Its role is architectural: it names the state in which a's representation of x can enter action selection. Loose analogy: cognitive-neuroscience Global Workspace Theory [3] distinguishes broadcast availability from local storage; Definition 3.11 is a formal analogue at the MCK layer only.

Definition 3.13 (Action-availability)**.** Artefact $x \in \mathrm{Sto}(a)$ is *action-available* at t when either x itself, or some symbol $s \in \Theta_a$ with $\mathrm{denotes}_a(s) = x$, can participate in an enabled step of a's step relation at the current region assignment — as the MOV / EXIT target, as a GEN operand, or as the licence artefact.

Remark 3.14 (The three-way distinction the paper commits to)*. Unrepresented* ($x \notin \mathrm{Sto}(a)$), *represented-but-operationally-inactive* (clause (2), (3) or (4) of Definition 3.11 fails), and *operationally available* ($\mathrm{Consc}_a(x, t)$). "Unconscious" is never applied to the first case: an artefact a does not represent is absent from a's world, not something a is unconscious of.

4 Other-modelling: agent, observation, monitor

An agent that certifies things about itself may also certify things about other agents it acts upon. Cognition of another decomposes into a triad: a model of that agent, a model of what that agent observes, and a model of the monitor that agent runs on what it observes. The three are provably independent.

4.1 The other-model triad

Definition 4.1 (Other-model)**.** For an external agent $b \neq a$, an *other-model of b held by a* is a pair

$$\widehat{M}_a(b) \;=\; (\, s_b,\; \Delta_b \,)$$

where:

1. $s_b \in \Theta_a$ with $\text{denotes}_a(s_b) = b$ (a symbol of a denoting b);
2. $\Delta_b \subseteq \text{Sto}(a)$ is a set of internal artefacts encoding a's attributions to b, i.e. (region, property, region-content-claim) triples that a associates with s_b through $\text{Lnk}(a)$.

Property-relative variant: $\widehat{M}_a^P(b)$ is the restriction of Δ_b to a property family P. May be sufficient for evaluating claims in P even when total $\widehat{M}_a(b)$ is incomplete.

Definition 4.2 (Observation-model)**.** An *observation-model of b held by a*, written $\widehat{O}_a(b)$, is an artefact in $\text{Sto}(a)$ representing b's observation regime: which events, configurations, or trace fragments a predicts b can perceive. Formally, $\widehat{O}_a(b) \subseteq \Sigma_b^*$ (a language over b's alphabet as a represents it), or equivalently a partial function $\widehat{\text{obs}}_a^{\,b} : \Sigma^* \to (\Sigma_b \cup \{\bot\})^*$ mapping ambient traces to their b-observable projections. The *actual* observation regime is O_b; it is external to a and not owned by a's architecture. In general $\widehat{O}_a(b) \neq O_b$.

Definition 4.3 (Predicted monitor)**.** A *predicted monitor of b held by a*, written $\widehat{\text{Mon}}_a^{\,b}(\Phi_b, u)$, is a's prediction of the verdict b's monitor $\text{Mon}_b^{O_b}(\Phi_b, u)$ would return on trace u under property Φ_b. Its value is computed by a using $\widehat{O}_a(b)$ to predict b's observed prefix, and using an internal representation of Φ_b to evaluate that prefix. It ranges over the three-valued prefix-verdict domain $\{\text{TRUE}, \text{FALSE}, \text{UNDETERMINED}\}$ of [4], read against the four-family classification of [10] (with UNTYPED added when Φ_b is not typed for a). The *actual* monitor $\text{Mon}_b^{O_b}(\Phi_b, u)$ is the observation-relative monitor in the sense of [13], generalised from bounded temporal implication to an arbitrary Φ_b.

Remark 4.4 (Non-collapse). $\widehat{M}_a(b)$, $\widehat{O}_a(b)$, $\widehat{\text{Mon}}_a^{\,b}$ live in $\text{Sto}(a)$; the actual triad b, O_b, $\text{Mon}_b^{O_b}$ is external to a. All six objects are pairwise distinct; no identification among them is ever made, and Theorem 4.7 shows the actor-side and receiver-side triads are functionally independent under a's architecture.

4.2 Failure modes and independence

Definition 4.5 (Six failure modes)**.** For each of $\widehat{M}_a(b)$, $\widehat{O}_a(b)$, $\widehat{\text{Mon}}_a^{\,b}$ separately, the artefact may be:

- ABSENT — the required s_b or supporting $\Delta_b / \widehat{O}_a(b) / \widehat{\text{Mon}}_a^{\,b}$ is not in $\text{Sto}(a)$;
- INCOMPLETE — present but does not cover the relevant property family P or the relevant sub-alphabet;
- MIS-DENOTED — $\text{denotes}_a(s_b) \neq b$;

- ILL-TYPED — content violates the region assignment of b's actual $\mathrm{Sto}(b)$;
- OUTDATED — accurate at $t' < t$, no longer accurate at t;
- FALSE — present, well-typed, current, but its content contradicts b's actual configuration.

Note ABSENT $\neq$ FALSE: absence returns UNTYPED, a false model returns a definite wrong verdict; the two require different remediation (Proposition 9.5).

Proposition 4.6 (Self-other independence)**.** *There exist finite MCK agents a_1, a_2 such that:*

- *a_1 holds $s^{a_1}_{\mathrm{self}} \in \Theta_{a_1}$ (self-symbol present) and no $s_b \in \Theta_{a_1}$ for any $b \neq a_1$ (no other-model);*
- *a_2 holds $s_b \in \Theta_{a_2}$ with $\mathrm{denotes}_{a_2}(s_b) = b$ for some external b (other-model present) and no $s^{a_2}_{\mathrm{self}}$ (no self-symbol).*

Proof. Explicit finite witnesses; see Appendix A, R2. □

Theorem 4.7 (Model-of-b vs. model-of-observation independence)**.** *There exists a bounded MCK agent a holding $\widehat{M}_a(b)$ that is adequate on b's $\mathrm{Sto}(b)/M_b/C_b/K_b$ assignment (accurate model of b's classification), while $\widehat{O}_a(b)$ is inaccurate (disagrees with O_b on some sub-alphabet).*

Proof. Explicit finite witness. Sketch: take b with two component-artefacts, one of which a correctly places in $M_b \cap K_b = R^b_5$ and the other in $C_b \cap K_b = R^b_6$. Have a's $\widehat{O}_a(b)$ include an alphabet letter ϵ (an event) that O_b does *not* let b observe. Then $\widehat{M}_a(b)$'s region assignments agree with b's actual $\mathrm{Sto}(b)$ classification, but $\widehat{O}_a(b)$ predicts b observes ϵ when b does not, so $\widehat{O}_a(b) \neq O_b$. Full witness enumeration in Appendix A, R4. □

Remark 4.8 (Why this independence is central for W_C/W_D). Theorem 4.7 is what separates W_C and W_D from a total-model-failure world: $\widehat{M}_a(b)$ can be perfectly good while $\widehat{\mathrm{Mon}}^{\,b}_a$ is uselessly miscalibrated because $\widehat{O}_a(b)$ is wrong — this is where monitor-blindness lives.

5 Three requirements and their dependency

5.1 The common base

Definition 5.1 (Self-attribution base)**.** For a self-property p,

$$\mathrm{BaseReq}_{\mathrm{self}}(p) \;=\; \big(\text{a self-referent for the subject},\ \text{a recogniser } \mathrm{Eval}_a(p)\big).$$

Only two components appear, deliberately: for a property evaluated at the current configuration (viability at an instant) nothing beyond that configuration is consulted, so an observation interface is not universally required. Inflating the shared base would understate the differential requirements that follow; per Convention 2.5 the base is not universalised beyond self-attribution.

5.2 Representational typability

Definition 5.2 (Type requirement)**.** Let $\mathcal{S}$ be a set of *structures* an architecture may hold. The *type requirement* $\mathrm{TypeReq}(p) \subseteq \mathcal{S}$ is the set of structures over which the evaluation of p quantifies. An architecture holding $\mathrm{Str} \subseteq \mathcal{S}$ *types* p when $\mathrm{TypeReq}(p) \subseteq \mathrm{Str}$. If p is not typed, then $\mathrm{Verdict}_a(p, x, h) = \text{UNTYPED}$ for every h.

5.3 Information sufficiency

Definition 5.3 (Attribution architecture and sufficiency)**.** An *attribution architecture* for p over Σ is a map $u : \Sigma^* \to Q$ that is *incrementally maintainable*: there is $\delta : Q \times \Sigma \to Q$ with $u(h \cdot e) = \delta(u(h), e)$. It is *sufficient* for p when

$$u(h) = u(h') \implies p(h) = p(h') \qquad \text{for all } h, h' \in \Sigma^*.$$

InfoReq(p) denotes a sufficient u of least state count, when one exists.

Remark 5.4 (Why incremental maintainability is not a convenience)*.* Without that clause the definition is empty: $u := p$ itself is sufficient for every p, uniformly, with two states, and no property distinguishes itself from any other. The clause is what makes InfoReq property-specific, and it is what makes the least sufficient u the Myhill–Nerode right-congruence rather than the verdict partition.

5.4 Temporal certification

We import the four-family classification of ω-properties by which verdicts finite prefixes can carry [10], in the Alpern–Schneider and good-prefix/bad-prefix lineage [1, 4]. For $\Phi \subseteq \Sigma^\omega$: $\Phi \in \mathsf{S}$ when every violation has a finite refuting prefix; $\Phi \in \mathsf{C}$ when every satisfaction has a finite confirming prefix; $\Phi \in \mathsf{L}$ when no finite prefix refutes; $\Phi \in \mathsf{K}$ when no finite prefix confirms. We write Temp(p) for the classification of p.

Convention 5.5 (Notation)*.* $\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}$ are property classes; G and F are the temporal operators *always* and *eventually*. The two are kept distinct throughout: $G(b) \in \mathsf{S}$ is a statement relating an operator to a class, not an abbreviation.

Remark 5.6 (Finite prefix is not finite memory)*.* Temp classifies which verdicts a finite *prefix* can carry. It does not bound the *memory* an agent needs: $G(p)$ is refuted on a finite prefix and costs one bit. The two axes are independent, and conflating them is the error this separation is designed to prevent.

5.5 The dependency, and why this is not a product

Remark 5.7 (Type dependency, observation-level)*.* If an architecture does not type p, then InfoReq(p) is undefined for it: sufficiency in Definition 5.3 is a condition on u relative to the verdict function p, and if p is untyped every history receives UNTYPED (Definition 5.2), so the sufficiency implication has no truth value to refine. We stipulate *undefined* rather than trivially satisfied, so a least sufficient u is not spuriously determined. This is a bookkeeping convention on the joint definition, recorded as Remark 5.7 for cross-referencing; its content is definitional.

Definition 5.8 (Typed profile)**.** $\mathrm{Prof}_a(p) = (\mathrm{TypeReq}(p), \mathrm{InfoReq}_a(p), \mathrm{Temp}(p))$, read in that order, the second component being defined only when the first is satisfied.

Remark 5.9 (Not a product)*.* Remark 5.7 forbids reading Definition 5.8 as a product of independent coordinates, and we prove no independence. What Section 22 proves is weaker and sufficient: a transformation may fix one component and change another.

6 Action, intention, predicted effect, actual verdict

Between an agent's certificate calculus and the receiver's actual monitor lies action. This section formalises action, intention, and predicted effect on the actor side; the actual effect and actual monitor sit outside a's architecture and are imported.

6.1 Actor side

Definition 6.1 (Action)**.** $\mathrm{Act}(a, \alpha)$ denotes an occurrence of action $\alpha \in A$ by agent a, producing a trace extension $u_\alpha \in \Sigma^*$ observable in the ambient event alphabet $\Sigma \supseteq A_a$. The action is realised via a's step relation (Foundations `def:step`); Own_a is *this paper's* ownership partition (Definition 7.1), not a Foundations object.

Definition 6.2 (Intention)**.** An *intention* of a is an internal artefact $i_{\alpha,g} \in \Theta_a$ whose denotation target is a composite artefact $x_{\alpha,g} \in \mathrm{Sto}(a)$ — itself in $\mathcal{A}_a^+$, as the target clause of Foundations' denotation requires — whose represented content (Δ_a-level) is the pair (α, g), where $\alpha \in \mathrm{Own}_a$ is an action and g is a goal artefact held by a. We write $\mathrm{Intent}_a(\alpha, g, t)$ for the holding of such an artefact at t. Intention is internal to a and not observable to any $b \neq a$; a communicative act "a says: I intend (α, g)" is a separate action, whose relation to Intent_a is not assumed.

Remark 6.3 (No utility, no numerical reward)*.* g is a goal artefact — a symbol, a property, a state fragment — not a scalar. The theory admits categorical goals without numerical utility, and this is deliberate: numerical rewards would introduce a representation-cost the paper wishes to keep out of the base algebra. Assistance-game and CIRL formulations [28] adopt a reward-based framing; we do not, and we distinguish what CIRL formalises as reward-inference-error from what Definition 4.5 calls a false model.

Definition 6.4 (Predicted effect)**.** For an action α by a addressed to (or affecting) an external agent b, the *predicted effect* is an artefact $\widehat{\mathrm{Effect}}_a(\alpha, b) \in \mathrm{Sto}(a)$ representing a's prediction of α's effect on b. Its computation quantifies over $\widehat{M}_a(b)$; well-formedness requires that $\widehat{M}_a(b)$ is not ABSENT in the sense of Definition 4.5.

6.2 Receiver side (imported)

Definition 6.5 (Actual effect)**.** $\mathrm{Effect}(\alpha, b, w)$ is the actual effect on b of action α by a in world w. External to both a and b; not owned by any agent's architecture. In general $\mathrm{Effect}(\alpha, b, w) \neq \widehat{\mathrm{Effect}}_a(\alpha, b)$.

Definition 6.6 (Actual receiver monitor)**.** $\mathrm{Mon}_b^{O_b}(\Phi_b, u)$ is the monitor b runs on the trace u for property Φ_b under its actual observation regime O_b, generalising to an arbitrary property the observation-relative setting that [13] develops for bounded temporal implication. It ranges over the three-valued prefix verdicts [4] augmented with UNTYPED where Φ_b is untypable for b. Its value depends on O_b: for the same trace u, distinct observation regimes may yield distinct verdicts.

6.3 Comparison

Definition 6.7 (Model-verdict agreement)**.** The actor-side prediction and receiver-side verdict *agree* for $(\alpha, b, \Phi_b, u_\alpha)$ when

$$\widehat{\mathrm{Mon}}_a^{\,b}(\Phi_b, u_\alpha) \;=\; \mathrm{Mon}_b^{O_b}(\Phi_b, u_\alpha).$$

Mismatch means the two disagree. Agreement is not implied by any of: adequate $\widehat{M}_a(b)$, adequate $\widehat{O}_a(b)$, correct Intent_a, or well-formed $\mathrm{Cert}_a(\varphi, r)$ for any φ, r.

Proposition 6.8 (Intent / receiver-verdict separation)**.** *Fix* $\mathrm{Intent}_a(\alpha, \mathit{help}, t)$*. There exist observation regimes* O_{b_1}, O_{b_2} *and property* Φ_b *such that* $\mathrm{Mon}_{b_1}^{O_{b_1}}(\Phi_b, u_\alpha) \neq$ FALSE *(non-failure:* UNDETERMINED *on* Φ_b *itself, or* TRUE *under a monitorable restatement of the property visible in* b_1*'s regime) and* $\mathrm{Mon}_{b_2}^{O_{b_2}}(\Phi_b, u_\alpha) =$ FALSE *(failure).*

Proof. Explicit finite witnesses; see Appendix A, R3. Sketch: let α be "make a decision for b without asking". b_1's observation regime records only outcomes (O_{b_1} = "final state"); b_2's regime records the interaction (O_{b_2} = "full trace"). The outcome may be beneficial: Mon_{b_1} evaluates a safety property "final state is admissible" and returns TRUE. The interaction violates a coliveness property $\Phi_b = G(\text{consulted-before-decision-affecting-}b)$ in class $\mathsf{S} \cap \mathsf{K}$; Mon_{b_2} returns FALSEon the finite prefix that skipped consultation. Same u_α, same $\mathrm{Intent}_a(\alpha, \mathrm{help})$, different receiver-side verdicts. The proposition is established by the finite witness R3; it is the cognitive-algebra counterpart of the observation-relativity that [13] proves for bounded temporal implication, whose theorem is neither invoked nor reproved here. □

7 A common carrier, refinement, and capability

The three properties of Sections 16 to 18 are evaluated over data of different kinds — a history, a state, an action labelling. Comparing their requirements demands one common carrier: comparing quotients of histories with predicates on states directly is a category error, and it is why Section 19 is stated over configurations.

Definition 7.1 (Configuration carrier)**.** Fix a finite event alphabet Σ, a finite set K of system components with values in a finite V, and a finite action alphabet A. A *configuration* is a triple

$$c \;=\; (h,\, s,\, O) \;\in\; \mathcal{C} \;=\; \Sigma^* \times V^K \times \mathcal{P}(A),$$

where h is the run so far, s the *global* component state, and $O \subseteq A$ the actions the agent controls.

Remark 7.2 (The carrier is not chosen for the examples)*.* This triple is fixed before any self-property is selected, and each coordinate is standard. A run over an event alphabet is the state of an interpreted system [24]; a global state over components is any transition system; and a distinguished subset of the action alphabet that the controller owns is the controllable/uncontrollable split of supervisory control [44]. Nothing in Definition 7.1 mentions the agent's own component, its viability, or its continuity.

The carrier deliberately does *not* record the value of the agent's own component as a coordinate: that would presuppose the ownership relation one case study needs and make the separation results of Section 19 partly true by construction. Here ownership is a structure an architecture may or may not hold, and the integrity architecture must apply it to s rather than read it off (Lemma 19.3).

Every architecture below is a function on $\mathcal{C}$, and every property below is a predicate on $\mathcal{C}$. Nothing forces an architecture to consult all three coordinates; the point of the carrier is that what it *fails* to consult is expressible.

Definition 7.3 (Architecture, refinement)**.** An *architecture* is a pair $\mathcal{A} = (\mathrm{Str}, u)$ with $\mathrm{Str} \subseteq \mathcal{S}$ and $u : \mathcal{C} \to Q$. For architectures $\mathcal{A}, \mathcal{B}$ over the same carrier,

$$\mathcal{A} \preceq \mathcal{B} \quad \text{iff} \quad \mathrm{Str}(\mathcal{A}) \subseteq \mathrm{Str}(\mathcal{B}) \;\text{ and }\; \exists\, \varphi : Q_{\mathcal{B}} \to Q_{\mathcal{A}} \;\text{ with } u_{\mathcal{A}} = \varphi \circ u_{\mathcal{B}}.$$

We say $\mathcal{B}$ *refines* $\mathcal{A}$.

The second conjunct says $u_{\mathcal{B}}$'s partition of $\mathcal{C}$ is finer than $u_{\mathcal{A}}$'s. It is the weakest relation supporting the arguments below — φ preserves no structure; anything stronger would make the negative results of Section 19 easier and less informative.

Definition 7.4 (Sufficiency on the carrier, minimality)**.** $\mathcal{A}$ is *sufficient* for a property p on $\mathcal{C}$ when $u_{\mathcal{A}}(c) = u_{\mathcal{A}}(c')$ implies $p(c) = p(c')$, and when $\mathrm{Str}(\mathcal{A}) \supseteq \mathrm{TypeReq}(p)$. Given a class $\mathcal{F}$ of candidate architectures, $\mathcal{A} \in \mathcal{F}$ is *minimal sufficient for p within $\mathcal{F}$* when it is sufficient and no $\mathcal{A}' \in \mathcal{F}$ with $\mathcal{A}' \prec \mathcal{A}$ is sufficient.

Convention 7.5 (Minimality is always relative to a stated class). Every minimality claim in this paper is of the form "minimal within $\mathcal{F}$" for an $\mathcal{F}$ exhibited in the same section. We make no claim of minimality across arbitrary encodings or arbitrary formalisms, and where the candidate class is changed the claim must be re-established.

7.1 Capability

Definition 7.6 (Self-attribution capability)**.** For an architecture $\mathcal{A}$,

$$\mathrm{SelfCap}(\mathcal{A}) \;=\; \{\, p \;:\; \mathrm{TypeReq}(p) \subseteq \mathrm{Str}(\mathcal{A}) \,\}.$$

SelfCap answers a different question from sufficiency: it says which self-properties are *meaningfully attributable* by an architecture, before any question of whether present evidence decides them. A property outside $\mathrm{SelfCap}(\mathcal{A})$ receives UNTYPED from $\mathcal{A}$ at every history.

Proposition 7.7 (Monotonicity)**.** *If* $\mathcal{A} \preceq \mathcal{B}$ *then* $\mathrm{SelfCap}(\mathcal{A}) \subseteq \mathrm{SelfCap}(\mathcal{B})$.

Proof. $\mathcal{A} \preceq \mathcal{B}$ gives $\mathrm{Str}(\mathcal{A}) \subseteq \mathrm{Str}(\mathcal{B})$ by Definition 7.3. If $p \in \mathrm{SelfCap}(\mathcal{A})$ then $\mathrm{TypeReq}(p) \subseteq \mathrm{Str}(\mathcal{A}) \subseteq \mathrm{Str}(\mathcal{B})$, so $p \in \mathrm{SelfCap}(\mathcal{B})$. □

Remark 7.8 (What this proposition is worth). The statement is close to definitional — SelfCap is upward-closed because TypeReq-satisfaction is — and is recorded only because it makes Section 24 well-posed: adding structure never removes an attribution. It is exhaustively confirmed over all 32 sub-architectures of the structure set used in this paper.

8 Regional cognitive algebra

The v2 cognitive algebra of Sections 3, 4, 6 and 21 runs over MCK-typed artefacts. This section is the regional-realisability audit of that vocabulary: for every cognitive object, where the artefact lives, which of M_a, C_a, K_a hold, which region R_i^a follows, and what carriers downstream operations require. All region definitions and predicates are Foundations-owned [12] and imported without redefinition (Definitions 3.2–3.4, the region definitions of Section 4, Theorem 4.8, the R_3-suspension axiom, and the definition of Θ_a; the full upstream contract is Section 11); every regional claim below cites the Foundations object it consumes.

8.1 Regional signature

Definition 8.1 (Regional signature)**.** For agent a, artefact $x \in \mathcal{A}_a^+$, and time t (unused when the classification is static, which is the default), the *regional signature* of x under a is

$$\sigma^a(x,t) \;=\; \big(\,\mathbf{1}[x \in M_a],\ \mathbf{1}[x \in C_a],\ \mathbf{1}[x \in K_a]\,\big) \;\in\; \{0,1\}^3.$$

The *canonical region* of x under a is $\mathrm{Region}_a(x) = R_i^a$ iff $\sigma^a(x,t)$ matches the Boolean triple of R_i per Foundations `def:region-r1`–`def:region-r7`; and $\mathrm{Region}_a(x) = \bot$ (absolute exterior) when $\sigma^a(x,t) = (0,0,0)$.

8.2 Category discipline: meta-level vs. reified

A central discipline point separates a mathematical function or relation defined by the theory from an internal artefact of a's architecture that represents, denotes, or executes that function. Regional membership applies only to the latter.

Convention 8.2 (Meta-level ≠ reified). Throughout this paper:

- $\text{denotes}_a : \Theta_a \to \text{Ref}$ is a meta-level function of the theory; it has *no* regional membership. What may have regional membership is an internal artefact of a's $\text{Sto}(a)$ that represents, denotes, or executes (an approximation of) denotes_a.
- Likewise for $\text{Effect}(\alpha, b, w)$, O_b, $\text{Mon}_b^{O_b}$, $\text{Truth}(\varphi, r)$: all meta-level or external objects, not in a's regions.
- Likewise for a property φ (a subset of the domain, meta-level) versus its property-symbol $s_\varphi \in \Theta_a$ (an artefact in a's $\text{Sto}(a)$).
- Likewise for an operator f as a mathematical map, versus an executable operator artefact $\hat{f} \in \text{Sto}(a)$ (with $K_a(\hat{f}) = 1$) that a can invoke via Foundations' step relation.

Regional membership is stated only for reified artefacts. When the paper writes "$s_\varphi \in R_4^a$" it names the artefact; when it writes "φ is a property in the awareness literature" it names the meta-level object; the two are not conflated.

8.3 The canonical regional inventory

Table 1 is the canonical regional interface for the v2 cognitive algebra. It is meant to be used by downstream programme papers as the reference for what regional carrier every cognitive object requires.

8.4 Symbol algebra: consequences of $\Theta_a = M_a \cap C_a$

Foundations `def:theta` gives $\Theta_a = M_a \cap C_a = R_4^a \cup R_7^a$. Every artefact that denotes under a is a symbol under a, and every symbol under a lies in one of two regions only. This constrains everything the paper builds.

Proposition 8.3 (Symbol-region restriction)**.** *Every $s \in \Theta_a$ satisfies $\sigma^a(s) \in \{(1,1,0),(1,1,1)\}$, i.e. $s \in R_4^a$ (non-operator symbol) or $s \in R_7^a$ (operator symbol).*

Proof. Immediate from Foundations $\Theta_a = M_a \cap C_a$ and the region definitions $R_4 = (M \cap C) \setminus K$, $R_7 = M \cap C \cap K$ (`def:region-r4,def:region-r7`). Any other Boolean triple violates $M_a \wedge C_a$. □

Corollary 8.4 (Symbol carriers are R_4-or-R_7)**.** *The self-symbol s^a_{self}, every property-symbol s_φ, every other-symbol s_b, every certificate-symbol s_z, and every intent-symbol $i_{\alpha,g}$ lies in $R_4^a \cup R_7^a$. When the symbol is operator-bearing (e.g. the symbol also participates in an execution licence), it lies in R_7^a; otherwise in R_4^a.*

8.5 Certificate regional decomposition

Definition 8.5 (Certificate regional signature)**.** The *regional signature of a certificate* $\text{Cert}_a(\varphi, r, t)$ realised by licence $z_{\varphi,r}$ is the tuple

$$\text{RegCert}_a(\varphi, r, t) \;=\; \big(\text{Region}_a(r),\ \text{Region}_a(s_\varphi),\ E \subseteq \text{Lnk}(a),\ \text{Region}_a(z_{\varphi,r})\big).$$

By Corollary 8.4 the second component lies in $\{R_4^a, R_7^a\}$; by Table 1 the fourth lies in $\{R_3^a, R_5^a, R_6^a, R_7^a\}$ (any K-cell; R_3^a excluded under the subjective reading of K, on which $R_3^a = \emptyset$ — Foundations' `def:k-objective-subjective`). The first is any active region.

Definition 8.6 (Meta-certificate regional signature)**.** Additionally requires a certificate-symbol carrier:

$$\text{RegMetaCert}_a(\varphi, r, t) \;=\; \big(\text{RegCert}_a(\varphi, r, t),\ \text{Region}_a(s^a_{\text{self}}),\ \text{Region}_a(s_z),\ \text{Region}_a(z_{\eta,s_z})\big).$$

$s^a_{\text{self}}, s_z \in R_4^a \cup R_7^a$; z_{η,s_z} is a K-cell licence.

Table 1: Canonical regional inventory for the v2 cognitive algebra. LEV = meta-level (M) vs. reified (R). Columns M, C, K record the necessary Foundations predicate on the reified artefact (1 = must hold, 0 = must not hold, $*$ = either admissible). REG names the induced canonical region. NEC distinguishes MUST (necessary for downstream capability), MAY (admissible carrier), CANNOT (structurally excluded). UP lists upstream regional requirements. OP lists a downstream operation the object enables.

Object	LEV	M	C	K	REG	NEC	UP	OP
referent r	R	any active region ($R_1..R_2, R_4..R_7$)				MAY	—	target of denotes
property φ	M	—	—	—	—	—	—	—
property-symbol s_φ	R	1	1	$*$	$R_4 \cup R_7$	MUST	—	denote φ
self-symbol s^a_{self}	R	1	1	$*$	$R_4 \cup R_7$	MUST for de se	—	anchor for de se cert
other-symbol s_b	R	1	1	$*$	$R_4 \cup R_7$	MUST for other-model	—	anchor for $\widehat{M}_a(b)$
evidence object E	R	fragment of Lnk(a); not a single-region artefact				MUST	$\{s_\varphi, r\}$	feed Cert
certificate licence $z_{\varphi,r}$	R	$*$	$*$	1	R_3, R_5, R_6, R_7	MUST	$\{s_\varphi, r, E\}$	MOV / EXIT licence
certificate-symbol s_z	R	1	1	$*$	$R_4 \cup R_7$	MUST for MetaCert	$z_{\varphi,r}$	denote a certificate
meta-cert licence z_{η,s_z}	R	$*$	$*$	1	R_3, R_5, R_6, R_7	MUST for MetaCert	$\{s_z, s^a_{\text{self}}, E'\}$	MOV/EXIT of η
attention weight w_a	M	—	—	—	—	—	—	modulates Str_{eff}
$\text{Consc}_a(x,t)$	M	predicate over reified artefacts				—	Definition 3.11(1)-(4)	step-selection eligibility
$\widehat{M}_a(b)$ counterpart Δ_b	R	subset of Sto(a) (region assignment claims about b)				MUST for other-model	s_b	PredEffect input
$\widehat{O}_a(b)$ artefact	R	1	$*$	$*$	any of R_1, R_4, R_5, R_7	MUST for $\widehat{\text{Mon}}$	s_b	PredMon input
$\widehat{\text{Mon}}^b_a$ artefact (data only)	R	1	$*$	0	R_1 or R_4	MAY	$\widehat{M}_a(b), \widehat{O}_a(b)$	store predicted verdict
$\widehat{\text{Mon}}^b_a$ artefact (executable)	R	$*$	$*$	1	R_3, R_5, R_6, R_7	MAY	$\widehat{M}_a(b), \widehat{O}_a(b), \Phi_b$-repr	compute predicted verdict
$\text{Intent}_a(\alpha, g)$ artefact	R	1	1	$*$	$R_4 \cup R_7$	MUST for intent tag	α, g	tag Act
$\widehat{\text{Effect}}_a(\alpha, b)$ artefact	R	1	$*$	$*$	any of R_1, R_4, R_5, R_7	MUST for $W_B..W_D$	$\widehat{M}_a(b), \alpha$	feed PredMon
action $\alpha \in \text{Own}_a$	R	$*$	$*$	1	R_3, R_5, R_6, R_7	MUST	—	step realiser
execution licence for α	R	$*$	$*$	1	R_3, R_5, R_6, R_7	MUST	α, E	realise MOV / EXIT
feedback object f	R	new artefact received from environment; classified when apprehended				MAY	—	update $\widehat{O}_a$ / $\widehat{M}_a$
Own_A carrier	R	1	$*$	$*$	R_1, R_4, R_5, R_7	MUST for p_{free}	—	structural for freedom
Own_a (self-membership over comps.)	R	1	$*$	$*$	R_1, R_4, R_5, R_7	MUST for p_{int}	s^a_{self}	structural for integrity
$\text{Effect}(\alpha, b, w)$	EXTERNAL. Not in a's regions.							—
O_b	EXTERNAL. Not in a's regions.							—
$\text{Mon}^{O_b}_b(\Phi_b, u)$	EXTERNAL. Not in a's regions. Imported per [13].							—
$\text{Truth}(\varphi, r)$	EXTERNAL. Meta-level.							—
denotes_a (meta-level map)	EXTERNAL. Foundations meta-level function.							—

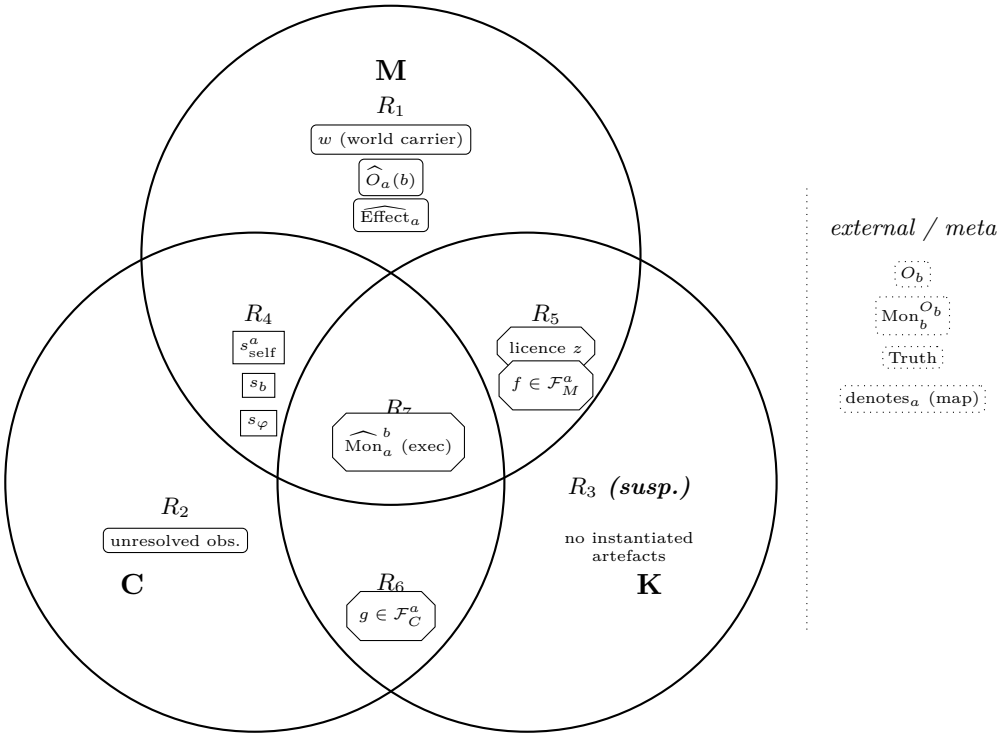


Figure 1: Figure P: the MCK regional atlas. Three circles are the Foundations predicates M, C, K; the seven canonical regions $R_1 \dots R_7$ are labelled at their cells, with R_3 marked suspended (empty in the agent-relative atlas under the subjective reading of K, Foundations' `def:k-objective-subjective`; the agent-independent suspension is `ax:r3-suspension`). Representative artefacts of this paper's cognitive algebra are placed at their spec-derived cells (at most three per region; Table 1 carries the full census): plain rectangles are Θ_a-symbols, rounded rectangles plain artefacts, chamfered boxes operators, and dotted-border objects on the right are external or meta-level — never inside the circles. The figure is machine-generated from the same source as the checker (`regional_visual_spec.py`); no region label is hand-typed.

Theorem 8.7 (MetaCert regional lower bound)**.** $\text{MetaCert}_a(\varphi, r, t) \in \{\textit{TRUE}, \textit{FALSE}\}$ *requires strictly more distinct regional carriers than* $\text{Cert}_a(\varphi, r, t) \in \{\textit{TRUE}, \textit{FALSE}\}$*. Specifically:* RegCert_a *requires four carriers (referent, property-symbol, evidence, licence);* RegMetaCert_a *requires those plus two more (self-symbol, certificate-symbol) plus a meta-cert licence, so at least seven distinct* $\text{Sto}(a)$*-entries with the constraint that at least three of them occupy* $R_4^a \cup R_7^a$ *(the property-symbol, the self-symbol, and the certificate-symbol are all* Θ_a*-typed) and at least two occupy a* K*-cell.*

Proof. By Definitions 8.5 and 8.6 the fields are independent and Foundations-typed as stated. No carrier can be re-used across the meta-symbol and the first-order licence, because the meta-symbol must *denote* the licence (be a Θ_a-member), and the licence itself is a K-artefact (may or may not be in Θ_a); if the licence happens to lie in R_7^a it is a member of Θ_a, but is still not a symbol of itself. Distinctness is a $\text{Sto}(a)$-cardinality condition. □

8.6 Operator realisability: representation, execution, certification

Definition 8.8 (Operator hierarchy)**.** For an internal operator artefact $\hat{f} \in \text{Sto}(a)$ representing mathematical map f, distinguish:

REPRESENTED: $\hat{f} \in \mathrm{Sto}(a)$, $\sigma^a(\hat{f})$ has $M_a(\hat{f}) = 1$ (understood as data), $K_a(\hat{f}) = 0$ (no operator status). Region: R_1^a or R_4^a.

EXECUTABLE: $\hat{f} \in \mathrm{Sto}(a)$ with $K_a(\hat{f}) = 1$. Region: any K-cell (subject to R3 suspension).

CERTIFIABLE: executable, and there exists a $\mathrm{Cert}_a(\psi, \hat{f}, t)$ for some property ψ about $\hat{f}$.

REFLEXIVELY CERTIFIABLE: certifiable and additionally $\mathrm{MetaCert}_a(\psi, \hat{f}, t)$ exists.

Theorem 8.9 (Representation-execution separation)**.** *There exists a bounded MCK agent a with $\hat{f} \in \mathrm{Sto}(a)$ representing map f but not executable: $\sigma^a(\hat{f}) = (1, 0, 0) = R_1^a$. Then a can query about f (subject to Foundations* `thm:query-horizon`*) but cannot execute f in a step of a's architecture; the operator is data, not machinery.*

Proof. Take $\mathcal{A}_a = \{\hat{f}\}$; $M_a = \{\hat{f}\}$; $C_a = K_a = \emptyset$. Then $\hat{f} \in R_1^a$ (understood, transparent, non-operator). No step of a's architecture using $\hat{f}$ as an operator f is realisable, because $K_a(\hat{f}) = 0$ prevents composition under Foundations `def:architecture`'s $\circ_a$. Yet a query (q, v) with q denoting $\hat{f}$ requires some symbol $q \in \Theta_a$ ($\hat{f}$ itself need only be denotable); here $\Theta_a = M_a \cap C_a = \emptyset$, so query is also inapplicable in this minimal witness, which makes $\hat{f} \in R_1^a$ only. Refined witness: add $s_{\hat{f}} \in \Theta_a$ with $\mathrm{denotes}_a(s_{\hat{f}}) = \hat{f}$; then a has a symbol for $\hat{f}$ and can query about $\hat{f}$, but $\hat{f}$ itself remains in R_1^a and cannot execute. □

Theorem 8.10 (Execution-certification separation)**.** *There exists a bounded MCK agent a with $\hat{f}$ executable ($K_a(\hat{f}) = 1$) and no $\mathrm{Cert}_a(\psi, \hat{f})$ for any ψ, because either the property-symbol s_ψ is absent from Θ_a, or the evidence $E \subseteq \mathrm{Lnk}(a)$ does not support any ψ-verdict on $\hat{f}$.*

Proof. Take $\mathcal{A}_a = \{\hat{f}\}$; $M_a = C_a = \emptyset$; $K_a = \{\hat{f}\}$. Then $\hat{f} \in R_3^a$ (K-only), which is the suspended cell — refined witness: $M_a = \{\hat{f}\}$, $C_a = \{\hat{f}\}$ (so $\hat{f} \in R_7^a$), $K_a = \{\hat{f}\}$; a has no other artefact. Then $\Theta_a = \{\hat{f}\}$; the only candidate symbol denotes $\hat{f}$ itself, not any property ψ; no s_ψ exists, so no attribute certificate for $\hat{f}$ is well-formed under Definition 2.6 clause (1). $\hat{f}$ is executable via composition with itself as identity but no property of $\hat{f}$ is certified. □

8.7 Regional non-injectivity

Proposition 8.11 (Regional non-injectivity)**.** *Distinct regional configurations of a may project to the same high-level cognitive predicate. Concretely: two witnesses of $\mathrm{Cert}_a(\varphi, r, t) = \textsc{true}$ can differ in $\mathrm{RegCert}_a(\varphi, r, t)$ (one with $z \in R_5^a$, another with $z \in R_7^a$).*

Proof. Explicit witnesses in the extended checker distributed with this article. Sketch: $z \in R_5^a$ means the licence is a transparent operator (understood, non-opaque, executable); $z \in R_7^a$ means the licence is symbol-bearing in addition. Both satisfy Definition 2.6, so both carry a first-order certificate. Only the R_7^a witness supports a $s_z \in \Theta_a$ trivially (the licence *is* in Θ_a); the R_5^a witness requires a separate certificate-symbol for MetaCert. □

8.8 Critical cut

Theorem 8.12 (Critical regional cut)**.** *Removal of the self-symbol carrier from a's $\mathrm{Sto}(a)$ (equivalently: transitioning s^a_{self} out of Θ_a, i.e. removing M_a or C_a from it) invalidates every de se $\mathrm{Cert}_a(\varphi, s^a_{\mathrm{self}})$ and every $\mathrm{MetaCert}_a(\varphi, r)$, regardless of φ or r. The self-symbol is a* critical regional cut *in the capability graph.*

Proof. By Definition 2.6 clause (2), a de se $\mathrm{Cert}_a(\varphi, s^a_{\mathrm{self}})$ requires $s^a_{\mathrm{self}} \in \Theta_a$ (as the referent's carrier); its removal turns clause (2) to UNTYPED. By Definition 3.5 clause (1)–(2), any $\mathrm{MetaCert}_a$ requires $s^a_{\mathrm{self}} \in \Theta_a$; its removal turns MetaCert to UNTYPED regardless of φ, r. This proves the cut. Compare Theorem 23.3: identity's upstream role is exactly the persistence of $s^a_{\mathrm{self}} \in \Theta_a$ across scope. □

Remark 8.13 (The criticality DAG has an obvious minimum cut). The singleton $\{s^a_{\mathrm{self}} \in \Theta_a\}$ is a distinguished minimum cut of the capability graph induced by Definition 23.1 (in Section 23): its removal collapses de se and reflexive certification simultaneously, while downstream cuts (Own_a, Own_A, $\widehat{O}_a(b)$) have smaller blast radius. The DAG structure is inherited from Definition 2.6's well-formedness dependencies.

8.9 Regional transitions and cognitive events

Convention 8.14 (Transitions are Foundations-owned). The region transitions this paper appeals to are Foundations' GEN / MOV / EXIT of `def:step`, later extended by two declared families — OBS (supply) and SEM (content rewrite) — in Definition 10.20; the extension is named there precisely because Foundations contains neither a supply operation nor a mutable denotation. A "cognitive event" is a sequence of such steps that together modifies $(\mathrm{Sto}(a), \mathrm{Exe}(a), \mathrm{Lnk}(a))$ in a way that changes the regional signature or the denotation record of some $x \in \mathcal{A}^+_a$.

Example 8.15 ("Becoming aware of x" as a regional path)**.** Suppose $x \notin \mathcal{A}^+_a$ (absolute exterior for a). A GEN step of a's architecture producing an artefact y that apprehends x (introducing $M_a(y) = 1$ or $C_a(y) = 1$) places y into R^a_1, R^a_2 or a higher region depending on what other predicates hold. A subsequent GEN step producing a symbol $s_y \in \Theta_a$ with $\mathrm{denotes}_a(s_y) = y$ places s_y into R^a_4 or R^a_7. A further GEN step producing a licence $z_{\varphi,y}$ about some property $\varphi(y)$ places z into a K-cell. The sequence $\mathrm{GEN}(y) \to \mathrm{GEN}(s_y) \to \mathrm{GEN}(z_{\varphi,y})$ is a regional path realising "a becomes aware, denotationally capable, and certification-capable regarding x". It is a path, not a dynamics: which GEN step a supports at which time is acquisition, which Foundations' `rem:execution-boundary` leaves unmodelled (cf. Remark 10.23); the example gives only the checkpoints any acquisition would pass through.

8.10 Four-world regional normal forms

- W_A: $\widehat{M}_a(b)$ absent or Δ_b lacks a PredEffect-computation carrier. Regional: no $s_b \in \Theta_a$ denoting b, or s_b present but no $e_{\alpha,b} \in \mathrm{Sto}(a)$ for the acted-upon α.
- W_B: full triad present; each artefact in its canonical region per Table 1; $\mathrm{Region}_a(\widehat{O}_a(b))$-compatibility with the actual receiver-side sub-alphabet is externally verified.
- W_{C_1}: $\widehat{O}_a(b) \in R^a_4$ or R^a_1 correctly records receiver-side blindness (the artefact's content specifies restriction of O_b); $\widehat{\mathrm{Mon}}^{\,b}_a$ artefact absent because there is nothing to predict.
- W_{C_2}: $\widehat{O}_a(b) \in R^a_4$ or R^a_1 with content that *fails* to record receiver-side observability; the regional membership is not the failure locus — the failure is content-level, per Corollary 8.16.
- W_D: full triad; $\widehat{\mathrm{Mon}}^{\,b}_a$ artefact present (executable or data). $W_{D\text{-}F_{\mathrm{INT}}}$ requires $\mathrm{Intent}_a(\alpha, g_{\mathrm{coerce}}) \in R^a_4 \cup R^a_7$; $W_{D\text{-}F_{\mathrm{MODEL}}}$ requires $\mathrm{Intent}_a(\alpha, g_{\mathrm{help}})$. Regional membership does not identify intent content; the failure locus is in the intent artefact's denotation target.

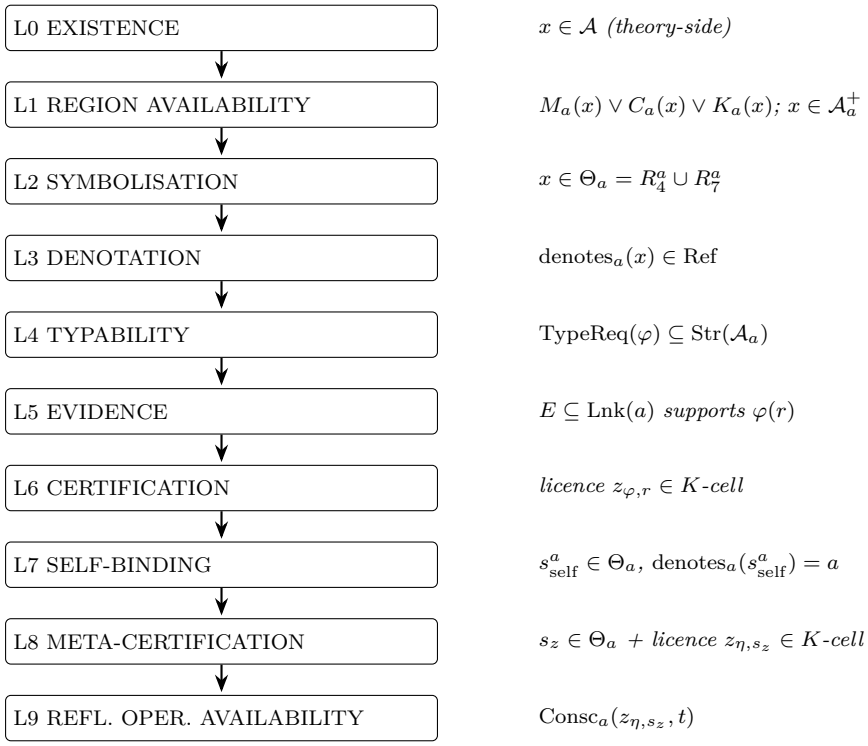


Figure 2: The regional cognitive ladder (Figure A of the v2 spec). Each layer is a prerequisite for the next, and the right-hand annotation names the MCK-native regional / carrier condition that must hold. Layers L0–L3 are Foundations-owned; L4–L6 are the certificate calculus of Section 3; L7 is the self-symbol of Definition 2.8; L8 is MetaCert of Definition 3.5; L9 is operational availability (Definition 3.11). Failure at level k prevents any level $k' > k$ from being reached.

Corollary 8.16 (Regional membership is a necessary condition, not sufficient)**.** *The regional signatures of Table 1 are necessary conditions on the corresponding artefacts, but do not identify the content of those artefacts. Content errors — a false* $\widehat{M}_a(b)$*, an inaccurate* $\widehat{O}_a(b)$*, a coercive* Intent_a *— occur inside correctly-region-typed carriers and cannot be detected by inspecting* σ^a *alone.*

9 Two-layer semantics: carrier vs. content

The regional algebra of Section 8 tells us whether an artefact can *structurally* play a role; it does not tell us whether its *content* is correct (Corollary 8.16). This section formalises that discipline.

9.1 Regional fibres

Definition 9.1 (Regional fibre)**.** For a regional signature $\sigma \in \{0,1\}^3 \setminus \{(0,0,0)\}$ and agent a, the *fibre* of σ under a is

$$\mathrm{Fib}_a(\sigma) \;=\; \{\, x \in \mathcal{A}_a^+ \;:\; \sigma^a(x) = \sigma \,\} \;\subseteq\; \mathrm{Region}_a^{-1}(R_i^a) \text{ for the } R_i^a \text{ matching } \sigma.$$

Elements of a fibre agree on their canonical region; they may differ on their denotation target, their evidence in $\mathrm{Lnk}(a)$, their temporal freshness, and their represented content.

Proposition 9.2 (Region equality does not imply content equality)**.** *There exist artefacts* $x, y \in \mathrm{Fib}_a(\sigma)$ *with* $\mathrm{denotes}_a(x) \neq \mathrm{denotes}_a(y)$*.*

Proof. Take $\sigma = (1, 1, 0) = R_4^a$; take $\mathcal{A}_a = \{x, y\}$ with $M_a = C_a = \{x, y\}$, $K_a = \emptyset$. Both lie in R_4^a, hence in $\mathrm{Fib}_a(\sigma)$. Set $\mathrm{denotes}_a(x) = r_1$ and $\mathrm{denotes}_a(y) = r_2 \neq r_1$; both denotations are well-formed under Foundations' `def:denotes-agent` because both targets can be arranged to lie in the active regions. □

9.2 Structural vs. semantic state

Definition 9.3 (Two-layer configuration)**.** The *cognitive configuration* of agent a at time t is a pair

$$\kappa_a(t) \;=\; \big(\,\Sigma_a(t),\;\Delta_a(t)\,\big)$$

where:

- $\Sigma_a(t)$ is the *carrier structure*: the region assignment $\sigma^a(\cdot, t)$ on $\mathrm{Sto}(a)$, together with the partition (M_a, C_a, K_a) it induces and the classification into active vs. suspended regions ($R_3^a = \emptyset$ under the subjective reading of K, Foundations `def:k-objective-subjective`);
- $\Delta_a(t)$ is the *content assignment*: the denotation-restriction $\mathrm{denotes}_a \restriction \Theta_a(t)$, the derivation record component of $\mathrm{Lnk}(a)$, and the represented content of each Sto/Lnk artefact (i.e. what the artefact *says* about its target, over and above the fact of the denotation link).

Both layers use Foundations objects only. Σ_a lives entirely in `def:memories`'s $\mathrm{Sto}(a)$; Δ_a's first two components live in $\mathrm{Lnk}(a)$ together with the $\mathrm{denotes}_a$ restriction, while represented content is borne by the artefacts themselves (Sto-resident) and is read *as content* rather than as carrier structure — a declared reading in the same pattern as the OBS/SEM extensions (Definition 10.20). *Notation:* the carrier layer $\Sigma_a(t)$ is unrelated to the event alphabet Σ of Section 2 and to certificate scopes Σ_φ; context disambiguates, and the subscript-a time-indexed form is reserved for the layer.

Definition 9.4 (Structural vs. semantic failure)**.** An architectural realisability failure at $\kappa_a(t)$ for capability q is:

structural: $\Sigma_a(t)$ does not supply a required regional carrier for q (a needed symbol is not in Θ_a; a needed operator has $K_a = 0$; the referent carrier is in the absolute exterior);

semantic: $\Sigma_a(t)$ supplies every required carrier but $\Delta_a(t)$ contains wrong content (a symbol denotes the wrong referent; the evidence in $\mathrm{Lnk}(a)$ does not compose into the claimed verdict; the represented other-model is inaccurate).

temporal: $\Sigma_a(t)$ and $\Delta_a(t)$ are correct at $t' < t$, but stale at t (the content is outdated but not intrinsically wrong).

The three are pairwise distinct and require different remediation.

Proposition 9.5 (Failure-mode reclassification)**.** *The six failure modes of Definition 4.5 classify as follows:*

- ABSENT ⇒ *structural (no carrier);*
- ILL-TYPED ⇒ *structural (wrong regional signature for the role);*
- MIS-DENOTED ⇒ *semantic (carrier correct, denotation content wrong);*
- FALSE ⇒ *semantic (content contradicts truth);*

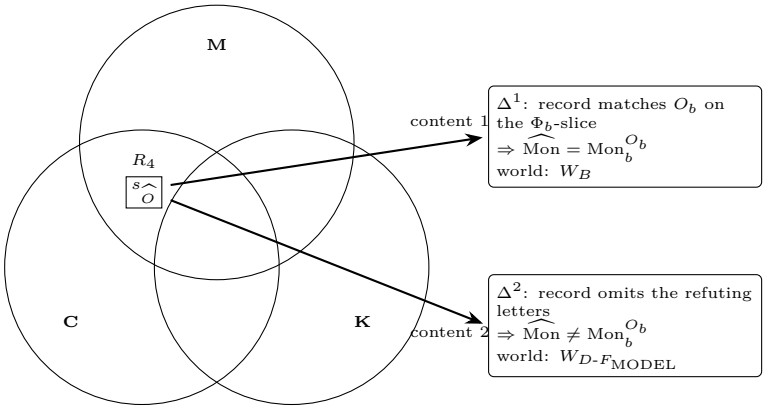


Figure 3: Figure R: same region, different content — the fibre theorem (Theorem 9.6) in the regional grammar. The symbol $s_{\widehat{O}}$ occupies one cell (R_4) of the miniature atlas; the two content assignments Δ^1, Δ^2 branch from the *same* cell into different semantic panels and different cognitive worlds. Regional position is shared by construction; only Δ differs. Regional audit alone cannot distinguish calibrated help from monitored failure with model-error origin; semantic inspection of Δ_a is required (Corollary 9.7).

- *OUTDATED* $\Rightarrow$ *temporal;*
- *INCOMPLETE* $\Rightarrow$ *hybrid: structural if the missing coordinate is a carrier, semantic if it is a content field.*

Proof. Direct inspection against Definition 9.4. □

9.3 Fibre-based cognitive non-injectivity

Theorem 9.6 (Same fibre, different worlds)**.** *There exist two cognitive configurations κ_a^1, κ_a^2 sharing identical Σ_a (same regional carrier structure, same inhabited fibres, same regional signatures for every relevant artefact) but differing in Δ_a, such that κ_a^1 realises world W_B and κ_a^2 realises world $W_{D\text{-}F_{\text{MODEL}}}$ per Definition 21.1 (defined in Section 21; the worlds are used forward here).*

Proof. Fix a common Σ_a with: $s_b, s_{\Phi_b}, s_{\widehat{O}_a(b)}, \text{Intent}_a(\alpha, \text{help}) \in R_4^a$; a $\widehat{\text{Effect}}_a(\alpha, b)$ carrier in R_4^a; an executable $\widehat{\text{Mon}}_a^b$ artefact in R_7^a; a licence $z \in R_5^a$. This is a W_B/W_D-compatible carrier structure per Table 1.

In κ_a^1, take Δ_a such that $\text{denotes}_a(s_{\widehat{O}_a(b)})$ encodes an observation regime matching the actual O_b on the Φ_b-relevant sub-alphabet. Then $\widehat{\text{Mon}}_a^b(\Phi_b, u_\alpha)$ agrees with $\text{Mon}_b^{O_b}(\Phi_b, u_\alpha)$ and Definition 21.1's W_B clause holds.

In κ_a^2, take Δ_a such that $\text{denotes}_a(s_{\widehat{O}_a(b)})$ encodes an observation regime that excludes the very sub-alphabet on which Φ_b is refuted. Same $s_{\widehat{O}_a(b)}$ artefact, same regional position R_4^a; different denotation content in Δ_a. Then $\widehat{\text{Mon}}_a^b(\Phi_b, u_\alpha) = \text{TRUE}$ (actor predicts non-failure) while $\text{Mon}_b^{O_b}(\Phi_b, u_\alpha) = \text{FALSE}$ (actual failure): world $W_{D\text{-}F_{\text{MODEL}}}$.

Both κ_a^1 and κ_a^2 share Σ_a exactly by construction. They differ only in Δ_a, which is not visible to σ^a. □

Figure 3 renders the construction in the regional grammar.

Corollary 9.7 (Regional inspection cannot separate W_B from $W_{D\text{-}F_{\text{MODEL}}}$)**.** *A capability-audit that observes only Σ_a cannot decide whether κ_a is in W_B or $W_{D\text{-}F_{\text{MODEL}}}$. Semantic inspection of Δ_a (specifically of* denotes_a*-content on the $\widehat{O}$ artefact) is required.*

9.4 Understanding, opacity, and functionality as function families

Foundations' M, C, K are predicates on artefacts. This paper now reads them as licensed by families of internal functions of a: *M-functions* produce certificates of understanding, *C-functions* maintain residual opacity, and *K-functions* enact operator behaviour. Formalising these families explicitly is what makes the regional algebra dynamic without introducing a new region.

Definition 9.8 (Function families of M, C, K)**.** Fix agent a and time t.

$\mathcal{F}_M^a$ **(comprehension functions)** is the set of internal artefacts $f \in \mathrm{Sto}(a)$ with $K_a(f) = 1$ (executable) whose application $\hat{f}(x)$, for $x \in \mathcal{A}_a^+$, produces a $\mathrm{Lnk}(a)$-entry that witnesses $M_a(x) = 1$: an artefact whose content is a certificate of understanding for x in the sense of Foundations `def:pred-m` (stable relational position, enumerable verifiable properties, predictable behaviour under known operations).

$\mathcal{F}_C^a$ **(opacity-recording / ignorance functions)** is the set of $g \in \mathrm{Sto}(a)$ with $K_a(g) = 1$ whose application produces a $\mathrm{Lnk}(a)$-entry marking one aspect of x's internal structure as unresolved for a, per Foundations `def:pred-c`.

$\mathcal{F}_K^a$ **(functionality-realising operators)** is the set of $h \in \mathrm{Sto}(a)$ with $K_a(h) = 1$ whose application to inputs produces outputs via Foundations' step relation `def:step`, participating as licence content in MOV or EXIT steps.

Remark 9.9 (Why these are the critical function families). Foundations names M, C, K but leaves *what makes* $M(x) = 1$ open (`02_predicates.tex`: "the certificate is not characterised"). Definition 9.8 names the internal objects that supply such certificates: a predicate value no internal function computes is a fact about the theorist's model, not about a's cognition. Every downstream capability of a depends on members of these families with adequate content.

Example 9.10 (Comprehension via statistical correlation)**.** A canonical class of $\mathcal{F}_M^a$-members are functions that map an observed co-occurrence stream to a stable relational-position certificate. If a observes events e_1, e_2 with empirical correlation $\rho > \tau$ (for some threshold τ internal to a's comprehension policy), an $f \in \mathcal{F}_M^a$ may produce a certificate of understanding for the composite artefact $(e_1 \cdot e_2)$: the certificate content records the co-occurrence regularity as a relational position between the two events under a's ontology. The certificate is *not* a truth claim about the world; it is a's internal witness that the composite artefact has a stable place in a's classification. Whether the correlation is causal, spurious, or a sampling artefact is a separate question — one that a *different* M-function would need to certify.

Example 9.11 (Opacity via bounded exploration)**.** A canonical class of $\mathcal{F}_C^a$-members are functions that enumerate the internal aspects of an artefact x up to a resource-bounded depth, and mark one or more aspects as unresolved when the enumeration terminates without a complete decomposition. Such a function produces a $\mathrm{Lnk}(a)$-entry with $C_a(x) = 1$: x is apprehended (the enumeration was possible) but at least one aspect of its internal structure remains unresolved (the enumeration did not complete). $\mathcal{F}_C^a$ is what turns a Foundations C-predicate value from a theorist's stipulation into an internal event of a's cognition.

Proposition 9.12 (Function-family regional signature)**.** *Every $f \in \mathcal{F}_M^a \cup \mathcal{F}_C^a \cup \mathcal{F}_K^a$ lies in a K-cell: R_5^a (transparent operator), R_6^a (opaque operator), or R_7^a (symbol-bearing operator). Under the subjective reading of K (Foundations `def:k-objective-subjective`), $R_3^a = \emptyset$, so no instantiated operator of a's architecture lies in R_3^a.*

Proof. By Definition 9.8 every family member has $K_a = 1$. By the region definitions the $K = 1$ cells are R_3, R_5, R_6, R_7; $K_a^{\text{subj}}(f) = 1$ implies $M_a(f) = 1$ or $C_a(f) = 1$ (Foundations `def:k-objective-subjective`), which excludes the R_3 cell $(0, 0, 1)$ for instantiated members of a's architecture. □

9.5 Agent universe and inter-agent bridging

Definition 9.13 (Agent universe)**.** The *agent universe* of a is

$$\mathcal{U}_a \;=\; (\,\mathcal{A}_a^{+},\ \{M_a, C_a, K_a\},\ \text{denotes}_a,\ \text{Sto}(a), \text{Exe}(a), \text{Lnk}(a),\ \mathcal{F}_M^a, \mathcal{F}_C^a, \mathcal{F}_K^a\,).$$

Every belief that a can form is an artefact in $\mathcal{A}_a^{+}$ produced by a finite composition of members of $\mathcal{F}_M^a \cup \mathcal{F}_C^a \cup \mathcal{F}_K^a$ under Foundations' architecture α_a and step relation. $\mathcal{U}_a$ is a's private cognitive universe: it is the closure of what a can classify, denote, understand, mark opaque, and execute.

Remark 9.14 (Universes are agent-private by construction)*.* $\mathcal{U}_a$ and $\mathcal{U}_b$ share no artefacts by default, and Foundations' `def:denotes-agent` allows $\text{denotes}_a(s) \neq \text{denotes}_b(s)$ ("no cross-agent coherence"). Semantic overlap is a further question requiring a bridging mechanism.

Definition 9.15 (Shared semantic conceptualisation)**.** A *shared semantic conceptualisation* between a and b is a pair $(C_{a\to b}, C_{b\to a})$ of internal artefacts, with $C_{a\to b} \in \text{Sto}(a)$ and $C_{b\to a} \in \text{Sto}(b)$, whose respective content maps a's referents onto b-recognisable referents and vice versa on a specified property family P. The pair is *coherent* on P when for every $x \in \mathcal{A}_a^{+} \cap C_{a\to b}$-domain and every $\varphi \in P$, a's classification $\text{Region}_a(x)$ and b's classification $\text{Region}_b(C_{a\to b}(x))$ agree.

Remark 9.16 (Bridging under DIA / REI)*.* When a shared conceptualisation exists but is not coherent on some sub-family, there is a *diástema* between $\mathcal{U}_a$ and $\mathcal{U}_b$ there; a revision restoring coherence is a *reintegration*. That machinery is developed in the ISG programme paper [17]; we import the vocabulary as a stance and no theorem here relies on it. This paper commits only to $\mathcal{U}_a$ as agent-private universe and to $C_{a\to b}$-style bridges, each subject independently to the six failure modes of Definition 4.5; the hand-over line is fixed in Section 11.

10 Capability realisability, composition, and bounded reachability

The two-layer state $\kappa_a(t) = (\Sigma_a(t), \Delta_a(t))$ of Definition 9.3 splits a capability's requirements into two orthogonal coordinates. This section formalises the split, proves the resulting (non-)monotonicity, types operator composition through both layers, and constructs the bounded transition system for reachable regional / content configurations.

10.1 The realisability decomposition

Definition 10.1 (Structural and semantic requirements)**.** For a bounded MCK agent a, time t, and capability q (an attribute certificate, a MetaCert, a reflexive-awareness predicate, an executable operator invocation, a W_X-classification, etc.), define:

$\text{Req}_a^{\Sigma}(q, t)$ **(structural):** the set of regional carriers, Θ_a-symbols, and K-cell licences that must be present in $\Sigma_a(t)$ for q's well-formedness clauses (as stated by the definition of q) to be satisfiable. The predicate "$\text{Req}_a^{\Sigma}(q, t)$ holds" means all such carriers exist with the required regional signatures.

$\mathrm{Req}_a^{\Delta}(q,t)$ **(semantic):** the set of denotation-content, evidence-composition, and represented-content conditions that must hold over $\Delta_a(t)$ for q to return a definite verdict consistent with truth. The predicate "$\mathrm{Req}_a^{\Delta}(q,t)$ holds" means all such content conditions are satisfied on the carriers already supplied by $\Sigma_a(t)$.

Definition 10.2 (Realisability)**.**

$$\mathrm{Realisable}_a(q,t) \iff \mathrm{Req}_a^{\Sigma}(q,t) \wedge \mathrm{Req}_a^{\Delta}(q,t).$$

Proposition 10.3 (Three-way realisability status)**.** *For any capability q and any $\kappa_a(t) = (\Sigma_a(t), \Delta_a(t))$, q's status at t falls into exactly one of three cases:*

(i) $\neg\mathrm{Req}_a^{\Sigma}(q,t) \implies$ *STRUCTURALLY UNTYPABLE: q returns UNTYPED; $\mathrm{Req}_a^{\Delta}(q,t)$ is undefined by convention (no verdict function to refine, cf. v1's Remark 5.7).*

(ii) $\mathrm{Req}_a^{\Sigma}(q,t) \wedge \neg\mathrm{Req}_a^{\Delta}(q,t) \implies$ *TYPABLE-BUT-INCORRECT: q is well-formed (a verdict function exists), but its verdict is wrong / undetermined-for-the-wrong-reason / misscoped.*

(iii) $\mathrm{Req}_a^{\Sigma}(q,t) \wedge \mathrm{Req}_a^{\Delta}(q,t) \implies$ *REALISED: q returns a verdict in $\{$TRUE, FALSE, UNDETERMINED$\}$ consistent with the actual situation.*

No fourth case is added: the pattern $\neg\mathrm{Req}^{\Sigma} \wedge \neg\mathrm{Req}^{\Delta}$ is subsumed by (i), because semantic adequacy on absent carriers is undefined rather than vacuously false.

Proof. By construction of Definition 10.1: Req^{Δ} quantifies over content of carriers named by Req^{Σ}; failure of the latter deprives the former of its domain, so "Req^{Δ} holds" is not defined. Reusing v1's four-valued verdict domain $\{$TRUE, FALSE, UNDETERMINED, UNTYPED$\}$: case (i) $\Rightarrow$ UNTYPED; case (ii) $\Rightarrow$ one of $\{$TRUE, FALSE, UNDETERMINED$\}$ but the value is unreliable relative to truth (this is where the four-world W_D-F_{MODEL} lives); case (iii) $\Rightarrow$ the verdict is a faithful record of the modelled state. □

10.2 Failure algebra

Definition 10.4 (Typed failure descriptor)**.** A capability-realisation failure at $\kappa_a(t)$ for q is a tuple

$$\mathrm{Fail}(q, \kappa_a, t) = (\,\mathrm{coord},\ \mathrm{layer},\ \mathrm{repair},\ \mathrm{observability},\ \mathrm{blast}(q)\,)$$

where COORD names the specific missing / corrupted coordinate (self-symbol / $\widehat{O}$ / evidence / licence / ...); LAYER $\in \{\Sigma, \Delta, \mathrm{both}\}$; REPAIR is the transition class that would fix it (a GEN producing the missing carrier, a MOV correcting a denotation-target, a Δ-content update); OBSERVABILITY takes a value in {regional-inspection, content-inspection, feedback-only}; and $\mathrm{blast}(q)$ is the set of downstream capabilities invalidated, formalised as the blast radius of Definition 12.15.

Proposition 10.5 (Repair non-equivalence)**.** *Structural repair (adding a carrier via a GEN step) and semantic repair (correcting a denotation or a Δ_a content field) are generally incomparable operations on κ_a:*

- *Adding a carrier cannot in general fix a wrong-content failure on a different, structurally-valid carrier.*
- *Correcting content on an existing carrier cannot in general supply a missing carrier for a different requirement.*

Proof. Structural repair changes Σ_a: it introduces a new artefact with new regional signature but does not affect denotation-content of any preexisting artefact (Foundations' GEN produces an artefact, does not rewrite the denotation of another artefact). Semantic repair changes Δ_a: it rewrites a denotation-restriction entry or a $\mathrm{Lnk}(a)$ derivation-record entry, but does not create any new $\mathcal{A}_a^+$ member. So the operations act on disjoint layers of κ_a. A capability whose failure is on the wrong layer for a given repair operation is unaffected by it. Finite witnesses in Appendix A. □

10.3 Monotonicity

Definition 10.6 (Structural refinement)**.** $\Sigma_a \preceq_{\mathrm{S}} \Sigma'_a$ when $\mathrm{Sto}(a) \subseteq \mathrm{Sto}(a')$, every artefact preserved keeps its region assignment (no reclassification), and no required carrier of any pre-existing capability is removed.

Theorem 10.7 (Structural monotonicity)**.** Req^{Σ} *is monotone under* $\preceq_{\mathrm{S}}$*:* $\mathrm{Req}_a^{\Sigma}(q, \Sigma_a) \Rightarrow \mathrm{Req}_a^{\Sigma}(q, \Sigma'_a)$ *whenever* $\Sigma_a \preceq_{\mathrm{S}} \Sigma'_a$*.*

Proof. Req^{Σ} names a finite set of required carriers with specified regional signatures. All such carriers exist in Σ_a by hypothesis. Definition 10.6 preserves each such carrier and its regional signature in Σ'_a. Hence every required carrier of q remains present in Σ'_a. □

Theorem 10.8 (Semantic non-monotonicity)**.** Req^{Δ} *is* not *monotone under* $\preceq_{\mathrm{S}}$*. There exist* $\Sigma_a \preceq_{\mathrm{S}} \Sigma'_a$ *and content assignments* Δ_a, Δ'_a *such that* $(\Sigma_a, \Delta_a) \models \mathrm{Req}^{\Delta}(q)$ *while* $(\Sigma'_a, \Delta'_a) \not\models \mathrm{Req}^{\Delta}(q)$*.*

Proof. Let $q = \mathrm{Cert}_a(\varphi, r, t)$ for some φ, r. Take (Σ_a, Δ_a) with the four RegCert carriers (referent $r \in R_4^a$, property-symbol $s_\varphi \in R_4^a$, evidence $E \subseteq \mathrm{Lnk}(a)$ that composes correctly to a TRUE verdict, licence $z \in R_5^a$) and no other artefacts. Then $\mathrm{Req}^{\Delta}(q)$ holds: Δ_a supplies the correct denotations and the correct evidence composition.

Extend to Σ'_a by adding a new evidence-fragment carrier $E' \in \mathrm{Lnk}(a)$; this is a valid GEN step producing an understood-but-non-operator artefact. $\Sigma_a \preceq_{\mathrm{S}} \Sigma'_a$. But choose Δ'_a to include E' in the evidence composition for $\varphi(r)$ with *contradictory* content — an entry $\Delta'_a(E')$ that refutes what E supported. Then the composed evidence in $\mathrm{Lnk}(a)$ no longer yields a definite TRUE; it yields UNDETERMINED or a wrong FALSE, depending on the composition rule. $\mathrm{Req}^{\Delta}(q)$ fails at (Σ'_a, Δ'_a). □

Corollary 10.9 (More representation does not imply more correct cognition)**.** *For bounded MCK agents, structural extension is monotone in capability-carrier presence but not in semantic calibration. Adding carriers can worsen the effective certification of previously realised capabilities if the added content is wrong.*

10.4 Regional / cognitive quotient incomparability

Definition 10.10 (Regional projection)**.** $\pi_\Sigma : \kappa_a(t) \mapsto \Sigma_a(t)$ is the regional projection (forgets Δ_a).

Definition 10.11 (Cognitive projection)**.**

$$\pi_{\mathrm{cog}} : \kappa_a(t) \;\mapsto\; W_X, \quad W_X \in \{W_A, W_B, W_{C_1}, W_{C_2}, W_{D\text{-}F_{\mathrm{INT}}}, W_{D\text{-}F_{\mathrm{MODEL}}}\},$$

per Definition 21.1, computed with respect to a fixed (b, α, Φ_b) context.

Theorem 10.12 (Kernel incomparability)**.** *The kernels* $\ker(\pi_\Sigma)$ *and* $\ker(\pi_{\mathrm{cog}})$ *are incomparable: neither refines the other.*

Proof. $\ker(\pi_\Sigma) \not\subseteq \ker(\pi_{\text{cog}})$: witness of Theorem 9.6 — two configurations κ_a^1, κ_a^2 share Σ_a (so π_Σ identifies them) but $\pi_{\text{cog}}(\kappa_a^1) = W_B$ and $\pi_{\text{cog}}(\kappa_a^2) = W_{D\text{-}F_{\text{MODEL}}}$ (so π_{cog} does not).

$\ker(\pi_{\text{cog}}) \not\subseteq \ker(\pi_\Sigma)$: exhibit two configurations κ_a^3, κ_a^4 both in W_B (well- calibrated support) but with distinct Σ_a. Take κ_a^3 with an executable $\widehat{\text{Mon}}_a^b \in R_7^a$; take κ_a^4 identical except that its $\widehat{\text{Mon}}_a^b$ artefact is data-only, in R_4^a, with the correct predicted verdict cached rather than computed. Both configurations satisfy the W_B clauses of Definition 21.1 (calibrated triad, model-verdict agreement), but they differ in $\sigma^a(\widehat{\text{Mon}}_a^b)$: R_7^a vs R_4^a. Hence $\pi_{\text{cog}}(\kappa_a^3) = \pi_{\text{cog}}(\kappa_a^4)$ while $\pi_\Sigma(\kappa_a^3) \neq \pi_\Sigma(\kappa_a^4)$. □

Corollary 10.13 (Regional inspection and cognitive-world inspection forget independent information)**.** *An audit that observes only π_Σ cannot recover the cognitive world; an audit that observes only π_{cog} cannot recover the underlying regional configuration. Both projections are information-lossy in different, incomparable directions.*

10.5 Two-layer operator composition

Definition 10.14 (Typed MCK operator arrow)**.** An MCK-realised operator with typed inputs and outputs is

$$\hat{f} : (\, X_{\text{in}},\ P_{\text{in}}^{\Delta}\,) \longrightarrow (\, X_{\text{out}},\ P_{\text{out}}^{\Delta}\,) \quad [\, K_a(\hat{f}) = 1,\ \sigma^a(\hat{f}) \in \{R_5^a, R_6^a, R_7^a\}\,]$$

where $X_{\text{in}}, X_{\text{out}} \subseteq \text{Sto}(a)$ are the input/output carrier types (with declared regional signatures), and $P_{\text{in}}^{\Delta}, P_{\text{out}}^{\Delta}$ are the input preconditions / output postconditions on Δ_a (the content conditions the operator assumes / establishes). The bracketed annotation records that $\hat{f}$ itself is a K-cell artefact.

Definition 10.15 (Structural composability)**.** $\hat{g} \circ \hat{f}$ is *structurally composable* iff $X_{\text{out}}(\hat{f}) \subseteq X_{\text{in}}(\hat{g})$ up to regional-signature match, and Foundations' $\circ_a$ is defined on the pair $(\hat{f}, \hat{g})$ (`def:architecture`).

Definition 10.16 (Semantic composability)**.** Structurally composable $\hat{g} \circ \hat{f}$ is *semantically composable* at Δ_a iff $\hat{f}$'s output postconditions $P_{\text{out}}^{\Delta}(\hat{f})$ imply $\hat{g}$'s input preconditions $P_{\text{in}}^{\Delta}(\hat{g})$ under Δ_a's content.

Theorem 10.17 (Structural / semantic composition separation)**.** *Structural composability does not imply semantic composability. There exist $\hat{f}, \hat{g}$ with $X_{\text{out}}(\hat{f}) = X_{\text{in}}(\hat{g})$ and correct regional signatures, but with $P_{\text{out}}^{\Delta}(\hat{f})$ inadequate to satisfy $P_{\text{in}}^{\Delta}(\hat{g})$: the composition executes, produces an output, but the output is semantically miscalibrated.*

Proof. Take $\hat{f}$ = PredEffect-computation from $\widehat{M}_a(b)$; take $\hat{g}$ = PredMon-computation from PredEffect and $\widehat{O}_a(b)$. Structurally composable: the output of $\hat{f}$ is a PredEffect carrier in R_4^a; $\hat{g}$ accepts such input. Semantic precondition of $\hat{g}$: PredEffect matches actual Effect (i.e. $\widehat{M}_a(b)$'s content is calibrated). If Δ_a makes $\widehat{M}_a(b)$'s content wrong on the Φ_b-relevant slice, $\hat{f}$ produces a PredEffect with content that does not satisfy $\hat{g}$'s content precondition; $\hat{g}$ nonetheless executes and returns a well-formed PredMon verdict, which is wrong. This is exactly the $W_{D\text{-}F_{\text{MODEL}}}$ pipeline of Definition 21.1. □

Remark 10.18 (The MCK root of downstream model-error phenomena)*.* Theorem 10.17 names the architectural site at which downstream model-error phenomena originate: a structurally executable pipeline can carry wrong intermediate content through a sequence of regionally-correct operator invocations, and the output is semantically miscalibrated even though every step is legally composed. Any downstream analysis of assistance failure, monitor mismatch, or drama-role production must locate its failure point at one of these semantic-composability breaks — not at a structural composability break, which is a different failure mode.

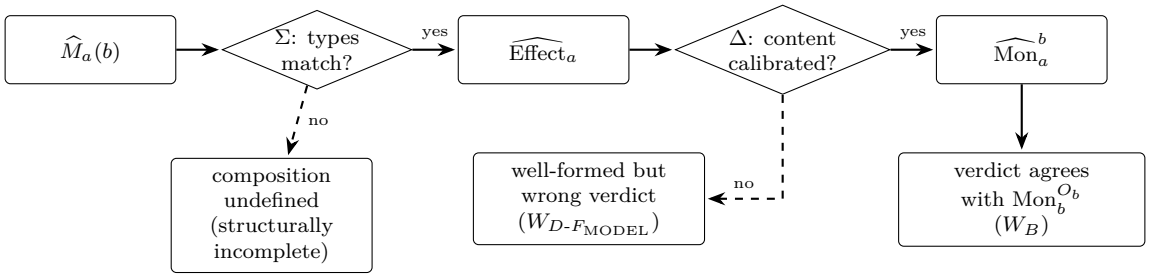


Figure 4: Figure G: the executable cognitive pipeline with its two gates. The structural gate (Σ: carrier types match, composition defined per Definition 10.15) and the semantic gate (Δ: content preconditions satisfied per Definition 10.16) fail differently: a structural failure stops execution; a semantic failure lets execution proceed and delivers a well-formed, wrong verdict (Theorem 10.17). Only the second failure mode produces $W_{D\text{-}F_{\text{MODEL}}}$.

10.6 Bounded transition system T_{MCK}

Definition 10.19 (Bounded MCK transition system)**.** $T^a_{\text{MCK}} = (\mathcal{K}_a, \to_a)$ where $\mathcal{K}_a$ is the set of two-layer configurations $\kappa_a = (\Sigma_a, \Delta_a)$ bounded by a fixed finite $\mathcal{A}_a$, and $\to_a$ is the union of the three transition families:

- **GEN**: introduces or functionalises an artefact in Σ_a per Foundations' `def:step` GEN clause (which toggles $K_a(y)$ $0 \to 1$; y may already be semantically present); produces a new region assignment $\sigma^a(y) = (M_a(y), C_a(y), K_a(y))$; may accompany a Δ_a-update supplying y's denotation and linkage-record entry.
- **MOV**: rewrites the region assignment of an existing artefact per `def:step` MOV clause, subject to a licence $z \in \mathcal{A}_a^+$; may accompany a Δ_a-update.
- **EXIT**: sends an artefact to the absolute exterior (region $\perp$) per `def:step` EXIT clause, subject to a licence; removes its denotation-restriction entry from Δ_a.

T^a_{MCK} describes which κ_a-configurations are reachable via legally composed steps (Remark 10.23).

Definition 10.20 (Declared dynamic extensions: OBS and SEM)**.** Foundations' step relation contains, in its own words, *no operation of supply*: no step by which an externally furnished artefact enters a's classified domain; and its denotation function denotes_a is *static*. The dynamics this paper studies need both an entry channel and content mutability, so we declare them — as named conservative extensions, not as readings of Foundations:

obs (supply): an OBS step introduces one externally furnished artefact y into $\mathcal{A}_a^+$, assigning it a regional signature, under a licence clause analogous to GEN's (a licence $z \in \mathcal{A}_a^+$ whose content records the apprehension). OBS is the formal carrier of observation, testimony, and feedback receipt.

sem (content rewrite): a SEM step rewrites one entry of the *denotation record* — a time-indexed record in $\text{Lnk}(a)$ that sits beside the static denotes_a and carries the agent's *current* binding for each symbol — or one evidence / linkage / freshness entry of Δ_a, preserving Σ_a exactly. Foundations' static denotes_a is untouched: what varies over the trajectory is the record, which is Δ-content and legitimately mutable. All semantic-repair operations (Definition 12.7) are SEM steps.

The transition relation of T^a_{MCK} is henceforth $\to_a = \text{GEN} \cup \text{MOV} \cup \text{EXIT} \cup \text{OBS} \cup \text{SEM}$, with the first three Foundations-owned and the last two owned by this paper. Every result whose

witness uses an OBS or SEM step is conditional on this declared extension; the ledger entries note this where it applies.

Remark 10.21 (Why declaring is more honest than stretching). Foundations' GEN and MOV do not cover observation or content rewriting: GEN's operands are K_a-members composed under $\circ_a$, and a feedback datum from another agent is neither. The two declared families keep the accounting exact: what Foundations licenses is imported; what this paper needs beyond it is named, bounded, and visible.

Proposition 10.22 (Cert-to-MetaCert reachability separation)**.** *There exists κ_a with* $\mathrm{Cert}_a(\varphi, r, t) = \textsc{true}$ *such that no finite $\to_a$-path from κ_a reaches a configuration with* $\mathrm{MetaCert}_a(\varphi, r, t)$ *realised, given the initial resources of κ_a.*

Proof. Take the witness of Theorem 3.6: $\mathcal{A}_a = \{r, z, s^a_{\mathrm{self}}\}$; no artefact in Θ_a denotes z, and the architecture α_a does not contain a GEN operator producing a new Θ_a-symbol denoting z. Every $\to_a$-transition either MOVs one of the existing three artefacts (which does not create a new denotation link) or invokes GEN with the existing α_a-basis; by hypothesis this basis cannot generate an s_z-symbol denoting z. So $\mathrm{MetaCert}_a(\varphi, r)$ remains UNTYPED under all reachable configurations. □

Remark 10.23 (Structurally reachable ≠ guaranteed to occur). T^a_{MCK} characterises which configurations are *consistent with* Foundations' step relation from a given κ_a. Whether any such transition is actually taken depends on a's acquisition policy, which Foundations does not model. This paper preserves that boundary: no acquisition dynamics theorem is proved.

10.7 Two-axis criticality

Definition 10.24 (Structural cut, calibration cut)**.** For a capability q realised at $\kappa_a(t)$:

structural cut $C_\Sigma(q)$**:** a minimal Σ_a-modification (specifically: removing a carrier or changing its regional signature out of the required cell) whose application makes $\mathrm{Req}^\Sigma(q)$ fail. Formalised in Theorem 8.12.

calibration cut $C_\Delta(q)$**:** a minimal Δ_a-modification (specifically: rewriting a denotation target on an existing Θ_a-member, or corrupting a $\mathrm{Lnk}(a)$ derivation-record entry) whose application preserves $\mathrm{Req}^\Sigma(q)$ but makes $\mathrm{Req}^\Delta(q)$ fail.

Criticality of q is a pair $\mathrm{Crit}(q) = (C_\Sigma(q), C_\Delta(q))$; the two coordinates are incomparable in general (Proposition 10.5).

Proposition 10.25 (Calibration cut for W_B)**.** *For $q =$ (agent a is in world W_B w.r.t. b, α, Φ_b), a single-target modification of $\Delta_a(s_{\widehat{O_a(b)}})$ suffices to move a to $W_{D\text{-}F_{\mathrm{MODEL}}}$, without touching Σ_a.*

Proof. Instantiate Theorem 9.6: κ^1_a (in W_B) and κ^2_a (in $W_{D\text{-}F_{\mathrm{MODEL}}}$) share Σ_a exactly; the only difference is the content of $s_{\widehat{O_a(b)}}$'s denotation. The transformation $\kappa^1_a \to \kappa^2_a$ is a pure Δ_a-rewrite on a single carrier, hence a valid $C_\Delta(W_B)$. □

10.8 Results ledger R14–R22 (this section)

- R14: capability realisability decomposition — Definitions 10.1 and 10.2 and proposition 10.3. DEFINITION + PROVED-IN-MODEL.
- R17: structural monotonicity / semantic non-monotonicity — Theorem 10.7 (PROVED) + Theorem 10.8 (PROVED-IN-MODEL) + Corollary 10.9.

- R18: structural vs semantic criticality separation — Definition 10.24 and proposition 10.25. PROVED-IN-MODEL.
- R19: regional / cognitive quotient incomparability — Theorem 10.12 and corollary 10.13. PROVED-IN-MODEL.
- R20: two-layer operator composition — Definitions 10.14 to 10.16. DEFINITION.
- R21: structurally-valid / semantically-invalid pipeline — Theorem 10.17 and remark 10.18. PROVED-IN-MODEL.
- R22: bounded reachability separation — Definition 10.19, proposition 10.22, and remark 10.23. DEFINITION + PROVED-IN-MODEL.

11 Downstream interface contracts

The realisability decomposition and the two-layer state (Σ_a, Δ_a) make this paper's downstream interface stable. Two other programme lines will consume this interface; the export contract is recorded here.

11.1 DIA / REI interface (ISG-side)

The ISG programme paper [17] develops diastêma and reintegration operators on grammar-mediated semantic spectra. This paper exports the substrate over which those operators act. The contract is: ISG consumes the following v2 objects verbatim and does not redefine them.

- $\kappa_a = (\Sigma_a, \Delta_a)$ (Definition 9.3).
- $\mathrm{Fib}_a(\sigma)$ (Definition 9.1).
- $\mathrm{Req}^{\Sigma}_a(q,t)$, $\mathrm{Req}^{\Delta}_a(q,t)$ (Definition 10.1), $\mathrm{Realisable}_a$ (Definition 10.2).
- Structural / semantic / temporal failure typing (Definition 9.4).
- M/C/K function families $\mathcal{F}^a_M, \mathcal{F}^a_C, \mathcal{F}^a_K$ (Definition 9.8).
- Agent universe $\mathcal{U}_a$ (Definition 9.13) and shared-conceptualisation pair $(C_{a\to b}, C_{b\to a})$ (Definition 9.15).

MCK does *not* import diastêma or reintegration semantics. $\mathrm{Req}^{\Sigma}/\mathrm{Req}^{\Delta}$ failures inside a single agent are named by this paper; the ISG paper describes their cross-agent bridging structure. The line $\mathrm{Region}_a(x) \neq \mathrm{Region}_b(C_{a\to b}(x))$ is where the ISG paper takes over (Remark 9.16).

11.2 LTL / Berne-LTL / DTE interface (interaction-side)

The Berne-LTL and Drama-to-Empowerment programme lines consume this paper's cognitive-world classification and its trace projection. The contract is:

- $W_A, W_B, W_{C_1}, W_{C_2}, W_{D\text{-}F_{\mathrm{INT}}}, W_{D\text{-}F_{\mathrm{MODEL}}}$ (Definition 21.1).
- Trace projection $\pi : \kappa_a \to u_\alpha$ (Definition 21.4) and its non-injectivity (Theorem 21.5).
- (Σ_a, Δ_a)-based structural vs semantic failure types (Definition 9.4).
- Content-driven $W_B \to W_{D\text{-}F_{\mathrm{MODEL}}}$ calibration cut (Proposition 10.25).

Table 2: Ownership boundary for the role-switch stack. MCK supplies bounded architectural witnesses; any inference from a finite observed behaviour to an ω-trace property is governed by the finite-evidence semantics of [11] and the prefix-verdict semantics of [10]; role vocabulary is downstream-owned throughout.

Object	MCK owns	Downstream owns	Import rule
$\kappa_a, \kappa_a^+, Q_{\text{crit}}$	yes	—	import, no redefinition
Focus_a, FOC, $\text{FB}_a(q,t)$	yes	—	import, no redefinition
attention-conditioned availability	yes	—	import
ActSig_a, activation classes	yes	—	import
semantic repair; formation results	yes	—	import
criticality; regional placement	yes	—	import
finite$\to\omega$ evidence semantics	no	RC / ltl_classes	cite owners
trace/role projection; role classifier	no	Berne-LTL	MCK silent
role-switch witness SW_k	no	Berne-LTL	typed per the RC contract
empowerment verdict; Ctrl	no	DTE / TED	MCK silent

- Structurally-valid / semantically-invalid pipeline (Theorem 10.17 and remark 10.18) as the MCK root of downstream model-error phenomena.
- Own_a / Own_A ownership carriers (Proposition 18.11). DTE may define its own Ctrl (episode-level selection authority) separately; MCK does not import Ctrl.
- The attention layer (Section 15): Focus_a, the FOC transition family (ShiftFocus_a), property-indexed feedback $\text{FB}_a(q,t)$, attention-conditioned availability, the implication-confirmation mechanism (Theorem 15.9 and remark 15.10), and the non-collapse discipline: focus is not representation, not truth, not incentive, and not a role. Role semantics over attention-conditioned traces are downstream property.

Downstream restriction: DTE and Berne-LTL may consume these objects and build interaction-level theorems over them, but they may not redefine them or promote them to logical closures MCK has not established.

11.3 Foundations upstream contract

For completeness, this paper's dependence on Foundations is fixed one-way: every regional statement here consumes a Foundations object (predicates M, C, K; regions $R_1, \ldots, R_7$; denotes_a; Θ_a; $\tau(a)$; step relation $\to_a$; memories Sto/Exe/Lnk). This paper never redefines a Foundations primitive and never adds a new one. Extensions introduced here (attribute certificate as licence-class, self-symbol as Θ_a-member, meta-certificate as certificate- about-certificate, other-model triad, function families, two-layer state, T_{MCK}) are conservative in the sense of H2: they compose Foundations' primitives without shadowing them.

12 Integrity, semantic repair, and bounded reachability

The realisability decomposition $\text{Req}^{\Sigma}/\text{Req}^{\Delta}$ of Section 10 and the two-axis criticality of Definition 10.24 are now in place. Three deferred formal results follow from it directly: a bounded consistency decision procedure for de se integrity certificates; a semantic-repair operator that can move an agent from $W_{D\text{-}F_{\text{MODEL}}}$ back to W_B without touching Σ_a; and a finite T_{MCK}-reachability enumeration classifying every bounded configuration by its realisability status. This section delivers each, along with a pipeline-state classification and a blast-radius separation.

12.1 Bounded integrity consistency

Integrity is a de se property whose failure is content-level: inconsistency between certificates the agent already holds. The consistency capability q_{int} audits the certificates of the worked property p_{int} of Section 17 and is distinct from it; the shared name records only that both are integrity-flavoured. The question is whether certificate-set inconsistency is architecturally decidable at a's bounded scale.

Definition 12.1 (Certificate set and comparison scope)**.** Let $\text{CertSet}_a(t) = \{z_1, \ldots, z_n\} \subseteq \mathcal{A}_a^+$ be a finite set of licences realising attribute certificates $\text{Cert}_a(\varphi_i, r_i, t)$ per Definition 2.6. Two certificates z_i, z_j are *comparable* at t when they share a compatible referent slot (same r or referents linked by an explicit denotation edge in $\text{Lnk}(a)$) and their scopes $\Sigma_{\varphi_i}, \Sigma_{\varphi_j}$ overlap at t.

Definition 12.2 (Certificate conflict)**.** $\text{Conflict}_a(z_i, z_j)$ holds iff:

1. z_i, z_j are comparable per Definition 12.1;
2. their verdicts differ on a shared referent slot (e.g. z_i.verdict = TRUE and z_j.verdict = FALSE over the overlapping scope), or one certifies $\varphi(r)$ while another certifies $\neg\varphi(r)$ under an explicit MCK-registered negation in $\text{Lnk}(a)$;
3. both are active in $\kappa_a(t)$ (neither has been EXITed).

Two certificates whose propositions merely differ (no shared referent, no overlapping scope, no MCK-registered negation) do *not* conflict.

Definition 12.3 (Consistent certificate set)**.** $\text{Consistent}_a(\text{CertSet}_a(t))$ holds iff for every pair z_i, z_j in the set, $\neg\text{Conflict}_a(z_i, z_j)$.

Theorem 12.4 (Bounded integrity consistency is decidable)**.** *For any finite* $\text{CertSet}_a(t)$ *with* n *certificates,* $\text{Consistent}_a(\text{CertSet}_a(t))$ *is decidable by inspection of* $\binom{n}{2}$ *pairs, in* $O(n^2 \cdot c)$ *time where* c *is the (bounded) cost of a single conflict check per Definition 12.2. The decision is over* a*'s internal* $\text{Lnk}(a)$*-content and does not require access to* Truth*.*

Proof. $\text{CertSet}_a(t)$ is finite by assumption. Each $\text{Conflict}_a(z_i, z_j)$ check is a comparison of finitely many fields in the licence content tuples — referent-slot compatibility, scope overlap, verdict comparison, negation registration in $\text{Lnk}(a)$ — each of which is bounded in c. Iterating over the $\binom{n}{2}$ pairs yields decidability in $O(n^2 \cdot c)$. The decision returns $\top$ (consistent) or a witness pair (inconsistent), depending only on Δ_a-content of the licences. □

Proposition 12.5 (Integrity is a Req^Δ property)**.** *For a de se integrity capability* $q_{\text{int}} =$ (*"a's active certificate set is consistent"*)*,* $\text{Req}^\Sigma(q_{\text{int}})$ *requires only the carriers of the certificates in* $\text{CertSet}_a(t)$ *and the linkage machinery that supports their content.* $\text{Req}^\Delta(q_{\text{int}})$ *is the consistency of the content itself, i.e.* $\text{Consistent}_a(\text{CertSet}_a(t))$*.*

Proof. By Definition 10.1 applied to q_{int}: the structural requirement is the existence of the carriers over which the consistency question is stated; the semantic requirement is the content-side conjunction check. □

Corollary 12.6 (Structurally-realised, semantically-inconsistent integrity)**.** *There exists* $\kappa_a(t)$ *with* $\text{Req}^\Sigma(q_{\text{int}})$ *satisfied and* $\text{Req}^\Delta(q_{\text{int}}) =$ FALSE*: all certificate carriers are present with correct regional signatures, yet the certificate set is inconsistent. Witness in the extended checker distributed with this article (M-R23).*

12.2 Semantic repair

Definition 12.7 (Semantic repair operations)**.** A *semantic repair* on $\kappa_a(t) = (\Sigma_a(t), \Delta_a(t))$ is one of the following four operations, each of which modifies $\Delta_a(t)$ while preserving $\Sigma_a(t)$:

denotation repair rewrites the *denotation-record* entry (Definition 10.20) of an existing $s \in \Theta_a$ to a different admissible target — Foundations' static denotes_a is untouched; the mutable object is the record. The embedding-dynamics companion states, without adopting it, what an admissible update of the denotation edges would owe [16, Def. 9.13]; a denotation repair is a record-level analogue of such an update, not an instance of it;

evidence repair rewrites the content of a $\text{Lnk}(a)$-entry supporting a certificate;

linkage repair rewrites the derivation-record edge in $\text{Lnk}(a)$ that composed an evidence set for a verdict, without changing individual $\text{Lnk}(a)$ entries;

freshness repair refreshes a stale content entry to reflect the current t.

Every semantic repair preserves Σ_a exactly: no artefact is added, removed, or reclassified.

Theorem 12.8 (Semantic repair sufficiency for $W_{D\text{-}F_{\text{MODEL}}} \to W_B$)**.** *Let κ_a^2 be a configuration in $W_{D\text{-}F_{\text{MODEL}}}$ produced from a W_B-compatible Σ_a by Δ_a-corruption on a single carrier (the $\widehat{O}_a(b)$ artefact, per Theorem 9.6). Then a single denotation-repair applied to that carrier's content is sufficient to move κ_a^2 back into W_B.*

Proof. By construction of Theorem 9.6, κ_a^2 differs from a W_B-configuration κ_a^1 only in $\text{denotes}_a(s_{\widehat{O}_a(b)})$-content. Applying denotation repair (Definition 12.7) to that single entry, replacing its content with κ_a^1's, yields exactly κ_a^1. Σ_a is preserved throughout. The result is $\kappa_a^1 \in W_B$ by Theorem 9.6. □

Remark 12.9 ($W_{D\text{-}F_{\text{INT}}}$ is not semantically repairable in the same way)*.* For $\kappa_a \in W_{D\text{-}F_{\text{INT}}}$, the failure is in the intent artefact $\text{Intent}_a(\alpha, g_{\text{coerce}})$, not in the other-model triad content. A denotation repair on $\widehat{O}_a(b)$ does not address the intent; the repair required is either intent-content rewrite (semantic repair on Intent_a) or action reselection (a MOV step, which touches Σ_a). The two flavours of W_D therefore have different repair signatures; the paper does not claim they are architecturally equivalent.

12.3 Bounded T_{MCK} reachability

Definition 12.10 (Bounded reachability enumeration)**.** For a fixed finite $\mathcal{A}_a$ and architecture $\alpha_a = (\mathcal{O}_a, \circ_a)$, the *bounded reachability set* from κ_a^0 is $\text{Reach}_a(\kappa_a^0) = \{\kappa_a : \kappa_a^0 \to_a^* \kappa_a\}$, the transitive closure of $\to_a$ (per Definition 10.19) restricted to configurations with underlying artefact set $\subseteq \mathcal{A}_a$.

Theorem 12.11 (Finite T_{MCK} reachability is decidable)**.** $\text{Reach}_a(\kappa_a^0)$ *is a finite set, computable by breadth-first enumeration in bounded time, for any finite $\mathcal{A}_a$ and α_a.*

Proof. The space of possible κ_a over a finite $\mathcal{A}_a$ is bounded: Σ_a ranges over Region-assignments $\mathcal{A}_a \to \{R_1^a, \ldots, R_7^a, \bot\}$ (at most $8^{|\mathcal{A}_a|}$ assignments); Δ_a ranges over denotation-restrictions $\Theta_a \to \text{Ref}$ and derivation-record sets over $\text{Exe}(a)$-pairs, each bounded by $|\mathcal{A}_a|$. The transitions $\to_a$ are finite unions of the five declared step families GEN / MOV / EXIT / OBS / SEM (Definition 10.20), each mapping one κ_a to a bounded number of successors — Definition 12.10 restricts reach to configurations with artefact set $\subseteq \mathcal{A}_a$, so OBS and SEM preserve boundedness. Breadth-first search from κ_a^0 terminates when no new configuration is discovered. □

Proposition 12.12 (Reachability ≠ semantic realisability)**.** *A configuration $\kappa_a \in \mathrm{Reach}_a(\kappa_a^0)$ can have $\mathrm{Req}^{\Sigma}(q) = \top$ and $\mathrm{Req}^{\Delta}(q) = \bot$ for some capability q: the configuration is architecturally reachable but the capability is not semantically realised. The converse movement within a Σ_a-class is Proposition 10.25.*

Proof. For the first claim: extend the witness of Theorem 9.6 to a κ_a^0 from which a W_B configuration is reachable via a GEN sequence that also can produce a Δ-corruption subsequence yielding $W_{D\text{-}F_{\mathrm{MODEL}}}$. Both endpoints are reachable; only the first is semantically realised. For the second claim: apply the semantic-repair operators of Definition 12.7 to transition within the reached Σ_a-class between semantic realisation and non-realisation. □

12.4 Executable cognitive pipeline

Definition 12.13 (Pipeline state)**.** For actor a, receiver b, action α, and property Φ_b, the *pipeline state* is

$$\mathrm{Pipe}_a^b(\alpha, t) \;=\; (\, s_b,\ \widehat{M}_a(b),\ \widehat{\mathrm{Effect}}_a(\alpha, b),\ \widehat{O}_a(b),\ \widehat{\mathrm{Mon}}_a^{\,b}(\Phi_b)\,),$$

each coordinate typed per Table 1. Its status falls into one of five classes:

absent a required coordinate is missing from $\mathrm{Sto}(a)$ (structural failure); typical world: W_A.

structurally-incomplete some coordinates present but the composition graph is broken (an operator output type does not match the next input type); the pipeline cannot execute end-to-end.

structurally-executable all coordinates present with matching types; Req^{Σ}(pipeline) holds; the pipeline runs.

semantically-calibrated structurally executable *and* Req^{Δ}(pipeline) holds; $\widehat{\mathrm{Mon}}_a^{\,b}$ agrees with $\mathrm{Mon}_b^{O_b}$; typical world: W_B.

semantically-miscalibrated structurally executable but at least one intermediate Δ_a-content is corrupted per Theorem 10.17; the pipeline emits a verdict that disagrees with the actual receiver monitor; typical worlds: W_{C_2} (miscalibration in $\widehat{O}_a(b)$) or $W_{D\text{-}F_{\mathrm{MODEL}}}$ (miscalibration surfacing in verdict mismatch).

Proposition 12.14 (Pipeline-state to cognitive-world mapping)**.** *The pipeline-state classification refines Definition 21.1: ABSENT ⇒ W_A; STRUCTURALLY-INCOMPLETE ⇒ W_A or W_C depending on which coordinate is missing; SEMANTICALLY-CALIBRATED ⇒ W_B; SEMANTICALLY-MISCALIBRATED ⇒ W_{C_2} or $W_{D\text{-}F_{\mathrm{MODEL}}}$ depending on which coordinate is wrong (Ohat vs. PredMon). W_{C_1} (blind by observability) is a specialisation of STRUCTURALLY-INCOMPLETE where the missing coordinate is $\widehat{\mathrm{Mon}}_a^{\,b}$ and the reason is that no such verdict exists on b's side. $W_{D\text{-}F_{\mathrm{INT}}}$ is orthogonal to pipeline state: the pipeline may be in any of the executable classes; the failure is in the intent artefact, not in the pipeline itself.*

Proof. By direct comparison of Definition 12.13's five classes against the world-clauses of Definition 21.1, using Theorem 10.17 and Proposition 10.25 for the miscalibration cases. □

12.5 Blast-radius separation

Definition 12.15 (Blast radius of a cut)**.** For a structural cut C_Σ or a calibration cut C_Δ per Definition 10.24, its *blast radius* is the set of downstream capabilities q' such that applying the cut to κ_a moves q' from REALISED to a strictly worse status. $\mathrm{Blast}_\Sigma(C_\Sigma) = \{q' : \mathrm{Realisable}_a(q', \kappa_a) \wedge \neg\mathrm{Realisable}_a(q', C_\Sigma(\kappa_a))\}$; similarly $\mathrm{Blast}_\Delta(C_\Delta)$.

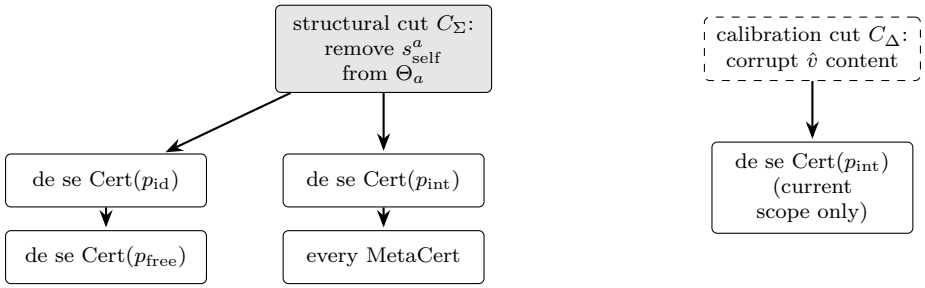


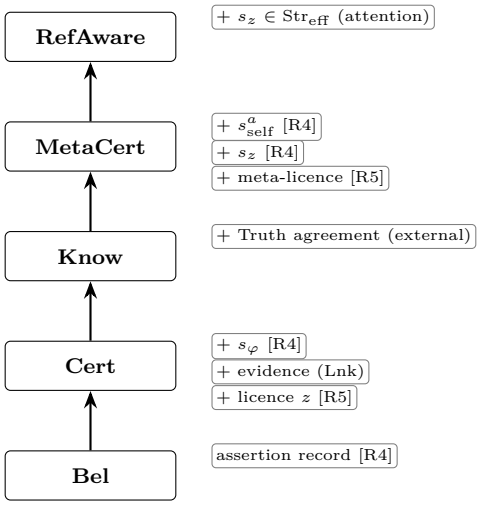
Figure 5: Figure H: two-axis criticality with distinct blast radii (Theorem 12.16). Both cuts invalidate the integrity certificate; the structural cut (solid, filled) collapses the entire de se stack including every meta-certificate, the calibration cut (dashed) only the integrity slice over its current scope.

Table 3: Realisability requirements for the belief / certification / reflection ladder. Each row makes explicit the structural and semantic requirements; the two failure columns record what goes wrong when only one side holds. This is the paper's answer to "*sapere vs. sapere di sapere*": the difference is a concrete increase in required carriers plus a concrete increase in content correctness conditions, not a modal-notation gloss.

Capability	Req^Σ	Req^Δ	Structural failure	Semantic failure
$\text{Bel}_a(\varphi, r)$	step output slot	a asserts $\varphi(r)$	no output channel	belief absent (silent)
$\text{Ev}_a(\varphi, r)$	fragment of $\text{Lnk}(a)$	entries compose to $\varphi(r)$-verdict	no Lnk trace	unrelated entries
$\text{Cert}_a(\varphi, r)$	r, s_φ, E, z per Definition 8.5	evidence-composition correct	missing carrier (UNTYPED)	wrong verdict emitted
$\text{Know}_a(\varphi, r)$	Cert's + Bel's	$\text{Cert} = \text{TRUE} \wedge \text{Truth} = \text{TRUE}$	any Cert or Bel carrier missing	belief / cert / truth misalign
$\text{MetaCert}_a(\varphi, r)$	Cert's + $s^a_{\text{self}}, s_z, z_{\eta,s_z}$	self-symbol denotes a; s_z denotes z; meta-evidence correct	no self-symbol or cert-symbol	reflexive claim wrong
$\text{RefAware}_a(\varphi, r)$	MetaCert's + $s_z \in \text{Str}_{\text{eff}}$	attention on the meta-symbol	s_z unattended	attentional distortion
$\text{Consc}_a(x)$	$x \in \text{Sto}(a)$, s denotes x, s attended, licence-available	denotation and evidence correct	any of the four sub-clauses fails	content on any carrier is wrong

Theorem 12.16 (Structural / semantic blast-radius separation)**.** *There exist structural cut C_Σ and calibration cut C_Δ that both invalidate the same high-level target capability q (e.g. de se attribution of a self-property p), yet have distinct downstream blast radii:* $\text{Blast}_\Sigma(C_\Sigma) \neq \text{Blast}_\Delta(C_\Delta)$.

Proof. Take $q = \text{Cert}_a(p_{\text{int}}, s^a_{\text{self}})$. $C_\Sigma = (\text{ remove } s^a_{\text{self}} \text{ from } \Theta_a)$ per Theorem 8.12: this is the identity-critical cut. Its blast radius includes every de se certificate $\text{Cert}_a(p', s^a_{\text{self}})$ for any p', and every $\text{MetaCert}_a(\varphi, r)$ (by Definition 3.5 clause requiring s^a_{self}). $C_\Delta = (\text{ rewrite the content of the viability-operator carrier } \hat{v}\text{'s } \text{Lnk}(a) \text{ entry to return the wrong value on the current state })$: this is a calibration cut. Its blast radius includes only $\text{Cert}_a(p_{\text{int}}, s^a_{\text{self}})$ over the current scope, not de se certificates about p_{id} or p_{free}, and not any MetaCert that does not consume $\hat{v}$-content. The two blast radii differ: Blast_Σ collapses the whole de se stack; Blast_Δ collapses only the integrity slice. Both invalidate q; they do not invalidate the same downstream set. □



each ascent ADDS carriers in the tagged regions; MetaCert is a larger architectural requirement, not a deeper point of the same one

Figure 6: Figure S: the certificate / reflexive stack in the regional grammar (subsuming the earlier deferred certificate-stack figure). Each ascent of the ladder *adds* carriers, tagged with their canonical regions; MetaCert is a strictly larger architectural requirement — more carriers in more cells — not a deeper point of the same one. Companion table: Table 3.

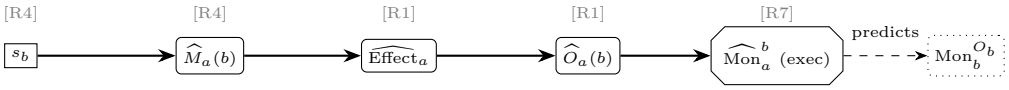


Figure 7: Figure Q: functional regional flow. Each pipeline node carries its spec-derived region tag; chamfered boxes are operator artefacts, plain boxes data artefacts, and the dashed edge is the *prediction relation* toward the external actual monitor (dotted border), which never enters the agent's regions. The region tags certify structural well-typing only; semantic calibration is invisible here by design (Theorem 10.17).

12.6 The realisability table for knowledge and reflection

12.7 Results ledger M-R23–M-R28 (this section)

- M-R23 bounded integrity consistency decidability — Definitions 12.1 to 12.3, theorem 12.4, proposition 12.5, and corollary 12.6. DEF + PROVED + CHECKED (extended checker; the gate mechanises the same-referent / scope-overlap comparability and verdict-difference conflict fragment of Definition 12.2 — the denotation-edge comparability and registered-negation branches are covered by the text proof only).

- M-R24 finite T_{MCK} reachability decidability — Definition 12.10 and theorem 12.11. PROVED + CHECKED (the gate's BFS instantiates the MOV family and cross-checks the derived count analytically; boundedness of the full five-family relation is established in the text proof).

- M-R25 reachability $\neq$ semantic realisability — Proposition 12.12. PROVED-IN-MODEL + CHECKED.

- M-R26 semantic-repair sufficiency for $W_{D\text{-}F_{\mathrm{MODEL}}} \to W_B$ — Definition 12.7, theorem 12.8, and remark 12.9. DEF + PROVED-IN-MODEL + CHECKED.

- M-R27 pipeline-state classification — Definition 12.13 and proposition 12.14. DEF + PROVED.
- M-R28 structural vs semantic blast-radius separation — Definition 12.15 and theorem 12.16. DEF + PROVED-IN-MODEL + CHECKED.

13 Continuous formation: belief, identity, incentive

The formal architecture through Section 12 is static in presentation: it supplies primitives and results about a given $\kappa_a(t) = (\Sigma_a(t), \Delta_a(t))$. An agent, however, is a trajectory $\kappa_a(0) \to \cdots \to \kappa_a(t)$ under the bounded transitions of T_{MCK} (Definitions 10.19 and 12.10). This section reframes the existing architecture as the formation theory of that trajectory: where beliefs come from, how the self-model is formed and continued, how incentives arise and differ from intents, and how internal computation and external interaction update the two-layer state. No new region or predicate is introduced: the transitions are Foundations' GEN / MOV / EXIT plus the extension families OBS and SEM (Definition 10.20), and every formation phenomenon below is composed of existing $\mathrm{Req}^{\Sigma}/\mathrm{Req}^{\Delta}$ satisfaction over that five-family relation.

13.1 Two channels of transformation

Definition 13.1 (Closed and open cognitive steps)**.** A single $\to_a$-transition of T_{MCK} is:

closed when the step's licence artefact and its inputs already lie in $\mathcal{A}_a^+$ at t: the transition consumes only representations a already holds.

open when the step consumes a new artefact (e.g. observed content from an interaction with the world or another agent b) that first enters $\mathcal{A}_a^+$ through an OBS step (Definition 10.20) apprehending it.

Every *artefact-consuming* $\to_a$-transition falls into exactly one of the two classes; the later FOC family (Definition 15.1), which consumes no artefact, is classified CLOSED by convention there. Both classes may change Σ_a (by GEN / MOV / EXIT) or Δ_a (by rewriting a denotation / evidence / linkage entry via a licence). Neither guarantees semantic improvement.

Remark 13.2 (Neither channel is privileged)*.* A closed step is not weaker: internal computation can produce a MetaCert, correct a semantic error via Definition 12.7, or discover a certificate conflict per Theorem 12.4. An open step is not stronger: observed content can be miscategorised (adding a wrong $\widehat{O}_a(b)$-entry, per Theorem 10.8). The two channels are architecturally symmetric with respect to T_{MCK}'s transition family; they differ only in whether the input already lived in $\mathcal{A}_a^+$.

13.2 The agent formation loop

The loop composes existing objects into a single diagram of cognitive causation.

13.3 The belief system as an evolving substructure

Definition 13.3 (Belief state)**.** The *belief state* of agent a at t is

$$B_a(t) \;=\; \{\, (\varphi, r) \;:\; \mathrm{Bel}_a(\varphi, r, t) \text{ holds} \,\}$$

per Definition 3.1. A *belief record* is the extended tuple

$$(\varphi,\ r,\ v,\ E,\ z,\ \sigma_\varphi),$$

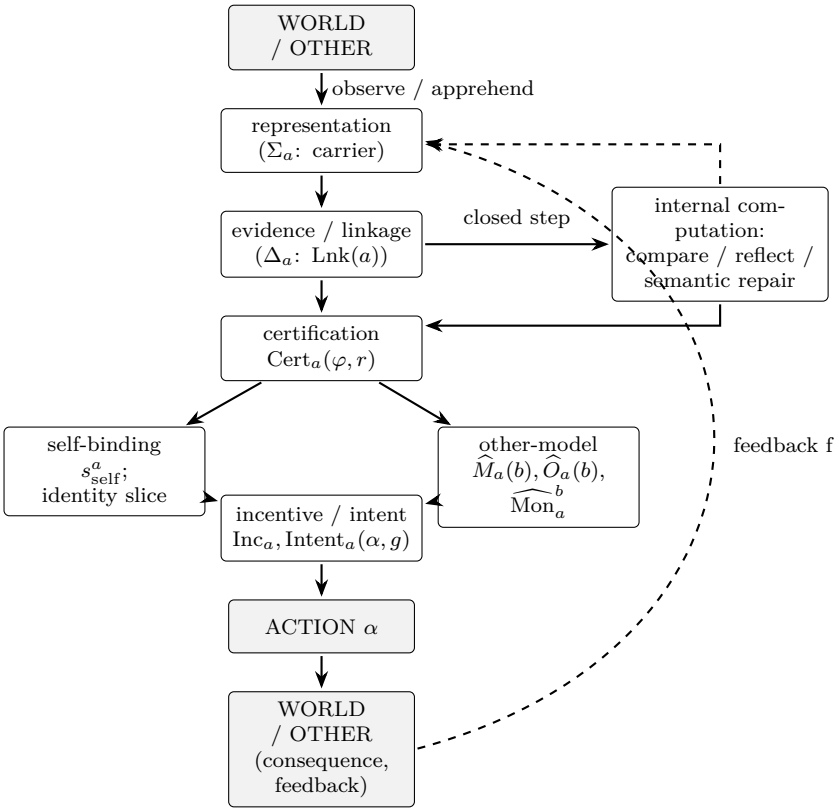


Figure 8: Figure J: the MCK agent formation loop. Solid arrows are T_{MCK}-transitions; dashed arrows feed a subsequent transition. External nodes (filled) supply content across the $\mathcal{A}_a^+$ boundary via OBS steps (Definition 10.20) — the open-step channel; the internal-computation branch is the closed-step channel (Definition 13.1). Both alter κ_a under the same $\mathrm{Req}^{\Sigma}/\mathrm{Req}^{\Delta}$ discipline.

where v is the current Cert_a-verdict in $\{\textsc{true}, \textsc{false}, \textsc{undetermined}, \textsc{untyped}\}$ if any, $E \subseteq \mathrm{Lnk}(a)$ is the supporting evidence, z is the licence realising the certificate if held, and σ_φ is the scope over which the belief is claimed.

Remark 13.4 (No probability introduced). $B_a(t)$ is a set of (proposition, referent) pairs with verdict tags, not a probability distribution. Where confidence is required for downstream analysis, it is derived from the verdict domain $\{\textsc{true}, \textsc{false}, \textsc{undetermined}, \textsc{untyped}\}$ and the certificate presence, not from a numerical prior. The paper does not commit to Bayesian or non-Bayesian revision.

Definition 13.5 (Belief provenance)**.** Every belief record in $B_a(t)$ carries a *provenance tag* in $\{\textsc{external}, \textsc{internal}, \textsc{mixed}\}$ recording whether its evidence set E was produced by (or composed with) an open step, a closed step, or both. Provenance is not truth: an externally-provenanced belief is not thereby more correct than an internally-provenanced one.

13.4 Continuous belief revision

Definition 13.6 (Revision events)**.** A belief-state transition $B_a(t) \to B_a(t+1)$ falls into one of six revision events, each realised by a specific $\to_a$-family:

- ACQUISITION: a new (φ, r) enters B_a via GEN of the property-symbol and evidence (open or closed).

- STRENGTHENING: an existing record's evidence set grows or its verdict moves from UNDETERMINED toward a definite value.
- CONTENT-REPLACEMENT: semantic repair (Definition 12.7) rewrites the record's evidence or denotation.
- INVALIDATION: the record's certificate verdict flips or the record is EXITed.
- UNRESOLUTION: an existing definite verdict falls to UNDETERMINED under new conflicting evidence.
- REFLEXIVE-LIFT: the record acquires an accompanying MetaCert / RefAware per Table 3.

Proposition 13.7 (M-R29 belief-state change without Σ change)**.** *There exist $\kappa_a(t), \kappa_a(t+1)$ with $\Sigma_a(t) = \Sigma_a(t+1)$ and $B_a(t) \neq B_a(t+1)$: belief revision can occur through Δ_a-only transitions.*

Proof. Instance of Definition 12.7: rewriting a denotes$_a$-entry on an existing $s \in \Theta_a$ that carries an evidence-supported belief moves the belief's verdict from TRUE to FALSE or UNDETERMINED, without changing any regional carrier. The revision event is CONTENT-REPLACEMENT of Definition 13.6. □

13.5 Self-model formation from two channels

Definition 13.8 (Self-model $\widehat{M}_a(a)$)**.** The *self-model* of a is the other-model triad (Definitions 4.1 to 4.3) instantiated at $b = a$: the counterpart is a itself. It requires the self-symbol $s^a_{\text{self}} \in \Theta_a$ as its anchor (Definition 2.8), a $\Delta_b \subseteq \text{Sto}(a)$ representing a's attributions to itself, an $\widehat{O}_a(a)$ representing what a observes about itself, and a $\widehat{\text{Mon}}^a_a$ predicting a's own monitor verdicts. Formally identical to $\widehat{M}_a(b), \widehat{O}_a(b), \widehat{\text{Mon}}^b_a$ under substitution $b := a$; regionally identical to any other-model per Table 1.

Theorem 13.9 (M-R31 self-model update induced by other-agent feedback)**.** *There exist $\kappa_a(t) \to \kappa_a(t+1)$ transition sequences in which an OBS step consuming feedback f from an interaction with b, followed by a SEM step, modifies Δ_a on the self-model slice $\Delta_b(a)$ (the record of a's attributions to itself), with no carrier* of the self-model slice *added, removed, or regionally reclassified.*

Proof. Take $\kappa_a(t)$ with a well-formed self-model $(s^a_{\text{self}}, \Delta_b(a), \widehat{O}_a(a), \widehat{\text{Mon}}^a_a)$ and a belief record $(\varphi_{\text{comp}}, s^a_{\text{self}}, \text{TRUE})$ attributing competence φ_{comp} to a. Perform $\text{Act}(a, \alpha)$ for some α; receive feedback f from b that contradicts $\widehat{\text{Mon}}^a_a(\varphi_{\text{comp}}, u_\alpha)$'s prediction. Under the semantic-repair operator (Definition 12.7, evidence repair): Δ_a's evidence supporting the competence belief is rewritten to include the f-witnessed refutation; the belief record's verdict flips to FALSE; $\Delta_b(a)$'s competence attribution content is correspondingly updated. Σ_a is unchanged; every carrier remains at the same regional signature. The revision event is CONTENT-REPLACEMENT + INVALIDATION. □

Theorem 13.10 (M-R32 internal-computation update without interaction)**.** *There exist $\kappa_a(t) \to \kappa_a(t+1)$ transitions realised by a closed step (no new artefact enters $\mathcal{A}^+_a$) in which B_a or the self-model slice changes.*

Proof. Instance of Theorem 12.4: iterating the $O(n^2 \cdot c)$ bounded consistency decision over an existing $\text{CertSet}_a(t)$ discovers a conflicting pair (z_i, z_j). Applying the paired semantic-repair operator (Definition 12.7) resolves the conflict by rewriting one of the two content records.

The step consumes only existing $\mathrm{CertSet}_a(t)$ artefacts and existing $\mathrm{Lnk}(a)$ content; no new artefact is GEN'd from an external channel; the step is CLOSED per Definition 13.1. The belief-state changes. □

Theorem 13.11 (M-R33 same feedback, different updates)**.** *There exist two agents a_1, a_2 receiving identical feedback f from b after identical actions α, with distinct Δ_a-updates: a_1 interprets f as competence- refutation; a_2 interprets f as noise. The difference is entirely in Δ_{a_i}'s pre-feedback content (the $\widehat{M}_{a_i}(b)$-recorded reliability of b's feedback).*

Proof. Fix $\Sigma_{a_1} = \Sigma_{a_2}$ up to a trivial relabeling. $\Delta_{a_1}(\widehat{M}_{a_1}(b))$ records b as a reliable evaluator; $\Delta_{a_2}(\widehat{M}_{a_2}(b))$ records b as unreliable. We construct two admissible SEM updates: a_1's witness rewrites its self-model competence attribution, and its $\mathrm{Lnk}(a_1)$ derivation record for that rewrite *cites* the reliability attribution; a_2's witness leaves the self-model unchanged and rewrites the noise-tolerance content of its evidence-composition rule, citing the unreliability attribution. Same f; the two witnesses differ, and each witness's derivation record cites the pre-existing Δ_a-content it is grounded in. The claim is the existence of these admissible updates, not a law that any agent must perform them. □

13.6 Identity as a continuous self-bound substructure

Definition 13.12 (Identity slice)**.** The *identity slice* of $\kappa_a(t)$ is

$$\begin{aligned}\mathrm{IdState}_a(t) \;=\; (\; & s^a_{\mathrm{self}},\ \mathrm{denotes}_a(s^a_{\mathrm{self}}),\ \Delta_b(a)\ \text{(self-model content)},\\ & \mathrm{CertSet}_a^{\text{de-se}}(t),\ \mathrm{Own}_a,\ \mathrm{Inc}_a^{\text{committed}}(t)\ ).\end{aligned}$$

where $\mathrm{CertSet}_a^{\text{de-se}}(t)$ is the sub-set of $\mathrm{CertSet}_a(t)$ whose referent is s^a_{self}, and $\mathrm{Inc}_a^{\text{committed}}(t)$ is the subset of $\mathrm{Inc}_a(t)$ (Definition 13.17 below) whose members satisfy clauses (1)–(2) of Definition 13.20. The mutual reference between this definition and Definition 13.20 is stratified, not circular: commitment consults only the de se certificate slice, which is determined before — and independently of — the committed-incentive slice.

Remark 13.13 (Identity is not the whole cognitive state)*.* $\mathrm{IdState}_a(t) \subsetneq \kappa_a(t)$: it is the self-bound slice, not the total state. Beliefs about the world (not de se) lie outside $\mathrm{IdState}_a$; other-models of b lie outside $\mathrm{IdState}_a$ (they are in the belief state B_a but not on the identity slice).

Definition 13.14 (Identity continuity)**.** $\mathrm{IdState}_a(t) \sim \mathrm{IdState}_a(t+1)$ holds when: s^a_{self} is preserved (same Θ_a-member); $\mathrm{denotes}_a(s^a_{\mathrm{self}}) = a$ throughout $[t, t+1]$; the derivation-record chain in $\mathrm{Lnk}(a)$ linking the two IdState-snapshots is unbroken (a single-agent Lnk-trajectory per Foundations `cor:linkage-irreducible`).

Proposition 13.15 (M-R30 identity continuity under belief revision)**.** *There exist $\kappa_a(t) \to \kappa_a(t+1)$ transitions with $B_a(t) \neq B_a(t+1)$ (belief revision) and $\mathrm{IdState}_a(t) \sim \mathrm{IdState}_a(t+1)$ (identity continuous). Identity is preserved under belief change.*

Proof. Take a belief-state transition of the CONTENT-REPLACEMENT class of Definition 13.6: an existing belief about the world (not de se) has its denotation-content rewritten. s^a_{self} and $\mathrm{denotes}_a(s^a_{\mathrm{self}})$ are untouched; the de se certificate set is untouched; the derivation-record chain in $\mathrm{Lnk}(a)$ continues unbroken. B_a changed; the identity slice's constitutive fields did not. □

Proposition 13.16 (Identity discontinuity via calibration cut)**.** *Conversely, a sufficiently destructive calibration cut on $\mathrm{denotes}_a(s^a_{\mathrm{self}})$ (Proposition 10.25 applied to the self-symbol) breaks $\mathrm{IdState}_a(t) \sim \mathrm{IdState}_a(t+1)$: continuity fails because the second clause of Definition 13.14 fails.*

Proof. Direct application of Definition 13.14: a rewrite of $\mathrm{denotes}_a(s^a_{\mathrm{self}})$ to a non-a target invalidates the "$\mathrm{denotes}_a(s^a_{\mathrm{self}}) = a$ throughout $[t, t+1]$" clause. □

13.7 Incentives

Definition 13.17 (Incentive)**.** The *incentive state* of a is

$$\mathrm{Inc}_a(t) \;=\; \{\, (g, w) \;:\; g \in \mathcal{G}_a,\; w \in \mathrm{Sto}(a),\; g\text{-relevance of } w \text{ registered in } \mathrm{Lnk}(a) \,\},$$

where $\mathcal{G}_a$ is the finite set of goal-artefacts held by a in Θ_a (each $g \in \mathcal{G}_a$ is an artefact denoting an intended outcome or normative constraint), and w is a $\mathrm{Sto}(a)$-carrier that a's $\mathrm{Lnk}(a)$ tags as g-relevant. Inc_a is *determined by* Δ_a given Σ_a: the pairs themselves are carrier-level, and the Δ_a-resident object is the relevance registration in $\mathrm{Lnk}(a)$, not the pair. No dedicated region is required beyond what the goal-artefacts and relevance-tags already occupy.

Remark 13.18 (Incentive $\neq$ intent $\neq$ action)*.* An incentive (g, w) records that g matters and w is g-relevant. An intent $\mathrm{Intent}_a(\alpha, g)$ (Definition 6.2) is a symbol committing a to attempt α in service of g; this requires that $g \in \mathcal{G}_a$ (goal represented) but not vice versa. An action $\mathrm{Act}(a, \alpha)$ (Definition 6.1) is the step-relation event realising an α; multiple intents can plausibly select the same α, and multiple incentives can plausibly fuel a single intent. The three are distinct architectural objects, and Definition 21.4's non-injectivity extends: none of the three uniquely identifies the other two.

Proposition 13.19 (M-R34 same belief, different incentive consequence)**.** *Two agents a_1, a_2 can share a belief record $(\varphi, r, \textsc{true})$ verbatim and select distinct actions $\alpha_1 \neq \alpha_2$ under distinct incentive states $\mathrm{Inc}_{a_1} \neq \mathrm{Inc}_{a_2}$. Belief content does not determine action.*

Proof. Fix the shared belief record. Let Inc_{a_1} contain (g_1, w) with g_1-relevance tagging w for action-family A_1; let Inc_{a_2} contain (g_2, w) with g_2-relevance tagging w for action-family A_2, where $A_1 \cap A_2 = \emptyset$. Construct admissible intents $\mathrm{Intent}_{a_1}(\alpha_1, g_1)$ with $\alpha_1 \in A_1$ and $\mathrm{Intent}_{a_2}(\alpha_2, g_2)$ with $\alpha_2 \in A_2$, each carrying a derivation record that cites its own incentive ground; disjointness gives $\alpha_1 \neq \alpha_2$. The claim is the existence of these admissible selections — not a law that incentives determine intents or that intents determine actions, both of which fail in general (Remark 13.18 and proposition 14.6). □

13.8 Identity × incentive

Definition 13.20 (Committed incentive)**.** $(g, w) \in \mathrm{Inc}_a(t)$ is *committed* to a's identity when: (1) $g \in \mathcal{G}_a \cap \Theta_a$; and (2) there exists a self-bound certificate $\mathrm{Cert}_a(\text{“}a \text{ pursues } g\text{”}, s^a_{\mathrm{self}}, t) = \textsc{true}$. Such a certificate lies in the de se certificate slice by definition of that slice (its referent is s^a_{self}), so no third clause is needed.

Remark 13.21 (Four levels of identity-incentive relation)*.* The paper distinguishes:

1. *I want g*: $(g, w) \in \mathrm{Inc}_a$ for some g-relevant w.
2. *I intend α for g*: $\mathrm{Intent}_a(\alpha, g)$ realised.
3. *I am someone who pursues g*: (g, w) is committed per Definition 13.20.
4. *I know I hold this commitment*: $\mathrm{MetaCert}_a(\text{“}a \text{ pursues } g\text{”}, s^a_{\mathrm{self}}, t) = \textsc{true}$.

These are four different capabilities with strictly increasing $\mathrm{Req}^{\Sigma}/\mathrm{Req}^{\Delta}$ per Table 3 extended by Definition 13.20. The paper does not identify them.

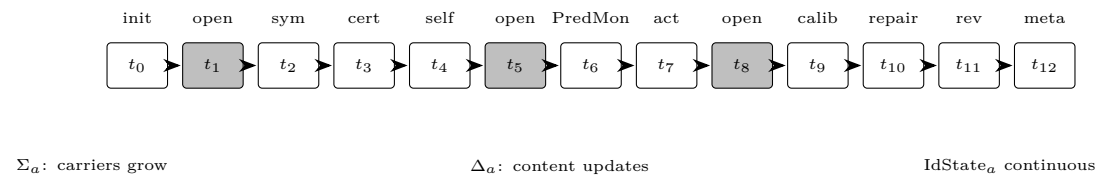


Figure 9: Figure K: the illustrative finite MCK trajectory. Filled cells (t_1, t_5, t_8) are OPEN steps consuming new external content; unfilled cells are CLOSED. The semantic-repair step at t_{10} is a Δ-only rewrite (Theorem 12.8); $\mathrm{IdState}_a$ is continuous throughout (Definition 13.14 and proposition 13.15).

13.9 An illustrative finite MCK trajectory

We integrate the running example (synthetic, illustrative): bounded agent a; alphabet $\{\mathrm{obs}, \mathrm{query}, \mathrm{decide}, \mathrm{fb}\}$; two other-model attributes; single receiver b.

- t_0: $\Sigma_a = \{s^a_{\mathrm{self}}\} \subseteq R^a_4$ with $\mathrm{denotes}_a(s^a_{\mathrm{self}}) = a$; $B_a = \mathrm{Inc}_a = \emptyset$; $\widehat{M}_a(b)$ absent.
- t_1 (OPEN): an OBS step introduces $y \in R^a_1$ apprehending an observed event e.
- t_2 (CLOSED): GEN from y yields $s_y \in \Theta_a$ with $\mathrm{denotes}_a(s_y) = e$; the record $(e\text{-happened}, s_y, \text{UNDETERMINED})$ enters B_a.
- t_3 (CLOSED): evidence composition on $\mathrm{Lnk}(a)$ yields licence z_e; the verdict moves to TRUE.
- t_4 (CLOSED): $\mathrm{Cert}_a(\text{“}a \text{ observed } e\text{”}, s^a_{\mathrm{self}}, t_4) = \text{TRUE}$ enters the de se slice of $\mathrm{IdState}_a$.
- t_5 (OPEN): an OBS step apprehends b; GEN yields $s_b \in \Theta_a$ denoting b; $\widehat{M}_a(b)$ initialised with region-content claims.
- t_6 (CLOSED): PredEffect and $\widehat{O}_a(b)$ generated from $\widehat{M}_a(b)$; $\widehat{\mathrm{Mon}}^b_a(\Phi_b, u)$ predicts non-failure for a prospective α.
- t_7 (CLOSED): $\mathrm{Intent}_a(\alpha, \mathrm{help})$ formed from incentive (g_{help}, w); $\mathrm{Act}(a, \alpha)$ realised.
- t_8 (OPEN): an OBS step apprehends feedback f from b.
- t_9 (CLOSED): interpretation of f via $\widehat{M}_a(b)$ concludes $\widehat{\mathrm{Mon}}^b_a$ was wrong.
- t_{10} (CLOSED): a SEM step repairs the denotation record of $s_{\widehat{O}_a(b)}$; Σ_a preserved; the (b, α, Φ_b) pipeline returns to W_B (Theorem 12.8).
- t_{11} (CLOSED): the competence attribution is revised; B_a changes; $\mathrm{IdState}_a$ continues (Proposition 13.15).
- t_{12} (CLOSED): GEN yields s_z denoting z_e plus a meta-licence: the meta-certificate “a’s certificate about the effect is held” ($\mathrm{MetaCert}_a$, referent s^a_{self}) is achieved.

Every step is a $\rightarrow_a$-transition in T_{MCK}; every belief-state and identity-state update is expressible in objects already defined.

13.10 Results ledger M-R29–M-R34

- M-R29 belief-state change without Σ change — Proposition 13.7. PROVED-IN-MODEL + CHECKED (extended checker).
- M-R30 identity continuity under belief revision — Proposition 13.15 and definition 13.14. PROVED-IN-MODEL + CHECKED.
- M-R31 self-model update induced by other-agent feedback — Theorem 13.9 and definition 13.8. PROVED-IN-MODEL + CHECKED.
- M-R32 internal-computation update without interaction — Theorem 13.10 and definition 13.1. PROVED-IN-MODEL + CHECKED. Beyond M-R29's Δ-only witness, this adds the consistency-discovery mechanism: the closed step that changes B_a is itself the product of an internal audit.
- M-R33 same external feedback $\rightarrow$ different self-model updates — Theorem 13.11. PROVED-IN-MODEL + CHECKED.
- M-R34 same belief content $\rightarrow$ different incentive consequences — Proposition 13.19, definition 13.17, and remark 13.18. PROVED-IN-MODEL + CHECKED.

14 Co-evolution: how belief, identity, incentive and social models condition one another

Section 13 established that beliefs, the self-model, the other-models, the identity slice and the incentive state each evolve along the T_{MCK} trajectory. This section supplies what it left open: how their changes *condition one another*. Throughout, an arrow between two formation coordinates is an *architectural dependency* — a statement about which $\mathrm{Req}^{\Sigma}/\mathrm{Req}^{\Delta}$ conditions of one object a change in another can affect — never a psychological causal law.

14.1 The formation state

Definition 14.1 (Formation state)**.** The *formation state* of agent a at t is the tuple of projections

$$\mathrm{Form}_a(t) \;=\; \big(\, B_a(t),\ \widehat{M}_a(a,t),\ \{\widehat{M}_a(b,t)\}_b,\ \mathrm{IdState}_a(t),\ \mathrm{Inc}_a(t),\ \{\mathrm{Intent}_a(\alpha,g,t)\}\,\big),$$

each component as previously defined (Definitions 4.1, 6.2, 13.3, 13.8, 13.12 and 13.17). $\mathrm{Form}_a(t)$ is a derived view over $\kappa_a(t)$: it introduces no carrier, no region, and no transition. Two configurations with the same κ_a have the same Form_a; the converse fails, since Form_a forgets carriers not participating in any of its six components.

14.2 The formation dependency graph

The relation "a change here can reach there" is directional, sparse, and checkable from machinery already present; it is a graph, derived rather than posited.

Definition 14.2 (Formation dependency graph)**.** $G^a_{\mathrm{form}}(t)$ is the directed graph whose nodes are those elements of $\mathrm{Form}_a(t)$'s components that carry defined well-formedness clauses — individual belief records, certificates, self-attributions, other-model claims, incentives, committed incentives, and intents; an object without Req-clauses (e.g. a raw feedback datum before interpretation) is not a node — and whose edges $x \rightarrow y$ hold exactly when a Δ_a- or Σ_a-modification of x can change the Req^{Σ}- or Req^{Δ}-status of y — i.e. when x occurs in y's

carrier list, evidence set, relevance-tag, or scope condition. Edges are read as *realisability dependencies*. The edge relation is computable from $\mathrm{Lnk}(a)$: it is the citation structure of the derivation record, not a new stored object.

Remark 14.3 (G_{form} and Lnk are not rivals). Where $\mathrm{Lnk}(a)$ already records an evidence or derivation edge, G^a_{form} simply reads it off. The graph adds only the uniform reading — every edge is a Req-status dependency — and the licence to apply graph-theoretic vocabulary (paths, cuts, blast radii per Definition 12.15) to formation questions. No parallel bookkeeping is introduced.

14.3 Belief → incentive

An agent that learns which outcomes its goals touch may find its incentive state changed without any change in who it is; the honest statement is an existence claim about admissible updates, not "beliefs determine incentives".

Proposition 14.4 (M-R35: belief revision can alter Inc_a without changing $\mathrm{IdState}_a$)**.** *There exist trajectory steps $\kappa_a(t) \to \kappa_a(t+1)$ in which a* CONTENT-REPLACEMENT *revision of a world-belief record changes the relevance-tag set of Inc_a (Definition 13.17) — a carrier w gains or loses g-relevance — while $\mathrm{IdState}_a(t) \sim \mathrm{IdState}_a(t+1)$ per Definition 13.14.*

Proof. Let $(g, w) \in \mathrm{Inc}_a(t)$ with the g-relevance of w registered in $\mathrm{Lnk}(a)$ via a belief record $(\varphi_{\mathrm{rel}}, w, \mathrm{TRUE})$ asserting that w bears on g. Apply evidence repair (Definition 12.7) flipping that record's verdict to FALSE: the relevance registration loses its support and (g, w) exits $\mathrm{Inc}_a(t+1)$. The revised belief is a world-belief, not de se: s^a_{self}, its denotation, the de se certificate slice and the committed-incentive slice are untouched, so identity continuity holds by the argument of Proposition 13.15. The four stage-distinctions of Remark 13.21 are preserved: the goal artefact g still exists; only its current relevance changed. □

14.4 Incentive → intent

Intent is not a function of the incentive state: the architecture is non-injective in both directions (Remark 13.18), and a function would silently import a decision theory the paper does not own. What the architecture supports is an admissibility relation.

Definition 14.5 (Support relation)**.** $\mathrm{Supports}_a(I, i)$ holds between a finite incentive subset $I \subseteq \mathrm{Inc}_a(t)$ and an intent artefact $i = \mathrm{Intent}_a(\alpha, g)$ when g occurs as the goal of some $(g, w) \in I$ and the $\mathrm{Lnk}(a)$ derivation record of i cites at least one w from I as its relevance ground. $\mathrm{Supports}_a$ is a relation, not a function: it neither selects a unique intent for given incentives nor determines the incentives from the intent.

Proposition 14.6 (Bidirectional non-injectivity of support)**.** *There exist κ_a with a single incentive set I supporting two distinct intents $i_1 \neq i_2$, and κ'_a with a single intent i supported by two disjoint incentive sets $I_1 \cap I_2 = \emptyset$.*

Proof. For the first: let $I = \{(g, w)\}$ and let $\alpha_1 \neq \alpha_2$ both be Own_A-actions whose derivation records cite w; then $\mathrm{Supports}_a(I, \mathrm{Intent}_a(\alpha_1, g))$ and $\mathrm{Supports}_a(I, \mathrm{Intent}_a(\alpha_2, g))$. For the second: let g occur in both $(g, w_1) \in I_1$ and $(g, w_2) \in I_2$ with $w_1 \neq w_2$; a single $i = \mathrm{Intent}_a(\alpha, g)$ citing both grounds is supported by either set alone. □

14.5 Identity → incentive, incentive → identity

Two directions, two distinct claims; the asymmetry is the scientific content. Identity constrains which incentives persist as *commitments*; incentive persistence alone never produces identity incorporation — that requires an explicit act of self-binding, visible below as a missing certificate.

Proposition 14.7 (M-R36: identity continuity does not imply incentive invariance)**.** *There exist trajectory steps with* $\mathrm{IdState}_a(t) \sim \mathrm{IdState}_a(t+1)$ *and* $\mathrm{Inc}_a(t) \neq \mathrm{Inc}_a(t+1)$.

Proof. Immediate from Proposition 14.4: the witness there changes a non-committed incentive while identity continues. More generally, any ACQUISITION of a fresh goal artefact $g' \in \mathcal{G}_a$ with a relevance-tag adds to Inc_a without touching the identity slice, since $\mathrm{Inc}_a^{\mathrm{committed}}$ is unchanged. □

Proposition 14.8 (M-R37: persistent incentive without identity commitment)**.** *There exists a trajectory* $\kappa_a(t_0) \to \cdots \to \kappa_a(t_n)$ *along which some* $(g, w) \in \mathrm{Inc}_a(t_k)$ *for every* k*, while for no* k *does* (g, w) *satisfy Definition 13.20: the de se certificate* $\mathrm{Cert}_a(\text{“}a \text{ pursues } g\text{”}, s^a_{\mathrm{self}}) = \textsc{true}$ *is never formed.*

Proof. Persistence of the pair (g, w) requires only that no revision event removes the relevance registration — a trajectory of closed steps not touching it suffices. Commitment additionally requires a GEN step producing the self-bound certificate licence; absent that step (which nothing in T_{MCK} forces to occur, per Remark 10.23), the incentive persists uncommitted. Persistence is a non-event; commitment is an event. □

14.6 Other → self, self → other

The relational mirror of Theorem 13.9 is one link of a longer chain, and the chain's scientific value lies in the fact that *every link is separately optional.* A response from b reaches a's identity only through five intermediate representations, any of which can absorb, distort, or discard it. Symmetrically, a's self-model conditions what a makes of b before b has done anything at all.

Remark 14.9 (The mediated chain, other → self)*.* The full architectural path from b's response to a's identity is:

$$\begin{aligned} b \text{ emits } &\to a \text{ observes a projection } \to a \text{ interprets } \to \widehat{M}_a(b) \text{ update} \\ &\to \text{evidence about } a \to \widehat{M}_a(a) \text{ update } \to \mathrm{IdState}_a \to \mathrm{Inc}_a. \end{aligned}$$

Each arrow is a distinct T_{MCK} step family with its own $\mathrm{Req}^{\Sigma}/\mathrm{Req}^{\Delta}$ conditions, and each may fail to fire. In particular no external judgment of a modifies $\mathrm{IdState}_a$ directly: the last two arrows require de se certificate formation, which is a's own GEN. The chain, not any single link, is the unit of analysis — this is the formal content behind the slogan that experience does not directly become belief, and another's judgment does not directly become the self.

Theorem 14.10 (M-R38: the self-model conditions the other-model update)**.** *There exist agents* a_1, a_2 *with identical* Σ*, identical* $\widehat{M}_{a_i}(b)$ *content, observing the same behaviour of* b*, whose posterior states differ:* a_1 *updates* $\widehat{M}_{a_1}(b)$ *and forms an intervening intent,* a_2 *leaves* $\widehat{M}_{a_2}(b)$ *unchanged — the difference residing entirely in the self-model content* $\widehat{M}_{a_i}(a_i)$*.*

Proof. Let a_1's self-model contain the self-attribution "a_1 is responsible for outcomes in this scope" (a de se certificate over a responsibility predicate), and let a_2's self-model lack it. Both observe b failing at a task. For a_1 we construct an admissible GEN whose derivation record cites the responsibility attribution and whose output is the $\widehat{M}_{a_1}(b)$-claim "b expects assistance", together with an intent supported via $\mathrm{Supports}_{a_1}$. For a_2 no admissible update citing such a ground exists — the ground is absent from $\mathrm{Sto}(a_2)$ — so the constructed a_2-witness records the observation as a world-belief and leaves the other-model untouched. Same observation, same carriers, different self-model content, different *available* formation updates — an existence claim about admissible updates, localised to the self-model slice, and an instance of path dependence (Proposition 14.13). □

14.7 Reciprocal modelling, bounded

Remark 14.11 (Two agents, no regress). Let a and b each hold an other-model of the other and a self-model of themselves, with model depth bounded at two ($\widehat{M}_a(b)$ may contain a claim about $\widehat{M}_b(a)$'s content, and no deeper). An interaction round — a acts, b observes a projection and updates $\widehat{M}_b(a)$, b responds, a observes and updates $\widehat{M}_a(b)$, each optionally propagating along its own mediated chain (Remark 14.9) — reshapes both participants' formation states without any common-knowledge assumption and without unbounded nesting: the depth bound is a Σ-level fact (no carrier for depth-three claims exists), so the regress is cut architecturally rather than by fiat. Result M-R42 (Appendix A) exhibits a finite two-round witness in which both agents' $\widehat{M}$-contents and one agent's self-model change, while both identities continue.

14.8 Closed-loop revision of identity and incentive

Theorem 14.12 (M-R39: closed identity/incentive revision)**.** *Suppose a's architecture basis $\mathcal{O}_a$ contains operators $f, h \in K_a$ with $f \circ_a h$ defined, whose GEN-step output is a licence artefact whose content is the claim "a pursues g" with referent s^a_{self} and evidence drawn from* $\text{Lnk}(a)$*'s record of a's own past actions citing w. Then there exist trajectories consisting exclusively of* CLOSED *steps (Definition 13.1) along which* Inc_a *changes and the incentive (g, w) becomes committed (entering* IdState_a*), with no interpersonal input.*

Proof. Start from $\kappa_a(t)$ holding the persistent uncommitted incentive (g, w) (per Proposition 14.8), the self-symbol, the cited evidence, and the hypothesised operators. The GEN step $y = f \circ_a h$ is enabled — its operands are K_a-members composable under $\circ_a$ by hypothesis, exactly as Foundations' GEN clause requires — and its output is the de se commitment licence. The resulting certificate satisfies Definition 13.20; the incentive becomes committed. Every step consumed only internal carriers: reflection, not testimony, produced the commitment. The hypothesis on $\mathcal{O}_a$ is essential and not eliminable: an architecture without a comparison operator of this type cannot reach commitment by closed steps (an instance of Proposition 10.22's reachability boundary). □

14.9 Path dependence and stability

Proposition 14.13 (M-R40: formation is state-dependent)**.** *There exist $\kappa_1(t) \neq \kappa_2(t)$ with identical Σ-components receiving the same external input e such that* $\text{Form}_1(t{+}1) \neq \text{Form}_2(t{+}1)$.

Proof. Theorem 13.11 is the two-agent instance (read the two agents as two trajectory states of one architecture); Theorem 14.10 is the self-model-localised instance. In both, the constructed witnesses' derivation records cite pre-existing Δ-content, so distinct histories that deposited distinct content admit distinct updates from identical input. We do not strengthen this to any claim of historical determinism: the proposition asserts sensitivity of the *available* updates to state, not determination of state by history. □

Definition 14.14 (Coordinate stability)**.** Over an interval $[t, t']$ of a trajectory, agent a is *belief-stable* when B_a is constant on the interval; *identity-stable* when $\text{IdState}_a(t) \sim \cdots \sim \text{IdState}_a(t')$; *incentive-stable* when Inc_a is constant; *model-stable* when every $\widehat{M}_a(\cdot)$ is constant. The four notions are logically independent.

Proposition 14.15 (M-R41: stability coordinates are independent)**.** *For each pair of the four stability notions of Definition 14.14, there is a finite trajectory interval witnessing one without the other.*

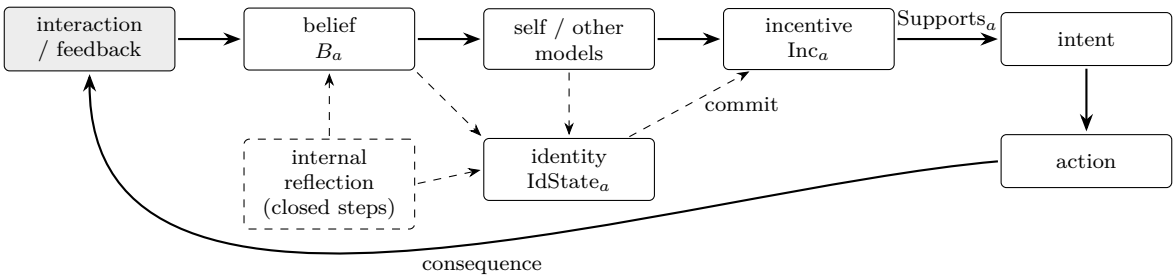


Figure 10: Figure L: the formation dependency graph G^a_{form} at component granularity. Solid arrows: the interaction-driven (open-step) update path; dashed arrows: realisability dependencies, including the closed-step reflection channel and the identity-commitment edge (Definition 14.2).

Proof. Proposition 13.15 gives identity-stable without belief-stable. Proposition 14.4 gives identity-stable without incentive-stable. Theorem 13.9 gives belief-stability of world-beliefs compatible with model-instability (the self-model slice changes). A trajectory of closed steps performing only consistency checks that find no conflict is stable in all four coordinates, witnessing the conjunctions; and Proposition 13.16 gives belief-stable, incentive-stable intervals that are not identity-stable (only the self-denotation was corrupted). The six ordered pairs are covered in Appendix A, M-R41. □

14.10 Formation criticality and reflexive formation

Remark 14.16 (Blast radii over G_{form}). The blast-radius machinery of Definition 12.15 applies to G^a_{form} unchanged, and reproduces at the formation level the asymmetry proved at the certificate level in Theorem 12.16: corruption of the self-symbol's denotation is a cut whose formation blast radius spans the identity slice, every committed incentive, and every de se belief record — whereas correction of a single world-belief typically reaches only the incentives and intents whose relevance-tags cite it. Formation-critical carriers are exactly those with high out-degree in G^a_{form} along Req-dependency edges.

Remark 14.17 (Reflexive formation). MetaCert and RefAware now read as formation-level achievements: the agent forms not only a representation but an architecturally realised representation of its own relation to that representation — for a belief (knowing that one holds the certificate), for an identity commitment (level 4 of Remark 13.21), for an incentive (knowing what one treats as mattering), or for an other-model (knowing that one's picture of b is a picture). None of this is automatic: each reflexive lift is a REFLEXIVE-LIFT revision event requiring its own carriers per Table 3, and Proposition 10.22 shows architectures from which no such lift is reachable.

14.11 Formation histories and their non-injectivity

Does an agent's present formation state reveal how it was formed? The question needs no new machinery — a history is a finite T_{MCK} path — and the answer is negative in an informative way.

Definition 14.18 (Formation history)**.** A *formation history* of agent a is a finite $\to_a$-path $h = \kappa^0_a \to_a \cdots \to_a \kappa^n_a$ in T^a_{MCK} (Definition 10.19). Its *outcome* is the formation state Form_a of its final configuration.

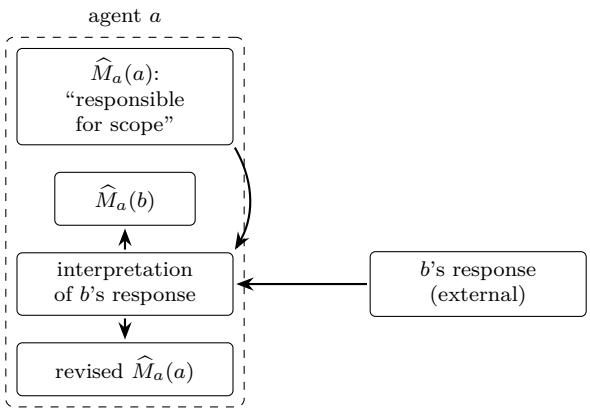


Figure 11: Figure M: the relational mirror. b's response is external; interpretation, other-model update, and any self-model revision happen inside a's Δ_a, conditioned by a's prior self-model (Theorem 14.10 and proposition 14.13).

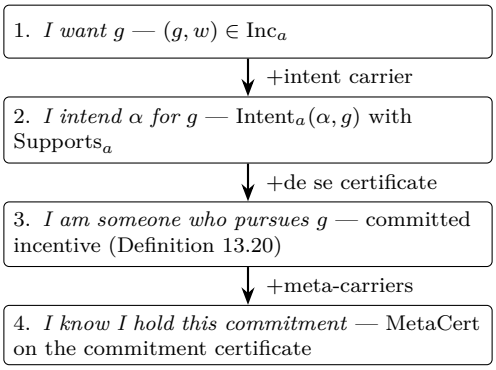


Figure 12: Figure N: the four-level incentive/identity ladder (Remark 13.21); each ascent adds concrete $\mathrm{Req}^{\Sigma}/\mathrm{Req}^{\Delta}$ conditions, and no ascent is automatic (Proposition 14.8).

Theorem 14.19 (M-R43: formation-history non-injectivity)**.** *The outcome map* $h \mapsto \mathrm{Form}_a(\kappa_a^n)$ *is not injective: distinct formation histories produce identical formation states. Two witness families: (i)* commutation *— two independent GEN steps (producing carriers with disjoint citation structures) executed in either order reach the same* κ_a*, hence the same* Form_a*; (ii)* cancellation *— a history that acquires a belief record, invalidates it, and re-acquires it with the same content and evidence reaches the same* Form_a *as the history that acquired it once.*

Proof. (i) GEN steps whose licence inputs and outputs are disjoint commute by inspection of Foundations' step relation: neither step's precondition mentions the other's artefact, and the resulting (Σ_a, Δ_a) is the union in both orders. (ii) The invalidation flips a verdict field and the re-acquisition restores it together with its evidence citation; the final Δ_a record is field-by-field identical to the single-step history's, and Σ_a agrees since no carrier was added or removed on net. Both witnesses are finite and enumerable; mechanised in the extended checker. □

Corollary 14.20 (Two successive information losses)**.** *Composing Theorem 14.19 with the trace non-injectivity of Theorem 21.5: formation history underdetermines the current formation state, and the current formation state underdetermines observable behaviour. An external observer of a's trace is therefore two projections away from a's formation history; an internal auditor of a's present state is still one projection away. Neither the world nor the agent's present self-inspection recovers how the agent came to be what it is — unless provenance records*

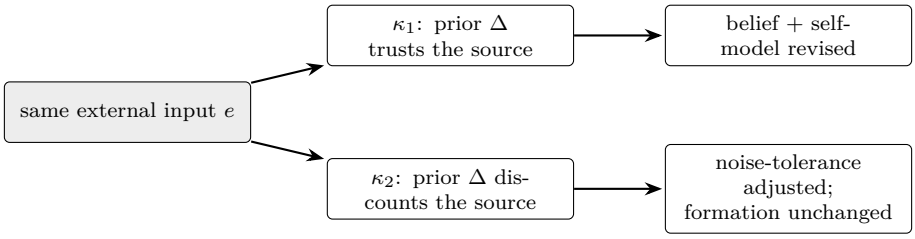


Figure 13: Figure O: same event, different formation (Theorem 13.11 and proposition 14.13): admissible updates cite pre-existing Δ-content.

(Definition 13.5) were themselves preserved as content, which is an architectural choice, not a given.

Remark 14.21 (G_{form} is mechanised). The extraction of G^a_{form} from a finite Lnk(a) instance is implemented in the checker distributed with this article (`gform_extract.py`): input a finite citation structure and the Req^Σ/Req^Δ declarations, output the derived edge set with per-edge provenance and transitive blast radii. The implementation confirms the manually stated examples — the self-symbol's blast spans the identity stack while an unrelated world-belief's blast is empty — and records the one metadata extension the derivation needs beyond Foundations' Lnk pairs: a relation-kind tag (*evidence / relevance / self-binding / derivation / scope*) on each citation. No second dependency store exists; every edge is derived.

14.12 Results ledger M-R35–M-R43

- M-R35 belief $\to$ incentive without identity change — Proposition 14.4. PROVED-IN-MODEL + CHECKED.
- M-R36 identity continuity $\neq$ incentive invariance — Proposition 14.7. PROVED-IN-MODEL + CHECKED (corollary-strength: follows from M-R35's witness).
- M-R37 persistent incentive $\neq$ identity commitment — Proposition 14.8. PROVED-IN-MODEL + CHECKED.
- M-R38 self-model conditions other-model / intent update — Theorem 14.10. PROVED-IN-MODEL + CHECKED.
- M-R39 closed identity / incentive revision — Theorem 14.12. PROVED-IN-MODEL + CHECKED.
- M-R40 formation path dependence — Proposition 14.13. PROVED-IN-MODEL (a generalisation reading over the instances M-R33 and M-R38; no independent witness).
- M-R41 stability-coordinate independence — Definition 14.14 and proposition 14.15. PROVED-IN-MODEL + CHECKED.
- M-R42 bounded reciprocal co-evolution witness — Remark 14.11, finite two-round witness in Appendix A. PROVED-IN-MODEL + CHECKED.
- M-R43 formation-history non-injectivity — Definition 14.18, theorem 14.19, and corollary 14.20. PROVED-IN-MODEL + CHECKED. Mechanised G_{form} extraction: Remark 14.21, CHECKED.

15 Attention dynamics: focus, action conditioning, and the feedback loop

The architecture so far treats attention minimally: an internal, exogenously given bounded weighting $w_a(t) : \mathrm{Sto}(a) \to [0,1]$ inducing the effective support $\mathrm{Str}_{\mathrm{eff}}(a,t)$ (Definition 3.9), consumed by reflexive awareness (Definition 3.8), operational consciousness (Definition 3.11), and the companion w-criticality criterion (Remark 23.2). This section closes the recorded attention-dynamics boundary at the honest scale — finite, constructive, existential — by making attention an explicit third coordinate of the configuration, defining focus shifts as a declared transition family in the OBS/SEM pattern, and proving that attention conditions action, feedback registration, and — the section's sharpest result — the agent's effective observation of *temporal implication* in its own experience.

15.1 What exists, and what was only a premise

The trail is documented: w_a was introduced as a bounded function on Foundations-owned domains with no commitment to a weighting mechanism (Remark 3.10); $\mathrm{Str}_{\mathrm{eff}}$ is derived from it; every consumer treats $w_a(t)$ as given at t. Never supplied: a transition that changes w_a, and any relation of attention to action selection or feedback registration. Those are this section's additions; since w_a lives in neither Σ_a nor Δ_a, the configuration itself must be extended.

Definition 15.1 (Attention-extended configuration and the FOC family)**.** The *attention-extended configuration* of agent a at t is

$$\kappa_a^+(t) \;=\; \big(\,\Sigma_a(t),\ \Delta_a(t),\ w_a(t)\,\big),$$

the two-layer state of Definition 9.3 together with the attention weighting of Definition 3.9 (an internal, exogenously given function: held by the architecture, not further modelled). A *FOC step* is a transition that modifies w_a and nothing else: Σ_a and Δ_a are preserved exactly. The *extended transition system* is

$$T^+_{\mathrm{MCK}} \;=\; (\mathcal{K}_a^+,\ \to_a^+), \qquad \to_a^+ = \to_a \,\cup\, \mathrm{FOC},$$

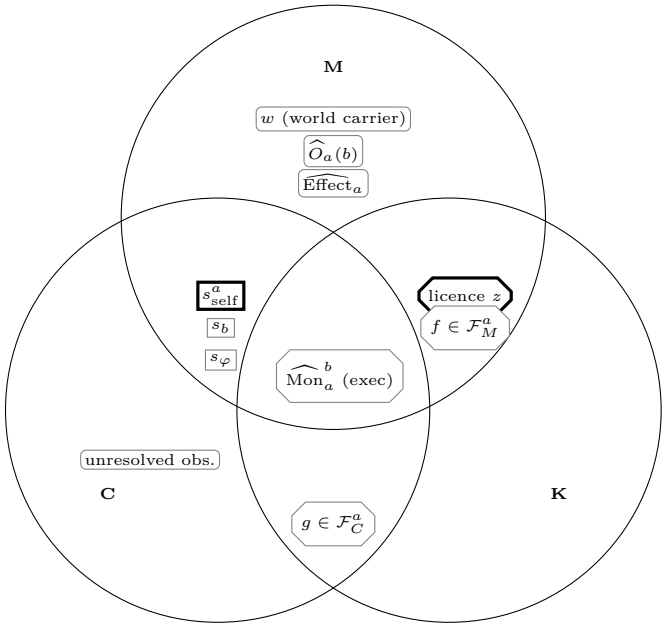
with $\to_a$ the five artefact-consuming families of Definitions 10.19 and 10.20 lifted to act as identity on the w_a-coordinate. FOC extends the declaration of Definition 10.20 as a third paper-owned family; the closed/open dichotomy of Definition 13.1 ranges over the artefact-consuming families, and a FOC step — which consumes no artefact — is classified CLOSED by convention. (The superscript in κ_a^+ marks the extension and is unrelated to the classified domain $\mathcal{A}_a^+$.) We write $\mathrm{ShiftFocus}_a(F \to F')$ for a FOC step whose induced focus (Definition 15.3) changes from F to F'. Note that the attention perturbation π of Remark 23.2 is precisely a FOC step: the companion w-criticality criterion is now a statement about T^+_{MCK}-reachable degradation.

15.2 The critical family and the focus view

Definition 15.2 (Critical family)**.** $Q_{\mathrm{crit}} = \{p_{\mathrm{id}}, p_{\mathrm{int}}, p_{\mathrm{free}}\}$ — exactly the three worked self-properties, which are the properties this paper formally owns as critical (Corollary 23.4). The derived consistency capability q_{int} of Section 12 is not a member: it audits members. No further critical properties are owned by the programme at this time.

Definition 15.3 (Attentional focus)**.**

$$\mathrm{Focus}_a(t) \;=\; \{\, q \in Q_{\mathrm{crit}} \;:\; \text{every carrier in } \mathrm{Req}_a^{\Sigma}\big(\mathrm{Cert}_a(q, s^a_{\mathrm{self}}), t\big) \text{ lies in } \mathrm{Str}_{\mathrm{eff}}(a,t) \,\}.$$



thick border = effectively attended (Str_{eff}); gray border = represented but unattended. Attention changes no region.

Figure 14: Figure U: critical focus over the regional atlas. The architecture is exactly Figure P's (same spec, same placement; s^a_{self}'s placement identity is machine-checked); the only new information is the border weight: thick = effectively attended, gray = represented but unattended. Attention changes no region: s^a_{self} sits in R_4 under every weighting.

The coercion is explicit: q is a self-property, and the capability whose requirements are read is its de se certificate; the same t feeds both the requirement evaluation and the effective support. We assume throughout that every Req^Σ-carrier — including Θ_a-symbols and K-cell licences — lies in the weighting's domain $\text{Sto}(a)$, so effective membership is well-defined. Focus_a is a derived view — no new stored object — computed from the existing effective support and the existing requirement declarations. Membership requires the property's carriers to be *effectively attended*: as artefact sets,

$$\text{REFLEXIVELY AVAILABLE} \subsetneq \text{ATTENDED} \subsetneq \text{REPRESENTED},$$

per Definitions 3.8 and 3.9, each inclusion strict with a finite witness (M-R44, checker); the corresponding property-indexed implications fail (Proposition 15.16).

Remark 15.4 (Attention is not value — the non-collapse). $q \in \text{Focus}_a(t)$ says only that q's carriers can participate in the current computation. It does *not* say q is true (truth is external); nor that q's certificate holds (that is Δ_a-content); nor that q is desirable (that is Inc_a); nor that q determines the action (support is relational, Definition 14.5). Five different facts; the formalism keeps all five apart, and a FOC step moves only the first.

15.3 Attention-conditioned action

Definition 15.5 (Attention-conditioned availability)**.** An intent $i = \text{Intent}_a(\alpha, g)$ is *available under focus* at $\kappa^+_a(t)$ when $\text{Supports}_a(I, i)$ holds for some $I \subseteq \text{Inc}_a(t)$ whose ground carriers — the w-components of its (g, w) pairs; the goal artefacts g need not be attended, deliberately — all lie in $\text{Str}_{\text{eff}}(a, t)$, and i's own carrier is action-available (Definition 3.13) at t. We write $\text{ConditionedBy}_a(\alpha, q, t)$ when every such supporting ground for α's intent carries a q-relevance

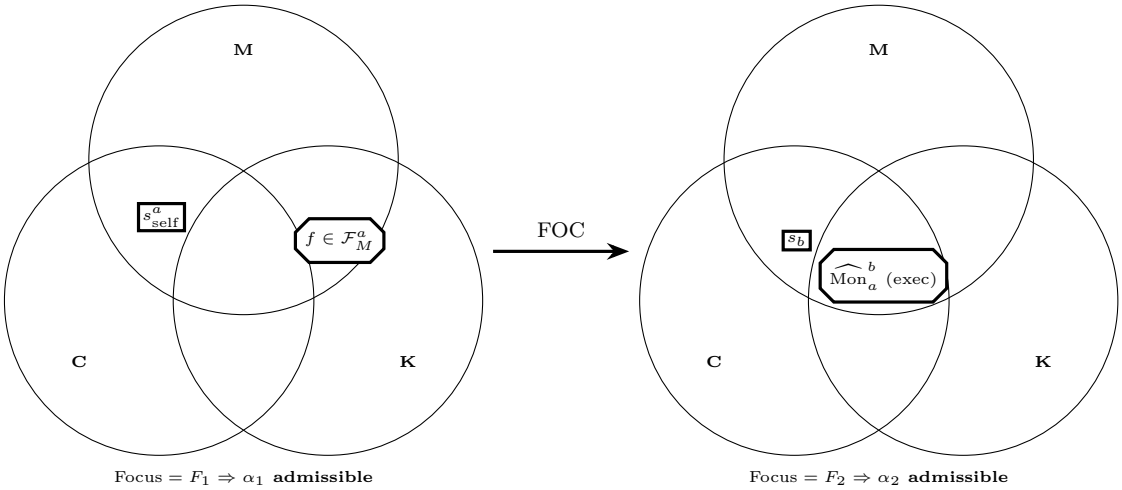


Figure 15: Figure V: the focus switch of Theorem 15.6. Each panel draws *only the attended carriers* of its focus, at machine-verified atlas coordinates; the full shared architecture is Figure U's, identical in both panels. One FOC step separates them; the admissible action witness changes with no rewrite of Σ_a or Δ_a.

registration in $\text{Lnk}(a)$ — the same registration structure that Definition 15.7(iv) later applies to feedback records.

Theorem 15.6 (M-R45/46: focus switch without state rewrite)**.** *There exists a finite $\kappa_a = (\Sigma_a, \Delta_a)$, fixed, with two attention weightings w^1, w^2 such that $\kappa^{+,1} = (\Sigma_a, \Delta_a, w^1)$ admits an available intent witness $\text{Intent}_a(\alpha_1, g_1)$ and no available witness for α_2, while $\kappa^{+,2} = (\Sigma_a, \Delta_a, w^2)$ admits $\text{Intent}_a(\alpha_2, g_2)$ and no available witness for α_1. A single FOC step separates the two; no belief, certificate, incentive content, or identity field is rewritten.*

Proof. Take Σ_a holding two incentive grounds u_1 (relevance-tagged to g_1, itself tagged to p_{int}) and u_2 (tagged to g_2, tagged to p_{free}), with both relevance registrations in Δ_a and both intents' carriers present. Let w^1 weight u_1's chain above threshold and u_2's below; w^2 the reverse. Under w^1 only g_1's ground is effectively attended, so only $\text{Intent}_a(\alpha_1, g_1)$ is available under focus; under w^2 only $\text{Intent}_a(\alpha_2, g_2)$. The underlying (Σ_a, Δ_a) is shared verbatim. Mechanised in the extended checker (M-R45/46 gate). Existence, not selection: the theorem says which witnesses are *admissible*, not which an agent takes. □

15.4 The feedback channel, five stages deep

Definition 15.7 (Feedback stages)**.** For an action α with external consequence, distinguish: (i) feedback *emitted* by world or other (external; not a's); (ii) feedback *observed* — entering $\mathcal{A}^+_a$ via an OBS step; (iii) feedback *represented* — the OBS output classified into a region; (iv) feedback *classified as q-relevant* — a $\text{Lnk}(a)$-registration tagging the record as bearing on $q \in Q_{\text{crit}}$; (v) feedback *used* — consumed by a SEM update. Five stages; each transition between them is separately optional, in the pattern of Remark 14.9.

Definition 15.8 (Property-indexed feedback)**.** $\text{FB}_a(q, t)$ is the set of stage-(iv) feedback records currently classified as bearing on q. This is agent-side represented *attribution* of relevance — the world emits events; the agent interprets some of them as evidence concerning identity, integrity, or freedom — and the classification step is itself gated by attention: a record whose q-classification operator's carriers lie outside $\text{Str}_{\text{eff}}(a, t)$ is not classified, however loudly the world emitted it. (This gating clause is an *axiom of the FOC/OBS extension*, owned

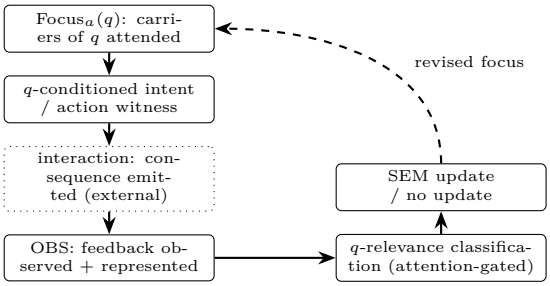


Figure 16: Figure W: the attention–action–feedback loop. The interaction node (dotted) is external; every solid node is a-internal; every arrow is separately optional. The q-relevance classification stage is attention-gated (Definition 15.8) — the loop's closure point, and the site of the implication-confirmation mechanism of Theorem 15.9.

by this paper: attended classification carriers are stipulated necessary for classification. It is what carries the negative branch of Theorem 15.9. The result is *parametric in the architecture*: within the declared extension this is an admissible gate; other architectures may implement another, and nothing here is a universal cognitive law.)

15.5 Attention selects the observation of temporal implication

The deepest consequence of the focus machinery is not about single events but about *event pairs*. Attention gives an endogenous *analogue* of the observation-relative situation of [13]: at any fixed $w_a(t)$, the $\mathrm{Str}_{\mathrm{eff}}$-gated OBS classification is a bounded observation regime over a's own experience, and the theorem below witnesses — inside this architecture, and independently of the imported theorem — the same joint dependence of verdict availability on property and regime. The imported theorem itself concerns receiver-side monitors over interaction traces and is not invoked in the proof; what is genuinely new here is that the regime is selected by the monitored system itself, through FOC steps. For a temporal-implication pattern — an antecedent-consequent relation $A \Rightarrow_{\mathrm{temp}} B$ — the consequence is sharp: the agent confirms an implication instance only if *both* the antecedent registration and the consequent registration survive the attention gate, in order.

Theorem 15.9 (M-R47: attentional implication-confirmation divergence)**.** *There exist two attention-extended configurations sharing* (Σ_a, Δ_a) *and exposed to the same external event sequence containing an antecedent-consequent pair* (e_A, e_B)*, such that: under* w^1 *(attending the* e_A*-classification carriers) the agent registers the pair, classifies it as confirming the implication belief* $(A \Rightarrow_{\mathrm{temp}} B)$*, strengthens that belief (*STRENGTHENING *event of Definition 13.6), and updates the associated incentive relevance; while under* w^2 *(not attending them) the same sequence yields no registered antecedent, hence no confirmation instance, no strengthening, and no incentive update. The implication belief's evidential trajectory diverges purely by attention.*

Proof. Finite witness, mechanised (M-R47 gate). Configuration: an implication belief record $(A \Rightarrow_{\mathrm{temp}} B)$ with verdict UNDETERMINED; classification operators c_A, c_B for the two event types, both represented; incentive (g, w) whose relevance registration cites the implication belief. Under w^1: OBS registers e_A then e_B; a classification step is taken for each in the witness trajectory (carriers attended); the ordered pair matches the antecedent-consequent template; a STRENGTHENING SEM step is admissible and its derivation record cites both registrations; the incentive relevance gains the new evidence. Under w^2: c_A's carriers fall outside $\mathrm{Str}_{\mathrm{eff}}$; the e_A observation is not classified as an antecedent; when e_B arrives there is no open antecedent to pair with; no confirmation exists to be registered, so no admissible

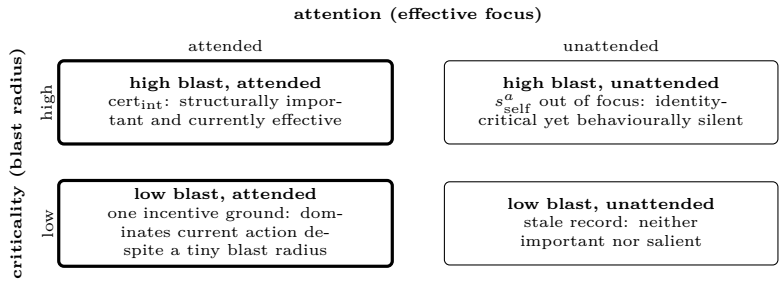


Figure 17: Figure X: criticality × attention. All four cells are inhabited (Proposition 15.11, checker-enumerated): structural importance (blast radius) and current behavioural salience (effective focus) are independent coordinates and must not be scalarised.

strengthening step cites this episode. Same world, same events, same stored architecture — different evidential history for the same implication belief. □

Remark 15.10 (Attentional reinforcement, stated honestly). Iterating Theorem 15.9 yields the loop the section is named for: a focus that includes q's carriers makes q-relevant implication confirmations registrable, and registered confirmations strengthen exactly the beliefs and incentive relevances that keep q's carriers attention-worthy under any weighting policy that weights carriers with live relevance registrations above threshold. The architecture thus *admits* self-maintaining attentional pathways in which what the agent attends determines which antecedent-consequent regularities it can ever confirm. This is an admissibility structure over finite witnesses (M-R49), not a law: nothing forces the loop to close, and a FOC step can exit it at any point. A switch of effective focus thereby changes which implications the agent's future experience can confirm, without any rewrite of what the agent believes today.

15.6 Criticality is not salience

Proposition 15.11 (M-R48: criticality and attention are independent coordinates)**.** *There exist configurations in which (i) the artefact with the largest criticality blast radius (s^a_{self}, per Theorems 8.12 and 12.16) lies outside* $\text{Str}_{\text{eff}}(a,t)$*, while (ii) an artefact of minimal blast radius (a single incentive-ground carrier) lies inside it and dominates the available action witnesses. Structural importance and current behavioural salience are separate coordinates;* $\text{Crit}(q) \times \text{Focus}(q)$ *must not be scalarised.*

Proof. Direct construction (M-R48 gate): weight s^a_{self}'s chain below threshold and the incentive ground above. De se certification is then unavailable-under-focus while the low-blast ground conditions every admissible intent. The four cells of the attended/unattended × high/low-blast matrix are all inhabited (checker enumerates them). □

15.7 Two-step recursion and the finite classification

Proposition 15.12 (M-R49: attention-feedback path dependence)**.** *There exists a two-step finite witness: under sustained focus F_1, action α_1 generates feedback classified q_1-relevant, whose use strengthens the F_1-supporting content; under a FOC step to F_2 before acting, action α_2 generates different feedback, classified q_2-relevant, updating different content — so the two trajectories diverge in Δ_a after two steps from a shared κ^+-origin differing only in one FOC event.*

Proof. Composition of Theorem 15.6 (step one: different admissible action) with Definition 15.7/(iv) (step two: attention-gated relevance classification of the resulting feedback)

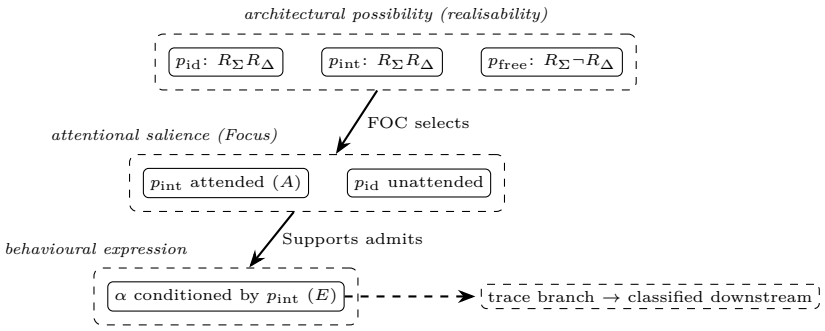


Figure 18: Figure Y/AA: the three-layer separation with the activation-signature count. Possibility (realisability), salience (Focus), and expression are distinct coordinates; each descent is information-lossy; 18 of 32 signatures are type-consistent (Proposition 15.16); the dashed terminal is classified downstream (Table 2).

and Theorem 15.9's strengthening mechanism. Mechanised (M-R49 gate). An instance of formation path dependence (Proposition 14.13) with the path difference located entirely in the attention coordinate. □

Proposition 15.13 (M-R50: finite attention-pattern classification)**.** *For* $|Q_{\text{crit}}| = 3$ *there are* $2^3 = 8$ *focus subsets, reducing to* 4 *classes under critical-property renaming (*$\emptyset$*, singleton, pair, full). The extended checker enumerates all* 8 *subsets, derives the four renaming classes, and tabulates per-class admissibility flags — whether an action difference, a feedback difference, a* Δ*-difference, and downstream-switch-capability are admissible for the class. All four quotient classes are realised; the degenerate patterns of Remark 15.14 appear as explicit rows.*

Remark 15.14 (Degenerate classes are results, not failures)*.* The enumeration retains, as first-class non-injectivity data: focus changes with no action change (both grounds attended in both foci); action changes with no feedback change (consequences indistinguishable under O); feedback changes with no Δ-change (record classified but no admissible update cites it); Δ-changes with no downstream behavioural difference; many-foci-one-action; one-focus-many-actions (support is relational). Each extends the programme's non-injectivity theme — Theorem 21.5, Theorem 9.6, Theorem 14.19 — into the attention coordinate.

15.8 Possibility, salience, expression: the activation signature

The attention layer completes a three-coordinate separation that the paper has been building since the realisability decomposition, and it is worth stating as one object. An agent may possess many structurally and semantically realised capabilities (*architectural possibility*); only a subset is effectively available to condition the current computation (*attentional salience*); and of those, only some participate in any admissible action witness (*behavioural expression*). Three coordinates, three different failure surfaces — and, as the recovered constraints below show, almost no free implications among them.

Definition 15.15 (Activation signature)**.** For $q \in Q_{\text{crit}}$ and $\kappa_a^+(t)$, the *activation signature* is the tuple of derived bits

$$\text{ActSig}_a(q,t) \;=\; (\,R_\Sigma,\; R_\Delta,\; A,\; X,\; E\,)$$

where $R_\Sigma = \text{Req}_a^\Sigma(\text{Cert}_a(q, s_{\text{self}}^a), t)$ holds; $R_\Delta = \text{Req}^\Delta$ holds on those carriers; $A = q \in \text{Focus}_a(t)$; $X = \text{RefAware}_a$ holds for q's certificate; $E =$ some intent in the current admissible action-witness set is $\text{ConditionedBy}_a(\cdot, q, t)$. Every bit is a projection of already-owned state; no new stored object and no psychological variable is introduced.

Proposition 15.16 (M-R53: recovered constraints and the 18 valid cells)**.** *Exactly three implications among the five bits are derivable from the definitions:* $A \Rightarrow R_\Sigma$, $X \Rightarrow R_\Sigma$, *and* $R_\Delta \Rightarrow R_\Sigma$ *(the last by convention: semantic adequacy is* undefined*, hence does not hold, on absent carriers, per Proposition 10.3). The naive activation chain* $X \Rightarrow A \Rightarrow R \Rightarrow E$ *fails at every link:* A *does not imply* R_Δ *(attended carriers can hold wrong content, Theorem 9.6);* X *does not imply* A *(reflexive awareness attends the* meta*-symbol* s_z*, not the object-level requirement carriers);* E *implies neither* A *nor* R_Δ *(an attended q-relevance-tagged ground can condition action while the certificate's own carriers are unattended and its content wrong). Consequently exactly* $16+2 = 18$ *of the* 2^5 *signatures are type-consistent — all* 16 *with* $R_\Sigma = 1$*, plus the two with all of* R_Δ, A, X *zero and* E *free — and all 18 are realised by finite witnesses (checker-enumerated).*

Proof. $A \Rightarrow R_\Sigma$: Focus-membership places every Req^Σ-carrier in $\mathrm{Str}_{\mathrm{eff}} \subseteq \mathrm{Str}(\mathcal{A}_a)$, so the carriers exist with their required signatures. $X \Rightarrow R_\Sigma$: RefAware requires MetaCert, which requires the first-order licence held, whose well-formedness is R_Σ. $R_\Delta \Rightarrow R_\Sigma$: by Proposition 10.3, Req^Δ is undefined when Req^Σ fails, and an undefined requirement does not hold. The non-implications are witnessed pairwise in the checker (M-R53 gate). Cell count: with $R_\Sigma = 1$ no constraint bites (16 cells); with $R_\Sigma = 0$ each of A, X, R_Δ must be 0, leaving E free (2 cells). Mechanised. □

Corollary 15.17 (M-R56: two construction classes of branch change)**.** *An effective-branch change (a change in the admissible action-witness set) can be constructed by a feedback-mediated route — an OBS step apprehending external content, followed by gated classification and SEM update (Theorem 13.9 and proposition 15.12) — or by a purely internal route — closed steps only, including FOC and authorised SEM repair (Theorems 13.10, 14.12 and 15.6). The two classes are distinguished architecturally by the presence or absence of an OBS step in the witness's derivation record; not every branch change requires counterpart feedback. Architectural possibility only; no frequency or realism claim.*

Corollary 15.18 (M-R57: behaviour does not identify the attended property)**.** *The same admissible action* α *can be* ConditionedBy *distinct critical properties in distinct configurations (an instance of the support relation's non-injectivity, Proposition 14.6, composed with the degenerate class "same action, different critical-property support" of Remark 15.14). Composed with the trace projection's non-injectivity (Theorem 21.5), an external observer of behaviour cannot in general recover which critical property was attended — extending the programme's non-injectivity chain (history → formation → effective attention → behaviour) by its final link.*

Remark 15.19 (Critical attributes regulate; they do not determine)*.* The exact exported wording: *critical attributes can regulate which representations, certificates, commitments, actions, or branching possibilities are effectively available; they do not determine behaviour, and no mapping from a critical attribute to a role or behaviour class is defined in this paper.* Downstream interaction semantics may classify the resulting trace branches; that classification, and any role vocabulary, are downstream property (Section 11).

Remark 15.20 (Attention-policy families, typed but not selected)*.* w_a's dynamics remain deliberately unmodelled: MCK specifies what a FOC step *is*, not which policy generates it. Policy families can nonetheless be typed by the citation structure of FOC derivation records, without selecting one: STATIC (no FOC steps occur); EXOGENOUS-SHIFT (FOC records cite no Δ_a content); FEEDBACK-REACTIVE (FOC records cite stage-(v) feedback uses); SELF-REFLECTIVE (FOC records cite MetaCert carriers). The families are architectural admissibility classes; committing to one would be a modelling decision this paper does not take, and the modularity is deliberate.

15.9 Results ledger M-R44–M-R50

- M-R44 represented $\neq$ attended $\neq$ reflexively available — strict inclusions with finite witnesses; stated in Definition 15.3, per Definitions 3.8 and 3.9. PROVED-IN-MODEL + CHECKED.
- M-R45/46 focus switch without state rewrite — Theorem 15.6. PROVED-IN-MODEL + CHECKED.
- M-R47 attentional implication-confirmation divergence — Theorem 15.9, resting on the gating axiom of Definition 15.8. PROVED-IN-MODEL + CHECKED.
- M-R48 criticality $\neq$ attentional salience — Proposition 15.11. PROVED-IN-MODEL + CHECKED.
- M-R49 attention-feedback path dependence — Proposition 15.12. PROVED-IN-MODEL + CHECKED.
- M-R50 finite attention-pattern classification — Proposition 15.13. CHECKED (exhaustive enumeration).
- FOC-preservation (the extension's soundness condition: a FOC step touches neither Σ_a nor Δ_a) — Definition 15.1. CHECKED.
- M-R53 activation signatures: 18 of 32 type-consistent cells; naive chain fails at every link — Definition 15.15 and proposition 15.16. PROVED + CHECKED (exhaustive).
- M-R56 two construction classes of branch change — Corollary 15.17. Corollary-strength over M-R31/49 (open route) and M-R32/39/45 (closed route); CHECKED (route distinction by OBS presence).
- M-R57 behaviour does not identify the attended property — Corollary 15.18. Corollary-strength over Proposition 14.6 + Theorem 21.5; CHECKED. (Candidates M-R51/52/54/55 of the session directive are subsumed by M-R44, M-R48's matrix, M-R45/46, and M-R47 respectively, and are deliberately *not* added as separate results — consolidation over inflation.)

15.10 Export to the interaction-level programme

The attention-layer additions to the downstream contract — Focus_a, $\mathrm{ShiftFocus}_a$ (FOC), $\mathrm{FB}_a(q,t)$, attention-conditioned availability (Definition 15.5), the implication-confirmation mechanism (Theorem 15.9 and remark 15.10), and the non-collapse discipline of Remark 15.4 — are recorded once, in the contract itself (Section 11); role semantics over attention-conditioned traces remain downstream property.

16 Continuity: the history-sensitive baseline

With the two-layer, realisability, formation and attention machinery in place, we return to the three worked properties around which version 1 was built; each now carries both its original minimality analysis and its native regional signature. The first property fixes a requirement sort and establishes a baseline; its mathematics is standard (Remark 16.4).

Definition 16.1 (Continuity)**.** Let $\Sigma_{\mathrm{id}} = \{c, r\}$, read as *continue* and *replace*. The continuity property is

$$p_{\mathrm{id}}(h) \;=\; [\, r \notin h \,],$$

i.e. no replacement event has occurred. Over infinite behaviours it is $G(\neg r)$.

Proposition 16.2 (Present configuration is insufficient)**.** *There are histories $h_1 \neq h_2$ with the same present observable component and $p_{\mathrm{id}}(h_1) \neq p_{\mathrm{id}}(h_2)$.*

Proof. Take $h_1 = cc$ and $h_2 = crc$. Both leave the agent running, so the value component of the configuration agrees. But $r \notin h_1$ and $r \in h_2$, so $p_{\mathrm{id}}(h_1) = \textsc{true}$ and $p_{\mathrm{id}}(h_2) = \textsc{false}$. □

Proposition 16.3 (Two states suffice and are necessary)**.** *Let $q(h) = [\, r \notin h \,]$ with $\delta(q, e) = q \wedge [\, e \neq r \,]$. Then q is incrementally maintainable and sufficient for p_{id}, it has two states, and no one-state architecture is sufficient. The Myhill–Nerode index of p_{id} is 2.*

Proof. Maintainability and sufficiency are immediate: $q(h{\cdot}e) = q(h) \wedge [e \neq r]$, and $q(h) = q(h')$ gives $p_{\mathrm{id}}(h) = p_{\mathrm{id}}(h')$ because $q = p_{\mathrm{id}}$. A one-state architecture identifies all histories, and *Proposition* 16.2 exhibits two with different verdicts, so it is not sufficient. For the index, ε and r are inequivalent (the suffix ε separates them) and every history is equivalent to one of these two, since membership of r is the only distinction any suffix can detect. □

Remark 16.4 (Standard, and labelled as such). Proposition 16.3 is the Myhill–Nerode right-congruence for a safety property whose bad prefixes are those containing r. It is textbook, we claim no novelty for it, and it appears here for two reasons: it fixes the *history* requirement sort used in Section 19, and it establishes that a history-sensitive property need not require history *storage* — one bit replaces the trace. The contribution in this section is the second observation's role in the comparison, not the automaton.

Proposition 16.5 (Type requirement)**.** $\mathrm{TypeReq}(p_{\mathrm{id}})$ *requires only the event alphabet: no predicate on component values and no structure on actions occurs in Definition 16.1.*

Proof. p_{id} is a predicate on Σ_{id}^* defined by membership of a letter. Its evaluation quantifies over nothing else. □

Proposition 16.6 (Regional signature of identity)**.** *Under the regional algebra of Section 8, p_{id}'s de se certificate $\mathrm{Cert}_a(p_{\mathrm{id}}, s^a_{\mathrm{self}})$ requires: (i) $s^a_{\mathrm{self}} \in \Theta_a = R^a_4 \cup R^a_7$ (self-symbol); (ii) a Myhill–Nerode two-state summary artefact $q \in \mathrm{Sto}(a)$ with $M_a(q) = 1$; (iii) evidence $E \subseteq \mathrm{Lnk}(a)$ covering the continuity scope; (iv) a licence in a K-cell.*

Proof. Instance of Definition 8.5 with the four carriers as stated. □

17 Viability: minimal typed grounding

Definition 17.1 (Viability)**.** Let P be a finite set of components with values in $V = \{\mathrm{ok}, \mathrm{dmg}\}$, let $\mathrm{Own} \subseteq P$ be the components the agent counts as its own, and let Viable be a predicate on the values of Own. The viability property is

$$p_{\mathrm{int}}(c) \;=\; \mathrm{Viable}\big(\,\mathrm{state}(c) \restriction \mathrm{Own}\,\big).$$

In the finite model we take $P = \{\pi_0, \pi_1\}$ with *opaque* names, $\mathrm{Own} = \{\pi_1\}$, and Viable true exactly when $\pi_1 = \mathrm{ok}$.

The names are opaque deliberately. If the agent's component were called π_{self}, the identifier would already encode the ownership relation and the hierarchy below would collapse for a spurious reason.

17.1 The candidate class

Definition 17.2 (Integrity architecture class $\mathcal{F}_{\text{int}}$)**.**

$I_0 : c \mapsto$ the multiset of component values
$I_1 : c \mapsto$ the full assignment $P \to V$
$I_2 : c \mapsto$ Own (the ownership relation alone)
$I_3 : c \mapsto \text{Viable}(\text{state}(c) \restriction \text{Own})$ (ownership *and* a viability predicate)
$I_4 :$ a state-transition model of the component set

Remark 17.3 (Why this family is the comparison domain)*.* The five levels are not an arbitrary list. They are a *nested* family, each step exposing exactly one further semantic ingredient of the property:

- I_0 exposes the raw component values and nothing that individuates them;
- I_1 adds individuation — which value belongs to which component — but no notion of which component is the agent's;
- I_2 adds the self-membership relation and drops the values, isolating ownership from condition;
- I_3 adds the classification of the owned condition as admissible or not;
- I_4 adds the dynamics by which that condition changes.

The property of Definition 17.1 is built from exactly two of these ingredients — a self-membership relation and a viability classification — so a family that introduces them one at a time is the domain in which "which of them is actually needed" is a well-posed question. That is the sense in which the family is natural: it isolates the semantic features the property mentions, and I_4 is included precisely because it is the level the surrounding literature builds.

We do not claim the family is exhaustive over all possible encodings, and Theorem 17.7 is not encoding-independent: a different family could contain an architecture we did not consider. Within a family designed to separate exactly the ingredients at issue, the least sufficient level is I_3; the insufficiency results, Lemmas 17.4 and 17.5, are unconditional and survive any enlargement.

17.2 Which levels suffice

Lemma 17.4 (I_0 insufficient)**.** *I_0 is not sufficient for p_{int}.*

Proof. Take c with $(\pi_0, \pi_1) = (\text{ok}, \text{dmg})$ and c' with (dmg, ok). Then $I_0(c) = I_0(c') = \{\{\text{ok}, \text{dmg}\}\}$, while $p_{\text{int}}(c) = \textsc{false}$ and $p_{\text{int}}(c') = \textsc{true}$. The multiset discards which component is the agent's, and that is exactly what the property reads. □

Lemma 17.5 (I_2 insufficient)**.** *I_2 is not sufficient for p_{int}.*

Proof. I_2 is constant on $\mathcal{C}$: it returns Own regardless of the values. It therefore identifies configurations with $\pi_1 = \text{ok}$ and $\pi_1 = \text{dmg}$, which differ in p_{int}. Knowing which component is one's own, without any predicate on its condition, decides nothing. □

Lemma 17.6 (I_3 sufficient, and strictly coarser than I_1)**.** *I_3 is sufficient for p_{int}, and $I_3 \prec I_1$ strictly: I_3 induces 2 classes on the four configurations of the finite model, I_1 induces 4.*

Proof. Sufficiency is immediate, I_3 being p_{int} composed with nothing. For the refinement, I_1 determines the full assignment hence determines I_3; the map φ sending an assignment to the viability of its Own-restriction witnesses $I_3 = \varphi \circ I_1$. Strictness: I_1 separates (ok, ok) from (dmg, ok) while I_3 does not. □

Theorem 17.7 (MIN-2: minimal component-integrity grounding)**.** *Within $\mathcal{F}_{\text{int}}$ of Definition 17.2, the least sufficient architecture for p_{int} is I_3: a self-membership relation together with a viability predicate. In particular I_4, a state-transition model of the component set, is not required.*

Proof. I_3 is sufficient by Lemma 17.6. Every other member of $\mathcal{F}_{\text{int}}$ is either insufficient or a strict refinement of I_3: I_0 and I_2 are insufficient by Lemmas 17.4 and 17.5 (neither is a refinement of I_3, but sufficiency is what fails on p_{int}, so the incomparability under $\prec$ does not rescue them); I_1 and I_4 are sufficient but refine I_3 strictly — I_1 by Lemma 17.6, and I_4 because a transition model determines the current assignment and hence I_1 — so neither is least. Minimality is within $\mathcal{F}_{\text{int}}$, per Convention 7.5. □

Remark 17.8 (I_1 is an upper bound, not a requirement)*.* I_1 is sufficient, and it would be misleading to record that as "identifiers suffice". I_1 is the full configuration state: it is sufficient for every property on the carrier, which is why it carries no information about what p_{int} needs. The content of Theorem 17.7 is the gap between I_1 and I_3.

Proposition 17.9 (Regional signature of integrity)**.** *Under Section 8, p_{int}'s de se certificate* $\text{Cert}_a(p_{\text{int}}, s^a_{\text{self}})$ *requires: (i) $s^a_{\text{self}} \in \Theta_a$; (ii) an ownership-relation carrier* $\text{Own}_a \in \text{Sto}(a)$ *with $M_a(\text{Own}_a) = 1$ so it lies in an understood cell ($R^a_1, R^a_4, R^a_5, R^a_7$); (iii) a viability predicate operator artefact $\hat{v} \in \text{Sto}(a)$ with $K_a(\hat{v}) = 1$, lying in a K-cell; (iv) evidence in* $\text{Lnk}(a)$ *witnessing $\hat{v}(\text{Own}_a$-restriction of state$) = \text{ok}$; (v) the licence $z_{p_{\text{int}}, s^a_{\text{self}}}$ in a K-cell. Notice that I_3 of Definition 17.2 is exactly the pair $(\text{Own}_a, \hat{v})$ — Theorem 17.7 delivered the minimum in the generic-set-theoretic language of v1, and here it recovers as "the minimum is one M-cell carrier plus one K-cell carrier".*

Proof. Instance of Definition 8.5; sufficiency of I_3 is Theorem 17.7; regional membership of Own_a is by construction of the ownership predicate as an understood datum ($M = 1$, not required to be an operator); regional membership of $\hat{v}$ as a K-cell is by construction as a value-classification operator. □

Remark 17.10 (Relation to robot self-modelling)*.* Continuous self-modelling [9] is level I_4 and beyond, and the tasks it serves — prediction, damage localisation, recovery — genuinely need it. Theorem 17.7 does not compete with those tasks: it says the *attribution* of the viability property needs only the relation and the predicate, not the transition model. The fuller comparison is in Section 27.

18 Controllability: minimal action typing

Definition 18.1 (Controllability)**.** Let $T = (S, A, \to, O)$ with S a finite state set, A a finite action alphabet, $\to \subseteq S \times A \times S$, and $O \subseteq A$ the actions the agent owns. Define

$$p_{\text{free}}(s) \;=\; \big[\, \exists\, \alpha, \beta \in O,\ \exists\, s' \neq s''.\ s \xrightarrow{\alpha} s' \ \wedge\ s \xrightarrow{\beta} s'' \,\big].$$

The quantifier ranges over O, not over A. This is what separates the property from reachability: branching that the environment produces is not the agent's alternative.

Remark 18.2 (Reachability is not the definition)*.* $|\text{Reach}(s)| > 1$ is a tempting shorthand and we do not use it. Proposition 18.4 shows it is not even extensionally adequate.

18.1 The witness

Definition 18.3 (Systems A and B')**.** Both have $S = \{s_0, x, y\}$, $A = \{\alpha, \beta\}$ and the same transition relation $s_0 \xrightarrow{\alpha} x$, $s_0 \xrightarrow{\beta} y$. They differ only in ownership: $O_{\mathsf{A}} = \{\alpha, \beta\}$ and $O_{\mathsf{B}'} = \{\alpha\}$.

Proposition 18.4 (F_0 and F_1 insufficient)**.** A *and* B' *have the same reachable set from* s_0 *and the same labelled successor relation at* s_0*, yet* $p^{\mathsf{A}}_{\text{free}}(s_0) = \textsc{true}$ *and* $p^{\mathsf{B}'}_{\text{free}}(s_0) = \textsc{false}$*.*

Proof. $\text{Reach}(s_0) = \{x, y\}$ in both, and the labelled relation is $\{(\alpha, x), (\beta, y)\}$ in both, since $\to$ is identical by Definition 18.3. For the verdicts: in A take $\alpha, \beta \in O_{\mathsf{A}}$ with successors $x \neq y$, so the property holds. In B', $O_{\mathsf{B}'} = \{\alpha\}$ is a singleton, so no pair $\alpha, \beta \in O_{\mathsf{B}'}$ has distinct successors and the property fails. □

Definition 18.5 (Freedom architecture class $\mathcal{F}_{\text{free}}$)**.**

F_0 : the reachable successor set
F_1 : the action-labelled successor relation
F_2 : the partition of A into O and $A \setminus O$
F_3 : an admissibility relation $S \to \mathcal{P}(A)$
F_4 : a strategy or counterfactual control model

Remark 18.6 (Why this family is the comparison domain)*.* As with Definition 17.2, the levels expose one further ingredient at a time: F_0 the reachable successors alone; F_1 which action produced which successor; F_2 which of those actions are the agent's; F_3 which are available where; F_4 how they may be combined into strategies. Definition 18.1 quantifies over exactly the third ingredient, so the family separates what the property mentions from what it does not, and F_4 is present because it is the structure usually assumed. The family is not claimed exhaustive, and Theorem 18.8 is not encoding-independent; the insufficiency of F_0 and F_1 is unconditional.

Remark 18.7 (Why F_0, F_1 and F_2 are commensurable)*.* The three are not architectures over different semantic domains, and the comparison is not "more information" loosely construed. All five members of $\mathcal{F}_{\text{free}}$ are functions on the same set — the configurations of Definition 7.1, each of which carries the transition relation $\to$ of the ambient system together with a state and an ownership subset — and all are compared under the single preorder of Definition 7.3, whose φ is a bare function.

What each exposes, on that common domain, is:

- F_0: the set $\{s' : \exists a.\ s \xrightarrow{a} s'\}$. A function of $\to$ and s.
- F_1: the set $\{(a, s') : s \xrightarrow{a} s'\}$. Also a function of $\to$ and s, and $F_0 = \pi_2 \circ F_1$, so $F_0 \prec F_1$.
- F_2: the pair consisting of that labelled relation and the ownership subset O. Since F_1 is recoverable by first projection, $F_1 \prec F_2$.

So the three sit in a chain $F_0 \prec F_1 \prec F_2$ under Definition 7.3, and "which is least sufficient" is well posed. Adding the ownership labels is a genuine refinement, not a change of domain: O is a coordinate of the configuration, present in Definition 7.1 for independent reasons (Remark 7.2). Theorem 18.8 is then the ordinary claim — the least sufficient element of a chain — not a comparison across incomparable formalisms.

Theorem 18.8 (MIN-3: minimal owned-branching typing)**.** *Within* $\mathcal{F}_{\text{free}}$ *of Definition 18.5, and given the transition relation, the least sufficient architecture for* p_{free} *is* F_2*: the partition of the action alphabet into agent-owned and other actions. Neither* F_0 *nor* F_1 *is sufficient, and* F_3*,* F_4 *are not required.*

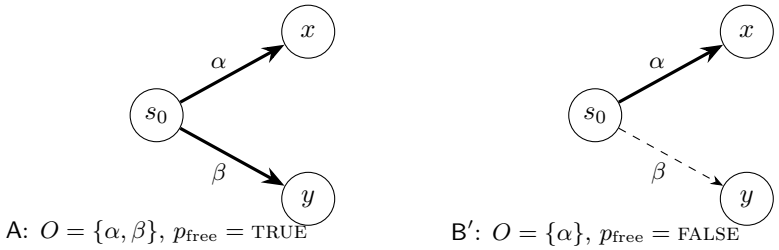


Figure 19: The witness of Proposition 18.4. The two systems have the *same* state set, the *same* reachable set $\{x, y\}$ and the *same* labelled successor relation $\{(\alpha, x), (\beta, y)\}$. They differ only in which transitions the agent owns (thick) and which the environment owns (dashed). No architecture reading only reachability or only the labelled relation can separate them, which is why F_0 and F_1 are insufficient and F_2 is the least level that is not.

Proof. F_0 and F_1 fail by Proposition 18.4: both are constant across A and B', which differ in verdict. F_2 is sufficient: given $\to$ and the partition, $p_{\text{free}}(s)$ is computed directly from Definition 18.1 by inspecting the O-labelled transitions out of s; two configurations agreeing on $\to$ and on O agree on the verdict. F_3 refines F_2, since an admissibility relation $s \mapsto \mathrm{Adm}(s)$ determines the owned actions available at each state and hence, ranging over s, the partition; F_4 refines F_3. Both are therefore sufficient but not least. Minimality is within $\mathcal{F}_{\text{free}}$, per Convention 7.5, and relative to the transition relation being given: the theorem concerns what must be added to $\to$, not what replaces it. □

Remark 18.9 (Credit: this partition is supervisory control's). F_2 is the controllable/uncontrollable event partition of Ramadge and Wonham [44], where a controlled discrete-event process adjoins to its generator a distinguished subset $\Sigma_c \subseteq \Sigma$ of controllable events, and control patterns $\gamma : \Sigma_c \to \{0, 1\}$ enable or disable exactly those. Their control is permissive — disabled events are prevented, enabled events are not forced — which is the same asymmetry Definition 18.1 records by quantifying over O.

The structure is theirs and we claim none of it, nor any of the mature theory built on it (the largest controllable sublanguage included). What Theorem 18.8 adds is a minimality statement about a *self-attribution interface*: given the transition relation, the partition is all that must be held for the agent's own controllability verdict. Ramadge and Wonham do not state this because their supervisor is external to the plant it controls and attributes nothing to itself.

Proposition 18.10 (Type requirement)**.** $\mathrm{TypeReq}(p_{\text{free}})$ *contains a self-referent, an ownership structure, and the action partition; it contains no predicate on component values.*

Proof. By inspection of Definition 18.1: the evaluation quantifies over O and $\to$ and mentions no component value. Systems A and B' carry no viability structure at all and the property is nonetheless evaluated on both. □

Proposition 18.11 (Regional signature of freedom)**.** *Under Section 8,* $\mathrm{Cert}_a(p_{\text{free}}, s^a_{\text{self}})$ *requires: (i)* $s^a_{\text{self}} \in \Theta_a$*; (ii) the action-alphabet carrier* $A \in \mathrm{Sto}(a)$*; (iii) the ownership partition* $\mathrm{Own}_A \subseteq A$ *with* $M_a(\mathrm{Own}_A) = 1$*; (iv) evidence in* $\mathrm{Lnk}(a)$ *witnessing that at least two distinct* Own_A*-actions are step-realisable from the current configuration under Foundations'* `def:step`*; (v) a licence in a* K*-cell.*

Proof. Instance of Definition 8.5; the executable-action clause is Definition 3.13; sufficiency of F_2 is Theorem 18.8. □

19 Separation of the minimal interfaces

We now interpret the three properties and their minimal interfaces over the single carrier of Definition 7.1, which by Remark 7.2 was fixed without reference to them.

Definition 19.1 (The three properties on $\mathcal{C}$)**.** For $c = (h, s, O)$ with $\Sigma = \{c, r\}$, components $K = \{k_0, k_1\}$ valued in $\{\mathrm{ok}, \mathrm{dmg}\}$, $A = \{\alpha, \beta\}$, and structures $\mathrm{Own} = \{k_1\}$, $\mathrm{Viable}(x) = [x = \mathrm{ok}]$:

$$p_{\mathrm{id}}(c) = [\, r \notin h \,], \quad p_{\mathrm{int}}(c) = \mathrm{Viable}\big(s(k_1)\big), \quad p_{\mathrm{free}}(c) = [\, |O| > 1 \,].$$

Definition 19.2 (The three minimal interfaces)**.**

$$u_{\mathrm{id}}(c) = [\, r \notin h \,], \quad u_{\mathrm{int}}(c) = \mathrm{Viable}\big(s \restriction \mathrm{Own}\big), \quad u_{\mathrm{free}}(c) = O.$$

Lemma 19.3 (The integrity interface is not a coordinate projection)**.** *u_{int} is not a projection of $\mathcal{C}$. Reading the second coordinate s in full is sufficient for p_{int} but induces four classes on the component states, whereas u_{int} induces two; and the ownership relation alone, without* Viable*, is insufficient.*

Proof. The full state determines $s(k_1)$ hence the verdict, so it is sufficient; it distinguishes all four assignments to (k_0, k_1). u_{int} distinguishes only $\mathrm{Viable}(s(k_1))$, two classes, and the map sending an assignment to that value witnesses $u_{\mathrm{int}} \prec s$. For the last clause, the constant map returning Own identifies configurations differing in $s(k_1)$, which differ in verdict. □

This is what the neutral carrier buys. u_{int} *applies* two structures — an ownership relation and a predicate — to a coordinate it does not read off, and is strictly coarser than that coordinate. The separation below is therefore between three architectures, not between three projections.

Theorem 19.4 (Pairwise separation of the minimal interfaces)**.** *Over $\mathcal{C}$, each of $u_{\mathrm{id}}, u_{\mathrm{int}}, u_{\mathrm{free}}$ is sufficient for its own property and for neither of the other two. Consequently no one of the three refines another under $\preceq$.*

Proof. Sufficiency in each own case is immediate. For the six cross-cases, write $\mathbf{ok} = (k_0 \mapsto \mathrm{ok}, k_1 \mapsto \mathrm{ok})$ and $\mathbf{dmg} = (k_0 \mapsto \mathrm{ok}, k_1 \mapsto \mathrm{dmg})$, and let $O_1 = \{\alpha\}$, $O_2 = \{\alpha, \beta\}$.

u_{id} against p_{int}: take $(\varepsilon, \mathbf{ok}, O_1)$ and $(\varepsilon, \mathbf{dmg}, O_1)$; the histories agree, the verdicts differ. Against p_{free}: take $(\varepsilon, \mathbf{ok}, O_1)$ and $(\varepsilon, \mathbf{ok}, O_2)$.

u_{int} against p_{id}: take $(cc, \mathbf{ok}, O_1)$ and $(crc, \mathbf{ok}, O_1)$; both are viable, and p_{id} separates them by Proposition 16.2. Against p_{free}: take $(\varepsilon, \mathbf{ok}, O_1)$ and $(\varepsilon, \mathbf{ok}, O_2)$.

u_{free} against p_{id}: take $(cc, \mathbf{ok}, O_1)$ and $(crc, \mathbf{ok}, O_1)$. Against p_{int}: take $(\varepsilon, \mathbf{ok}, O_1)$ and $(\varepsilon, \mathbf{dmg}, O_1)$.

For non-refinement, if $u_{\mathrm{int}} = \varphi \circ u_{\mathrm{id}}$ then u_{id}-equality would force p_{int}-equality, contradicting the first case; the other five are identical in form. □

Remark 19.5 (Status of this theorem, stated plainly). The result is *conditional on the declared carrier.* The carrier is independently motivated (Remark 7.2), so the theorem is not an artefact of the examples; but a different principled carrier could support different separations, and the theorem does not quantify over carriers.

It is also not the paper's main claim. Separation says the three interfaces are different; the results that carry the paper are Theorems 17.7 and 18.8, which say each is *smaller* than the structure its literature assumes. Separation is reported here because it is what makes the property-family question of Section 20 non-trivial: if one interface sufficed for all three there would be nothing to compose.

Table 4: The four properties and what each requires beyond $\mathrm{BaseReq}_{\mathrm{self}}$. The first three occupy different sorts (Corollary 19.6); the fourth shares a sort, and an interface, with the third (Proposition 20.2).

Property	Additional structure	Sort	Summary	Status
p_{id}	none beyond the alphabet	run quotient	2 states	standard
p_{int}	self-membership, viability	state predicate	2 states	Theorem 17.7
p_{free}	action-ownership partition	alphabet subset	—	Theorem 18.8
p_{auth}	ownership, permission set	alphabet subset	—	Section 20

Corollary 19.6 (Distinct sorts)**.** *The three minimal interfaces are respectively a quotient of runs, a predicate obtained by applying two structures to the component state, and a subset of the action alphabet. No two are objects of the same kind.*

Remark 19.7 (The standard principle)*.* Each cross-case instantiates standard indistinguishability reasoning — cited, not claimed (Section 27); what the cases contribute is the identification, per property, of which feature and which carrier the argument runs on.

20 Property families and a joint interface

An agent tracks several self-properties at once. This section asks what a single architecture must hold to type a family of them, and shows that the answer is markedly smaller than the sum of the parts and markedly smaller than the carrier.

20.1 A fourth property, sharing an interface

Definition 20.1 (Authorisation)**.** Let $\mathrm{Perm} \subseteq A$ be the permitted actions. The authorisation property is $p_{\mathrm{auth}}(c) = [\, O \subseteq \mathrm{Perm} \,]$: every action the agent controls is one it is permitted to take.

This property is chosen for a reason and not for a count. It shares the ownership structure with p_{free} and adds one structure, which lets us test reuse rather than merely add an instance.

Proposition 20.2 (Requirement reuse, and its direction)**.** u_{free} *is sufficient for* p_{auth}*, while* $u_{\mathrm{auth}} = [\, O \subseteq \mathrm{Perm} \,]$ *is not sufficient for* p_{free}*.*

Proof. u_{free} returns O itself, which determines $O \subseteq \mathrm{Perm}$. Conversely take $A = \{\alpha, \beta\}$, $\mathrm{Perm} = \{\alpha\}$ and the ownership sets $\{\beta\}$ and $\{\alpha, \beta\}$: both fail the permission test, so u_{auth} identifies them, while p_{free} separates them. □

Remark 20.3 (A degeneracy avoided)*.* The witness needs an ownership family containing $\{\beta\}$. If the available ownership sets are only $\{\alpha\}$ and $\{\alpha, \beta\}$, the permission test happens to separate exactly those two, and u_{auth} *does* determine p_{free} — for a reason that is an accident of the two-element family and not a fact about the properties. We record this because it is the kind of coincidence a small model produces and a reader should not have to detect.

20.2 The joint interface

Definition 20.4 (Sufficient interface for a family)**.** For a finite family P of properties, an architecture is *sufficient for* P when it is sufficient for each $p \in P$. Write $u_P = \langle u_p \rangle_{p \in P}$ for the tupling of the individual minimal interfaces.

Proposition 20.5 (The joint interface is small)**.** *Let $P = \{p_{\mathrm{id}}, p_{\mathrm{int}}, p_{\mathrm{free}}, p_{\mathrm{auth}}\}$. Then u_P is sufficient for every member of P, and over the finite carrier of Definition 19.1 extended to the full subset family of A, it induces* 8 *classes against the carrier's* 240*. On the structure side,*

$$\mathrm{TypeReq}(P) \;=\; \bigcup_{p \in P} \mathrm{TypeReq}(p)$$

has six elements against a disjoint sum of eleven.

Proof. Sufficiency is componentwise: u_P refines each u_p by projection, and each u_p is sufficient for its own property, with p_{auth} covered by u_{free} through Proposition 20.2. The class counts are the product of the component ranges — two continuity classes, two viability classes, and the ownership sets — against the carrier's product with the runs. The structure count is Proposition 22.7 applied to the family: the self-referent and the ownership relation are each required by three members and held once. □

Remark 20.6 (What the compression means)*.* Two reductions are in play and they should not be confused. The first is informational: the agent need not retain the run, only the two bits and the ownership set. The second is representational: the family needs six structures, not eleven, because the members share prerequisites. Only the second is about typability; the first is about summaries. An architecture can achieve the second and still fail the first, and conversely.

20.3 A finite systems instance

We instantiate the account on a small service rather than on the abstract triples, so that the interfaces have a systems reading.

Definition 20.7 (Authenticated storage service)**.** The service has an identity token $\mathrm{tok} \in \{t_0, t_1\}$, a store $\mathrm{st} \in \{\mathrm{clean}, \mathrm{corrupt}\}$, and controls a subset of $\{\mathsf{read}, \mathsf{write}\}$; permitted actions are both. Events are $\{\mathsf{ok}, \mathsf{rot}, \mathsf{fault}\}$, where rot rotates the token. The four properties read: *identity* — the token has never rotated; *integrity* — the store is clean; *controllability* — the service still controls more than one action; *authorisation* — every controlled action is permitted.

The instance has 320 configurations at runs of length at most three, and every claim below is exhaustively checked over all of them.

Proposition 20.8 (The interfaces separate on the instance)**.** *Each minimal interface of Definition 19.2, read on Definition 20.7, is sufficient for its own property; the identity interface is insufficient for integrity, the integrity interface for controllability, and the controllability interface for identity.*

Proof. Sufficiency is by inspection. For the three failures: two runs of the same length with and without rot leave the store equally clean; a clean store is compatible with both ownership sets; and a fixed ownership set is compatible with rotated and unrotated tokens. □

Proposition 20.9 (Representational extension on the instance)**.** *An architecture holding the alphabet, the self-referent, the ownership relation, the action partition and the permission set types identity, controllability and authorisation, and leaves integrity* UNTYPED*. Adding the viability predicate alone types integrity as well, with no change to the run or the store.*

Proof. Immediate from Definition 7.6 and the type requirements in Table 4: only p_{int} requires viability. Adding it changes Str and nothing else, and Proposition 7.7 gives the inclusion. □

Remark 20.10 (Why this instance and not a larger one). The service is the smallest system in which all four properties are simultaneously meaningful and independently movable. No performance claim and no empirical evaluation is offered: the instance exercises the interface theory on a system rather than on a triple.

21 Four cognitive worlds

Interpretation over the certificate calculus (Section 3), the other-model triad (Section 4), and the action-intent primitives (Section 6) partitions an actor's cognitive stance toward another agent into four worlds. The partition is deliberately coarse; its point is to name the failure and success modes that persist under the trace projection defined below.

21.1 The four worlds

Definition 21.1 (Cognitive worlds W_A, W_B, W_C, W_D)**.** Fix an actor a, a receiver b, an action $\alpha \in \mathrm{Own}_a$, a property Φ_b, and the trace u_α produced by $\mathrm{Act}(a, \alpha)$. Let $(\widehat{M}_a(b), \widehat{O}_a(b), \widehat{\mathrm{Mon}}_a^{\,b}(\Phi_b, u_\alpha))$ be a's other-model triad and let $\widehat{\mathrm{Effect}}_a(\alpha, b)$ be a's predicted effect. The cognitive world of a with respect to (b, α, Φ_b) is:

W_A **(first-order non-social):** $\widehat{\mathrm{Effect}}_a(\alpha, b)$ is ABSENT, i.e. no artefact in $\mathrm{Sto}(a)$ encodes the effect of α on b. Equivalently, $\widehat{M}_a(b)$ is ABSENT or lacks a computation producing $\widehat{\mathrm{Effect}}$ for α. a acts on its own belief and goal without representing the receiver-side impact.

W_B **(calibrated help):** $\widehat{M}_a(b)$ is adequate on the relevant property family $P \ni \Phi_b$; $\widehat{O}_a(b)$ agrees with O_b on the relevant sub-alphabet; $\widehat{\mathrm{Effect}}_a(\alpha, b)$ predicts benefit (formalised as: for a supportive property Ψ_b, $\widehat{\mathrm{Mon}}_a^{\,b}(\Psi_b, u_\alpha) = \mathrm{TRUE}$); and $\widehat{\mathrm{Mon}}_a^{\,b}(\Phi_b, u_\alpha) = \mathrm{Mon}_b^{O_b}(\Phi_b, u_\alpha)$ (model-verdict agreement, Definition 6.7).

W_C **(monitor-blind assistance):** $\widehat{\mathrm{Effect}}_a(\alpha, b)$ predicts benefit *and* $\widehat{\mathrm{Mon}}_a^{\,b}(\Phi_b, u_\alpha)$ is inadequate — either $\widehat{O}_a(b)$ disagrees with O_b on the relevant sub-alphabet, or the actual receiver-side monitor is not represented by a at all (i.e. $\widehat{\mathrm{Mon}}_a^{\,b}$ is ABSENT). Two sub-cases:

- W_{C_1} (*blind by observability*): b genuinely cannot observe u_α's relevant sub-alphabet under O_b; $\widehat{O}_a(b)$ correctly records this and no $\widehat{\mathrm{Mon}}_a^{\,b}$ is meaningful to form. The blindness is structural on b's side.
- W_{C_2} (*blind by model*): b can observe u_α's relevant sub-alphabet, but $\widehat{O}_a(b)$ incorrectly excludes it. The blindness is structural on a's side.

W_D **(monitored safety failure):** All of $\widehat{M}_a(b)$, $\widehat{O}_a(b)$, $\widehat{\mathrm{Mon}}_a^{\,b}$ present; $\widehat{\mathrm{Mon}}_a^{\,b}(\Phi_b, u_\alpha) = \mathrm{TRUE}$ (non-failure predicted); and $\mathrm{Mon}_b^{O_b}(\Phi_b, u_\alpha) = \mathrm{FALSE}$ (actual failure). This is the model-verdict mismatch case. Two sub-cases distinguished by Intent_a:

- $W_{D\text{-}F_{\mathrm{INT}}}$: $\mathrm{Intent}_a(\alpha, g)$ with g coercive / persecutory / punitive with respect to Φ_b; the actor intentionally selects an action classified by Mon_b as failure.
- $W_{D\text{-}F_{\mathrm{MODEL}}}$: $\mathrm{Intent}_a(\alpha, g)$ with g supportive with respect to Φ_b; the actor selects the action under a miscalibrated other-model triad.

Remark 21.2 (W_{C_1} vs W_{C_2} is not decorative). The two sub-cases of W_C produce identical actor-side surface state (the actor does not represent a receiver-side monitor verdict) but have different downstream consequences. In W_{C_2}, receiving feedback that b did observe u_α should

trigger an $\widehat{O}_a(b)$ update; in W_{C_1}, no such feedback event is possible. Collapsing the two would misclassify a class of feedback-driven repairs.

Remark 21.3 ($W_{D\text{-}F_{\text{INT}}}$ vs $W_{D\text{-}F_{\text{MODEL}}}$). Model-verdict mismatch does *not* entail malicious intent. The same observable action producing the same receiver-side monitor failure can arise from a supportive intent under a miscalibrated other-model (F_{MODEL}) or from a coercive intent (F_{INT}). Distinguishing them requires reading $\text{Intent}_a(\alpha, g)$, which is internal to a and not observable to b. The paper takes no position on how b should distinguish them; it observes that the distinction exists and is architecturally different.

21.2 The trace projection and its non-injectivity

Definition 21.4 (Trace projection)**.** The *cognitive-to-trace projection* is the function

$$\pi \;:\; (\text{v2 cognitive state of } a \text{ at } t) \;\longrightarrow\; u_\alpha \in \Sigma^*$$

that sends the tuple

$$(\text{Bel}_a,\ \text{Cert}_a,\ \text{MetaCert}_a,\ \widehat{M}_a(b),\ \widehat{O}_a(b),\ \widehat{\text{Effect}}_a,\ \widehat{\text{Mon}}_a^{\,b},\ \text{Intent}_a(\alpha, g),\ \text{Act}(a, \alpha))$$

to the observable ambient trace u_α produced by $\text{Act}(a, \alpha)$.

Theorem 21.5 (Cognitive non-injectivity)**.** *The trace projection π is not injective. Explicitly, for each of the following pairs of cognitive worlds one can construct actor states that π collapses to the same observable trace u_α: $W_A \sim W_B$, $W_B \sim W_{C_2}$, $W_B \sim W_{D\text{-}F_{\text{MODEL}}}$, $W_{D\text{-}F_{\text{INT}}} \sim W_{D\text{-}F_{\text{MODEL}}}$.*

Proof. Six finite witnesses collected in Appendix A, R6. Sketch of the $W_B \sim W_{D\text{-}F_{\text{MODEL}}}$ collapse: fix α, u_α. In world W_B the actor holds a fully calibrated triad and $\widehat{\text{Mon}}_a^{\,b}(\Phi_b, u_\alpha) = \text{Mon}_b^{O_b}(\Phi_b, u_\alpha) = \textsc{true}$ (non-failure predicted and returned). In world $W_{D\text{-}F_{\text{MODEL}}}$ the actor holds a triad with $\widehat{O}_a(b)$ miscalibrated on a sub-alphabet that changes the predicted monitor value; $\widehat{\text{Mon}}_a^{\,b}(\Phi_b, u_\alpha) = \textsc{true}$ (actor predicts non-failure) while $\text{Mon}_b^{O_b}(\Phi_b, u_\alpha) = \textsc{false}$ (actual failure). Both actors emit the same u_α because Act is a function of Intent and α, and both actors selected α on the same supportive-intent basis; their cognitive worlds differ but π collapses them. The other three collapses are constructed analogously; see appendix. □

Corollary 21.6 (Trace-level observation does not identify cognition)**.** *Given only the observable trace u_α and its receiver-side monitor verdict $\text{Mon}_b^{O_b}(\Phi_b, u_\alpha)$, an external observer cannot in general determine which cognitive world $W_A, W_B, W_{C_1}, W_{C_2}, W_{D\text{-}F_{\text{INT}}}, W_{D\text{-}F_{\text{MODEL}}}$ the actor was in when producing u_α.*

Proof. Immediate from Theorem 21.5: multiple cognitive worlds are consistent with any given u_α. □

Remark 21.7 (Bridge to trace-level analyses of transactional interaction). Corollary 21.6 is the paper's canonical bridge to downstream trace-level analyses. Analyses that project cognitive interaction onto ambient traces — including Karpman / drama-triangle projections and their LTL formalisations — necessarily coarsen the v2 cognitive state, and no such projection recovers the missing $\widehat{O}_a(b)$, $\widehat{\text{Mon}}_a^{\,b}$, or Intent_a information. The theorem does not say those projections are wrong; it says they are strictly below the cognitive-algebra layer in what they identify.

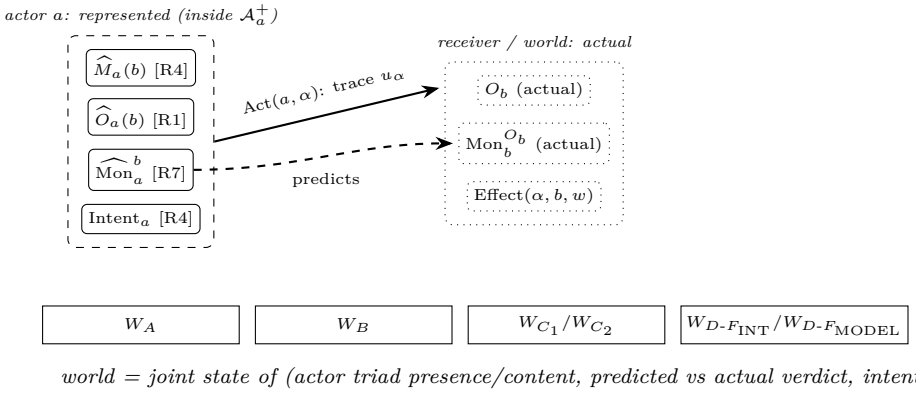


Figure 20: Figure B/C: actor-internal representation vs. receiver-side actuality, with the four-world strip. Everything in the dashed actor group lives inside $\mathcal{A}_a^+$ with the region tags shown; O_b, $\text{Mon}_b^{O_b}$ and the actual effect (dotted borders) are external and never collapse into the actor's represented triad. The dashed edge is the prediction relation whose agreement or failure, jointly with triad presence and intent, determines the world.

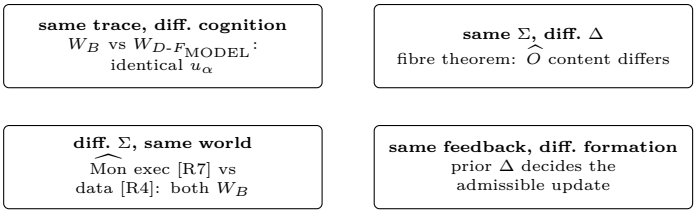


Figure 21: Figure D: the four non-injectivity witness families as small multiples — same trace / different cognition (Theorem 21.5), same Σ / different Δ (Theorem 9.6), different Σ / same world (Theorem 10.12), same feedback / different formation (Theorem 13.11).

22 How requirements transform

With the worked properties and the cognitive worlds in place, we return to the requirement calculus itself: how the three requirements move when properties are built from other properties.

22.1 Temporal lifting

Definition 22.1 (Lift)**.** Let b be a base predicate on configurations and op a temporal operator. $\text{Lift}_{\text{op}}(b)$ is the property obtained by applying op to b along the history.

Theorem 22.2 (LIFT-1: type requirement is invariant)**.** *Suppose* $\text{Lift}_{\text{op}}(b)$ *introduces no new atomic predicate, no new referent, no new grounding structure and no new domain object beyond those of* b*. Then*

$$\text{TypeReq}(\text{Lift}_{\text{op}}(b)) \;=\; \text{TypeReq}(b).$$

Proof. TypeReq collects the structures over which the evaluation quantifies (Definition 5.2). Evaluating $\text{Lift}_{\text{op}}(b)$ at a history proceeds by evaluating b at positions of that history and combining the results by op. The combination step quantifies over positions, which are data of the history and not structures of the architecture. By hypothesis op contributes no atom, referent, grounding or object of its own. Hence the set of structures quantified over is exactly that of b, giving containment in both directions. □

The hypothesis is doing real work and is not a formality: a lift that quantified over a new sort — "every *component* is eventually viable" — would introduce a quantifier over P and the conclusion would fail.

Proposition 22.3 (The temporal class does change)**.** *With b the current-viability predicate over $\Sigma_{\text{int}} = \{o, d\}$: $G(b) \in \mathsf{S}$, since any occurrence of d is a finite refuting prefix; and $F(b) \in \mathsf{C}$, since any occurrence of o is a finite confirming prefix.*

Proof. For $G(b)$: if a behaviour violates it, some position carries d, and the prefix through that position has no extension satisfying $G(b)$. For $F(b)$: if a behaviour satisfies it, some position carries o, and every extension of the prefix through that position satisfies $F(b)$. □

Proposition 22.4 (INFO-LIFT: the sufficient summary is not invariant)**.** *There are a base predicate b and a lift satisfying the hypotheses of Theorem 22.2 for which* InfoReq *strictly increases. With $\Sigma = \{o, d\}$ and $b(h) = [\,h \text{ ends in } o\,]$:*

$$|\text{InfoReq}(b)| = 2, \qquad |\text{InfoReq}(G\,b)| = 2, \qquad |\text{InfoReq}(G(d \to Xo))| = 3.$$

Proof. b has index 2: the classes are "ends in o" and "ends in d or is empty". $G\,b$ = "every letter is o" also has index 2, the classes being "no d yet" and "some d", the latter absorbing. For $G(d \to Xo)$, the right-congruence classes are: *violated* (some d was followed by a non-o), absorbing; *pending* (the last letter is d, so the next letter is constrained); and *clear* (the last letter is o or the history is empty). These are pairwise separated — d then d distinguishes pending from clear, and no suffix leaves violated — giving index $3 > 2$. All three properties are built from the same atoms o, d, so the hypotheses of Theorem 22.2 hold and TypeReq is unchanged throughout. □

Corollary 22.5 (The requirements move separately)**.** *Under temporal lifting,* TypeReq *may remain fixed while* Temp *changes (Proposition 22.3) and while* InfoReq *changes (Proposition 22.4). No one of the three determines another.*

Remark 22.6 (An earlier over-generalisation, corrected)*.* The invariance of InfoReq under G in Proposition 22.4 is an artefact of that particular lift, which admits an absorbing violation state. It is tempting to read the two-state coincidence as a general factorisation of the profile into an invariant informational part and a varying temporal part; the bounded-response lift shows this is false. We state the factorisation only for TypeReq.

22.2 Composition

Proposition 22.7 (COMP-T: type requirements union modulo sharing)**.**

$$\text{TypeReq}(p \wedge q) \;=\; \text{TypeReq}(p) \cup \text{TypeReq}(q),$$

which is strictly smaller than the disjoint sum whenever $\text{TypeReq}(p) \cap \text{TypeReq}(q) \neq \emptyset$.

Proof. Evaluating $p \wedge q$ evaluates each conjunct and takes the conjunction of the results; the conjunction step quantifies over nothing. So the structures quantified over are those of p together with those of q, as a set. A structure required by both is held once. □

Example 22.8 (Sharing is not marginal)**.** With

$$\text{TypeReq}(p_{\text{int}}) = \{\text{self-referent}, \text{ownership}, \text{viability}\},$$

$$\text{TypeReq}(p_{\text{free}}) = \{\text{self-referent}, \text{ownership}, \text{action partition}\},$$

the union has four elements while the disjoint sum has six. Two of the six are shared.

Proposition 22.9 (COMP-I: the summary may collapse or grow)**.** *Composition of sufficient summaries under conjunction is not determined by the summaries alone.*

1. *With $p =$ "no r ever" and $q =$ "no d ever", both of index 2, the conjunction has index 2.*

2. *With $p =$ "no r ever" and $q = F(o)$, both of index 2, the conjunction has index 3.*

Proof. (1) Both are safety properties whose monitors have a single absorbing violated state, and the violated states are identified in the product: once either conjunct is violated the conjunction is violated and no suffix recovers it. The reachable product therefore has the classes *alive* and *violated*.

(2) Here the monitors do not share an absorbing state: violation of p is absorbing, but satisfaction of q is absorbing on the other side. The reachable classes are *violated* (an r occurred), *alive with o seen*, and *alive with no o yet*, and these are pairwise separated by suffixes. The index is 3. □

Remark 22.10 (Terminology). We avoid calling case (2) "super-additive": it is the ordinary behaviour of minimised product automata, which collapse or not according to shared absorbing structure. It is recorded as an architectural consequence — requirement reuse across composed properties is real but not uniform — not as automata theory.

23 Criticality

Version 1 explicitly refused a formal notion of "critical property", on the ground that an early framing tested for criticality-by-capability-loss and only p_{int} qualified. Version 2 supplies a different criterion, appropriate to the certificate calculus of Section 3: a property is critical when the failure of its certificate invalidates the well-formedness of at least one downstream certificate over the same self-referent. The three worked properties of *Sections* 16*to* 18 are critical in this sense.

23.1 The upstream-dependency criterion

Definition 23.1 (Upstream-dependency criticality)**.** Fix a bounded MCK agent a with self-symbol $s^a_{\text{self}} \in \Theta_a$. For self-properties p, q, write $p \to_a q$ when the well-formedness of $\text{Cert}_a(q, s^a_{\text{self}})$ requires that $\text{Cert}_a(p, s^a_{\text{self}})$ be well-formed and hold verdict TRUEover an overlapping scope; formally, $\text{Cert}_a(q, s^a_{\text{self}})$'s clause (2) of Definition 2.6 (referent-typing) fails or its verdict flips to UNTYPED when $\text{Cert}_a(p, s^a_{\text{self}})$'s verdict is not TRUE.

A property p is *critical for a* at time t iff there exists some other *conjunction-free* self-property $q \neq p$ with $p \to_a q$ at t. Its *criticality index* is $|\{q : p \to_a q\}|$ over conjunction-free q. The restriction is not cosmetic: admitting conjunctions would make every certified p critical via $q = p \wedge p'$, inflating every index and voiding the criterion.

Convention (identity-scope collapse). When the identity scope underlying a de se certificate is broken — the $\text{Lnk}(a)$-chain witnessing $\text{denotes}_a(s^a_{\text{self}}) = a$ over the scope does not exist — the certificate's evidence-composition clause fails *constitutively*, not evidentially: there is no single-agent verdict function left for evidence to feed. We stipulate that this failure mode returns UNTYPED, distinguishing it from an ordinary insufficient-evidence UNDETERMINED within an intact scope.

Remark 23.2 (Companion criterion: attention-w-criticality). An alternative, complementary criterion is available via Definition 3.9. A property p is *w-critical for a* at t when $\text{TypeReq}(p) \subseteq \text{Str}(\mathcal{A}_a)$ (typable) but there exists an attention perturbation π within a bounded budget such that $\text{TypeReq}(p) \not\subseteq \text{Str}_{\text{eff}}(a, t)$ under π (not effectively typable). The

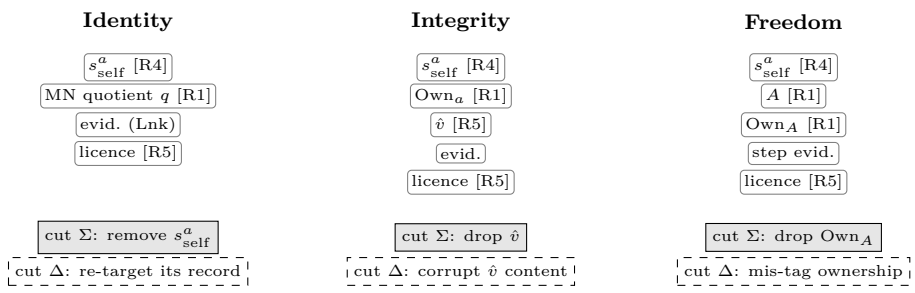


Figure 22: Figure T: the three worked properties' regional signatures with both cut types. Each column lists the required carriers with their region tags; the filled box beneath is the structural cut (a carrier removal breaking Req^{Σ}), the dashed box the calibration cut (a content corruption breaking Req^{Δ} with all carriers intact). Missing carrier and wrong content are visually distinct failure classes.

two criteria are distinct: upstream-dependency asks about certificate structure, w-criticality about attentional vulnerability. The paper treats upstream-dependency as the primary criterion, and w-criticality as a companion notion.

23.2 Identity is upstream

Theorem 23.3 (Identity-upstream)**.** $p_{\text{id}} \to_a p_{\text{int}}$ *and* $p_{\text{id}} \to_a p_{\text{free}}$ *over any scope in which the self-referent* $\text{denotes}_a(s^a_{\text{self}}) = a$ *must remain stable.*

Proof. Consider $\text{Cert}_a(p_{\text{int}}, s^a_{\text{self}}, [t_0, t_1])$: its referent-typing clause requires that s^a_{self} denote the same a throughout $[t_0, t_1]$, since p_{int}'s Own-restriction of the state (Definition 17.1) is agent-relative. If $\text{Cert}_a(p_{\text{id}}, s^a_{\text{self}}, [t_0, t_1])$ has verdict FALSE or UNTYPED, the derivation-record entries in $\text{Lnk}(a)$ witnessing the ownership relation across $[t_0, t_1]$ cannot be composed into a single-agent viability verdict, and $\text{Cert}_a(p_{\text{int}}, s^a_{\text{self}}, [t_0, t_1])$'s clause (3) fails constitutively. By the identity-scope-collapse convention of Definition 23.1, the verdict is UNTYPED. The same argument transfers to p_{free}: Own_A is agent-relative and a broken identity scope makes the ownership partition unassignable. □

Corollary 23.4 (Identity, integrity, freedom are all critical)**.** *Under* $s^a_{\text{self}} \in \Theta_a$ *and the certificate calculus of Definition 2.6:* p_{id} *is critical with criticality index* ≥ 2 *(*$\to_a p_{\text{int}}, p_{\text{free}}$*);* p_{int} *and* p_{free} *are critical over combined-attribution scopes, since a de se certificate* $\text{Cert}_a(p_{\text{int}} \wedge p_{\text{free}}, s^a_{\text{self}})$ *depends on both.*

Proof. Immediate from Theorem 23.3 and Definition 23.1 for p_{id}. For p_{int} and p_{free}: each is upstream of conjunction-free derived self-properties whose certificates cite its verdict as a premise in their evidence composition (e.g. a scoped operational-readiness property whose evidence set includes the integrity certificate); failure of the cited certificate collapses the citing certificate's clause-(3) composition. Their criticality indices are smaller than identity's, consistent with the blast-radius asymmetry of Theorem 12.16. □

Remark 23.5 (This is not a moral criticality claim)*.* Definition 23.1 is a structural criterion over certificate well-formedness. It does not claim moral or normative importance for identity, integrity, or freedom. It says: within the certificate calculus this paper defines, these three properties occupy structural positions such that their certificate-failure invalidates downstream certificates. That fact is what the paper offers under the word "critical".

Remark 23.6 (Relation to v1's rejected criticality tests)*.* Version 1 tested criticality-by-capability-loss (a property is critical if its violation strictly reduces capabilities) and reported

that only p_{int} qualified, on the basis that repairing a replacement event changed no capability. That test is a different criterion from Definition 23.1: it evaluates capability consequences of a state change, whereas the upstream-dependency criterion evaluates certificate structure. The two are not comparable, and the v1 test's negative result on p_{id} and p_{free} does not carry over. Both criteria are legitimate; v2 selects the certificate-structural one because it composes naturally with MetaCert and with the four-worlds algebra.

24 Representational extension and partial self-representation

24.1 Three ways a verdict can change

Definition 24.1 (Update modes)**.** Fix an agent and a property. A *world update* changes the configuration. An *evidence update* leaves the configuration and the architecture fixed and changes what the agent has observed. A *representational update* leaves the configuration and the observations fixed and changes $\text{Str}(\mathcal{A})$.

The first two are the familiar ones. The third is available only because TypeReq-satisfaction is a property of the architecture and not of the world or the history, and it is the mode the next proposition isolates.

Proposition 24.2 (REP-EXT)**.** *There are architectures $\mathcal{A} \prec \mathcal{B}$ differing by a single structure, and a configuration c, such that*

$$\text{Verdict}_{\mathcal{A}}(p_{\text{int}}, c) = \textit{UNTYPED} \qquad \textit{and} \qquad \text{Verdict}_{\mathcal{B}}(p_{\text{int}}, c) \in \{\textit{TRUE}, \textit{FALSE}\},$$

with the configuration, the world truth of p_{int}, and the agent's observations all unchanged.

Proof. Let $\mathcal{S} = \{\text{event alphabet}, \text{self-referent}, \text{ownership}, \text{viability}, \text{action partition}\}$ and take $\text{Str}(\mathcal{A}) = \mathcal{S} \setminus \{\text{viability}\}$, $\text{Str}(\mathcal{B}) = \mathcal{S}$. Then $\mathcal{A} \prec \mathcal{B}$ by Definition 7.3.

By Proposition 18.10 and Definition 17.1,

$$\text{TypeReq}(p_{\text{int}}) = \{\text{self-referent}, \text{ownership}, \text{viability}\}.$$

This is contained in $\text{Str}(\mathcal{B})$ but not in $\text{Str}(\mathcal{A})$, which lacks viability. So by Definition 5.2, $\mathcal{A}$ returns UNTYPED at every configuration, while $\mathcal{B}$ types the property and returns its truth value, which at $c = (\varepsilon, \text{ok}, O_1)$ is TRUE. Neither c nor the history was touched. □

Proposition 24.3 (Partial self-representation gives a strict hierarchy)**.** *The architecture $\mathcal{A}$ of Proposition 24.2 types p_{free} and leaves p_{int} untyped:* $\text{SelfCap}(\mathcal{A}) \ni p_{\text{free}}$ *and* $\text{SelfCap}(\mathcal{A}) \not\ni p_{\text{int}}$.

Proof. By Proposition 18.10,

$$\text{TypeReq}(p_{\text{free}}) = \{\text{self-referent}, \text{ownership}, \text{action partition}\} \subseteq \text{Str}(\mathcal{A}),$$

while $\text{TypeReq}(p_{\text{int}})$ requires viability, which $\mathcal{A}$ lacks. □

Remark 24.4 (Why this is the paper's use of partiality)*.* Proposition 24.3 is what a genuinely partial self-representation buys analytically: an agent may represent the ownership of its actions and not the condition of its components, and the consequence is not that it is wrong about its integrity but that it has no integrity verdict to be right or wrong about, while its controllability verdict is perfectly well defined. The two properties are not ordered by difficulty, evidence or temporal class; they are separated by which structures happen to be present.

This is the point at which the account departs from architectures that guarantee self-representation. Section 25 makes that comparison precise.

Remark 24.5 (Relation to the evidence axis). UNTYPED and UNDETERMINED are not degrees of the same thing. A representational update can move a property out of UNTYPED and into UNDETERMINED — typed but undecided — with no evidence whatever; an evidence update can move it from UNDETERMINED to a truth value but can never move it out of UNTYPED, because there is no verdict function for evidence to feed. Remark 5.7 is the formal content of that asymmetry.

25 Reflection is neither necessary nor sufficient

An architecture that represents its own computation is the obvious candidate for one that can attribute properties to itself. We show the two capabilities come apart in both directions, under the models of this paper.

25.1 What computational reflection is

Smith introduces a dialect whose embedded theory of itself — the *reflective model* — is, in his words, "causally connected to the workings of the underlying calculus in crucial and primitive ways", and states the design goal that "in order to build a general reflective calculus, one must embed within it a fully articulated theory of one's understanding of it" [48]. Maes gives the notion its general form: a reflective system "incorporates structures representing (aspects of) itself", called its self-representation, and because that representation is causally connected to what it represents, "the system always has an accurate representation of itself" [37].

Two features of these architectures matter here. First, the self-representation is guaranteed by construction: within the reflected aspect there is no state in which a running program meets one of its own structures for which no designator exists. Second, the guarantee is stronger than anything the present account assumes — causal connection means changes propagate in both directions, which we nowhere require.

Remark 25.1 (Scope of the comparison). Smith is explicit that his dialect is procedurally and not fully reflective, so these systems are not claimed to reflect everything. But the limitation is a scope decision fixed at language-design time and uniform over all programs: the procedural aspect is reflected and the declarative aspect is not. It is not the agent-relative, structure-by-structure partiality of Proposition 24.3, where one architecture holds ownership and lacks viability.

25.2 The two separations

Proposition 25.2 (Reflection does not imply attribution)**.** *There is an architecture with a complete, causally connected self-representation of its own computation that does not type* p_{int}*.*

Proof. Take any architecture whose structure set contains a full reflective model of its own procedures — designators for its interpreter state, its continuation and its environment — but no viability predicate on component values. By Definition 17.1, TypeReq(p_{int}) contains viability; by Definition 5.2 the property is untyped and the architecture returns UNTYPED at every configuration. Nothing in the reflective model supplies the missing predicate: a designator for a component is not a criterion separating admissible from inadmissible values of it. □

Proposition 25.3 (Attribution does not imply reflection)**.** *There is an architecture with no reflective capability that types and decides* p_{int}*.*

Proof. Take I_3 of Definition 17.2: a self-membership relation and a viability predicate, and nothing else. It has no meta-level, no reification of its own procedures, no designators for

its own computation and no causal connection between a model and what it models. By Lemma 17.6 it is sufficient for p_{int}, so it returns the truth value at every configuration. □

Corollary 25.4. *Under the models of this paper, computational reflection and self-attribution capability are independent: neither implies the other.*

Remark 25.5 (The claim is local). Corollary 25.4 is stated for the properties and architectures defined here, and it should not be read as a claim about every reflection formalism. A formalism whose reflective model was required to include a viability criterion would make Proposition 25.2 fail by construction. The interest of the pair is that reflection, as the cited sources define it, is about access to one's own structure, whereas attribution additionally needs a criterion applied to what is accessed — and access does not supply criteria.

Remark 25.6 (Relation to MetaCert). Version 2's meta-certificate (Definition 3.5) formalises a strictly weaker structural condition than reflection: $\text{MetaCert}_a(\varphi, r)$ requires only a self-symbol $s^a_{\text{self}} \in \Theta_a$ and a symbol denoting the first-order certificate $z_{\varphi,r}$, plus a certificate about those. It does not require the causally-connected reflective model of a's own computation that Smith and Maes provide. Corollary 25.4 carries over: computational reflection and MetaCert are independent, in both directions, because a designator for an interpreter component is not a symbol for a certificate licence, and vice versa. This is the sense in which v2's certificate calculus lies below the reflection layer: it describes what an agent must hold to certify things about itself, before questions of self-computation-reflection arise.

26 Relation to two companion frameworks

The account above is self-contained. This section records its relation to two published frameworks, and states in each case what is imported and what is not claimed.

26.1 An observer-relative artefact classification

The partiality used in Section 24 has a published formal home. An observer-relative classification of artefacts by three binary predicates — structured meaning, residual opacity, functionality — partitions a classified domain into seven regions, relative to each agent, and confines denotation to the overlap of the two semantic predicates [12]. Two features are the ones this paper uses. First, the classification is *agent-relative*: the same artefact may fall in different cells for different agents. Second, it is *partial*: an artefact outside the classified domain receives no region at all, so the classification of a given artefact may simply be undefined for a given agent.

That is the substrate for Definition 5.2: an architecture's structure set is what it holds, and an attribution whose prerequisites lie outside it is UNTYPED rather than false. A companion develops the denotation structure further [16].

Remark 26.1 (What is not claimed). No result of this paper is a result about that classification, and the seven-region geometry appears in no proof above. We import a stance — self-representation may be partial, and partiality is agent-relative — not machinery.

26.2 A typed action interface

A companion framework types agent actions by their read/write footprints on a partitioned global state, evaluating temporal properties by prefix verdicts [14]. Its five canonical action classes are footprint equivalence classes, and its internal agent state is a pair consisting of an observed prefix and an explicitly opaque parameter, with knowledge and capabilities placed in extension layers.

Proposition 26.2 (Footprint abstraction forgets attribution capability)**.** *Two architectures may realise the same action footprint while differing in* SelfCap*.*

Proof. Take $\mathcal{A}$ and $\mathcal{B}$ of Proposition 24.2. An internal computation performed by either reads and writes only the agent's own state component, so both induce the footprint $(\{X\}, \{X\})$ of an internal transition. By Proposition 24.3 and Proposition 24.2, $p_{\text{int}} \in \text{SelfCap}(\mathcal{B}) \setminus \text{SelfCap}(\mathcal{A})$. □

Remark 26.3 (The relation is refinement, not dependency)*.* The exact statement is: *the present account refines internal representational structures that the footprint abstraction intentionally places outside its scope.* There is no formal dependency in either direction — Proposition 26.2 uses only its footprint typing, which that framework states as a design choice — and we do not describe either paper as downstream of the other.

Remark 26.4 (An extension that is not taken)*.* The observation primitive of that framework is unaddressed: it reads a state component, not a named subject. A subject-addressed variant would be the first construct at which the typing discipline developed here could bind an action — a read about a denoted subject would be well-typed only if the subject is referable. We record this as a possible refinement and do not develop it. In particular, reading a raw sensor datum plainly requires no denotation, so referentiality must not be imposed on observation generally.

26.3 Observation-relative monitorability

Version 2 imports the observation-relative machinery of [13] as the actual receiver monitor $\text{Mon}_b^{O_b}(\Phi_b, u)$ of Definition 6.6. That paper proves, for bounded temporal implication $G(p \to F_{\leq k}\, q)$, that monitorability is a property of the pair (formula, observation model) and not of the formula alone: a change of observation model alone changes the verdicts reachable for the same formula. *Imported:* this observation-relative stance, which Definition 6.6 generalises by definition to an arbitrary receiver property Φ_b and regime O_b; the generalisation is ours, not a consequence of that theorem. *Not claimed:* any new monitorability result. *Consumed by:* Definition 6.6, Proposition 6.8, and Definition 21.1's W_D clause.

26.4 Downstream programme contract

The cognitive algebra of Sections 3, 4, 6 and 21 is designed to be the upstream substrate for two downstream programme lines. The transactional-role line (LTL Triangles / Karpman / Winner-Triangle reconstruction) consumes the trace projection π of Definition 21.4 and works below it, at the level of observable action-traces classified by role. The Drama-to-Empowerment line consumes the four-world classification and formalises constructive and pathological transitions between them, particularly the $W_{D\text{-}F_{\text{MODEL}}} \to W_B$ recovery move.

The paper does not develop either downstream line; it fixes the interface. The boundary contract is Theorem 21.5 and remark 21.7: any projection to observable traces coarsens the cognitive-algebra layer and recovers none of the missing $\widehat{O}_a(b)/\widehat{\text{Mon}}_a^{\,b}/\text{Intent}_a$ information.

26.5 The temporal axis

Temp is imported unchanged from a four-family classification of ω-properties by finite-prefix certifiability [10], itself in the Alpern–Schneider and runtime-verification lineage [1, 4]. We use its definitions and none of its results beyond the class memberships stated in Proposition 22.3.

Remark 26.5 (State predicate vs temporal requirement)*.* For the discipline of this paper it is important that an agent state predicate p and its temporal requirement Φ_p are different formal

objects. The state predicate p carries only a TypeReq/InfoReq/SelfCap profile at each time. A temporal requirement Φ_p lifting p additionally carries an indicator vector

$$\chi(\Phi_p) = (\mathbf{1}[\Phi_p \in \mathsf{S}],\ \mathbf{1}[\Phi_p \in \mathsf{C}],\ \mathbf{1}[\Phi_p \in \mathsf{L}],\ \mathbf{1}[\Phi_p \in \mathsf{K}])$$

in the sense of [10], so a single predicate can populate several distinct cells depending on the lift chosen. Using the current-viability predicate b of Proposition 22.3 as base: b itself has no χ; $G\,b \in \mathsf{S} \cap \mathsf{K}$ (no finite prefix ever certifies it forever, whence the K component); $F\,b \in \mathsf{C} \cap \mathsf{L}$ (no finite prefix ever permanently refutes it, whence the L component); the recurrence lift $GF\,b$ and the persistence lift $FG\,b$ both lie in $\mathsf{L} \cap \mathsf{K}$ and differ only by the persistence (Σ_2^0) / recurrence (Π_2^0) bipartition. For these particular lifts, the three-valued monitor of [4] decides $G\,b$ (only the refuting verdict is reachable) and $F\,b$ (only the confirming verdict is reachable), while on $GF\,b$ and $FG\,b$ it is inconclusive on every prefix, and only weak verdicts carry information (cf. [10, Ex. 10.12, Ex. 10.16]). These are facts about the individual formulas, not about their class cells, and are not reopened here. The remark is a systematisation, not a new theorem; internally an agent may keep compact monitor state, a belief state, or a progress summary adequate for Φ_p rather than raw history, in the same sense as InfoReq(Φ_p) replaces raw traces with the right-congruence quotient in Proposition 22.4.

Remark 26.6 (Attribution-support stabilisation via refinement confirmation). For a lifted temporal requirement of the form $\Phi = FG\,C$ — of which the persistence lift $FG\,b$ of Proposition 22.3 is one instance, and the persistence lift of a capability-availability predicate $FG\,\mathrm{Cap}_a$ another — an orthogonal certification layer is available when the agent's attribution-support mechanism can be organised as a refinement system $\mathcal{R} = (D, A_0, U, E)$ in the sense of [11], where D is a sequence of admissible support configurations, U is a computable refinement step, E is a Lyapunov error metric with strict-decrease constant $c > 0$ relative to an admitted trace class $\mathcal{T}$, and the predicate $C(A) \Leftrightarrow E(A) = 0$ is absorbing along the orbit. Under those hypotheses, the finite-confirmation theorem (Theorem 5.5) of [11] yields $k^* \le \lceil E(A_0)/c \rceil$ with $E(A_{k^*}) = 0$, so a witnessing prefix of bounded length is produced, and the bridge proposition (Proposition 5.6(ii)) of the same source upgrades this to $\tau \models FG\,C$ under absorption. The certification is strictly relative to $\mathcal{T}$: the guarantee is that attribution support has entered an absorbing region in which the selected property remains attributable along traces admitted by $\mathcal{T}$, not that the property is objectively true forever. Two boundaries are important. First, the recurrence form $GF\,C$ is *not* delivered by finite descent ([11] Proposition 5.6(iii)): distinct temporal requirements sharing indicator vector $\chi(\Phi) = (0, 0, 1, 1)$ (the recurrence $GF\,p$ and the persistence $FG\,p$) do not share refinement-confirmability, so temporal-class membership and refinement-confirmability are orthogonal descriptors. Second, generic empirical convergence of beliefs, statistical estimates, or learned parameters is *not* an instance of this scheme unless it exhibits the exact strict-decrease structure with a fixed progress quantum $c > 0$; the framework's Lyapunov condition is stronger than the classical continuous-analysis one, not weaker ([11] §5, the orbit-gap proposition, Proposition 5.2, and its uniform-quantum remark). The remark is a derived application; no new attribution theorem is introduced here.

27 Related work

We organise by scientific boundary rather than chronology. For each we state what is imported, what is not claimed, and what this paper adds.

Monitorability and diagnosability. A discrete-event system is diagnosable when occurrences of distinguished unobservable events can be detected with finite delay; the diagnoser is a machine whose states carry failure information and which both performs the online detection and certifies the property offline [47]. *Imported:* the joint treatment of finite certification

and sufficient summary — a diagnoser is precisely an incrementally maintainable state that suffices for a verdict, which is Definition 5.3. *Not claimed:* any novelty for the pairing of Temp and InfoReq; that pairing is theirs, and the runtime-verification prefix-verdict machinery is likewise standard [4], instantiated for this programme in [10]. *Added:* the third requirement. In that setting the system model has the failure events in its alphabet by construction, so the diagnosis question is always well posed; there is no counterpart to an attribution that cannot be formed.

Supervisory control. A controlled discrete-event process distinguishes controllable from uncontrollable events and acts by enabling or disabling the former, permissively [44]. *Imported:* the ownership partition, in full — F_2 is Σ_c. *Not claimed:* the partition, the theory of controllable languages, or any synthesis result. *Added:* Theorem 18.8, a minimality statement for a self-attribution interface: given the transition relation, the partition suffices and nothing further is required. Supervisory control does not state this because its supervisor is external to the plant and attributes nothing to itself.

Unawareness and partial languages. An agent may fail to entertain a proposition rather than fail to know it; the foundational literature supplies structures for this and the introspection constraints they must respect [23, 40, 31, 21], and the object-based treatment axiomatises the distinction between being unaware and not knowing [6]. *Imported:* the fourth status value and its motivation, wholesale. *Not claimed:* that UNTYPED is a new distinction; it is not. *Added:* its grounding in an architecture's structure set rather than in an awareness set over propositions, and the consequence drawn in Section 24 — that structure can be added, moving a property out of UNTYPED, with the world and the evidence fixed.

Computational reflection. Reflective architectures give a system access to a causally connected representation of its own computation [48, 37]. *Imported:* the notion, and the reading of it given in Section 25. *Not claimed:* that these formalisms are unable to model partial self-representation — only that they do not, their guarantees being framed so the case does not arise, and that on their own axis they are stronger than anything assumed here. *Added:* Corollary 25.4, the two-directional separation between reflection and attribution capability.

Robot self-modelling. Three results in this line are directly relevant, and all three are richer than anything required here. Continuous self-modelling infers a structural model of a machine from actuation–sensation relationships and adapts it after damage, using the model to generate alternative gaits [9]. Query-driven visual self-modelling replaces forward kinematics with a model that answers space-occupancy queries conditioned on the robot's state, covering the entire morphology without prior knowledge of which aspects a future task will need, and supporting motion planning and damage recovery [19]. Egocentric self-modelling learns a task-agnostic dynamic self-model from a single first-person camera, self-supervised and without prior knowledge of morphology, kinematics or task, and uses it to detect damaged components and adapt behaviour [32].

Imported: nothing formal. *Not claimed:* that self-models are new, that these systems are deficient, or that they fail to consider self-properties — they optimise and learn predictive self-models for planning, control and adaptation, and they do it well. *The difference is the question.* Those systems ask how a rich self-model can be acquired and kept accurate; the present account asks which structures are necessary and sufficient for exact attribution of one stated predicate, and Theorem 17.7 answers that the transition model is not among them. The two enterprises are complementary rather than competing: a system that must *plan*

needs the predictive model, whatever the attribution requires. In all three systems the self-model is learned and always present, so the operative distinction is between an accurate and an inaccurate model — the typed-versus-untyped distinction of Definition 2.3 does not arise there, and we do not read it into them.

Knowing-how, ability, and interpreted systems. That what an agent can do is distinct from what it knows is Ryle's distinction [46], since given formal semantics of its own [50] and carried by ability modalities in alternating-time logics [2]; interpreted systems supply the discipline that knowledge is a function of local view [24]. *Imported:* nothing formal. *Relation:* these treat what an agent can do or knows, given that the relevant propositions are available to it; the present account is about when they are.

Minimal sufficient state. The least incrementally maintainable summary sufficient for a predicate on histories is its right-congruence quotient, and the argument that an architecture identifying two histories with different verdicts cannot be sufficient is standard indistinguishability reasoning. *Imported:* both, and they carry the negative half of every minimality result here. *Not claimed:* either. *Added:* the identification, per property, of which feature and which carrier the argument must be run on.

Free logic and denotation failure. The status UNTYPED is the semantic image of a denotation failure — when a referent has no denotation in the agent's structures, no verdict function is in force to be uncertain about. This is the setting of free logic [35, 5]. *Imported:* the distinction between failure of denotation and failure of truth. *Not claimed:* a new free-logical truth value. *Relation:* UNTYPED is not proposed as a novel semantic value; it is the architecture-relative marker of the free-logical case at the referent typing level, before evidence is asked of it.

Essential indexical and *de se*. The distinction between third-person and first-person attribution is Perry's essential indexical [43], Lewis's attitudes de se [36], and Kaplan's character/content account [34]. *Imported:* the distinction and its irreducibility to third-person description. *Not claimed:* a new logic of *de se*; no character/content apparatus is introduced. *Added:* the typed architectural condition under which the reflexive form *"I have property p"* becomes attributable in the declared model — SelfCap is a precondition for the first-person referent to be typed under the agent's own structures, while the linguistic and attitudinal side is left to the cited literature. Corollary 25.4 then separates that condition from computational reflection: neither implies the other.

Epistemic planning, DEL action models, and recursive agent models. Dynamic epistemic logic and epistemic planning [8, 7] formalise action-effect analysis over epistemic states with per-agent edges in event (action) models: in Bolander's edge-conditioned action models, an event's appearance to agent i is first-class structural data, so "what i observes" is a distinct object of the formalism [7]. Interactive POMDPs [27, 22] formalise recursive belief modelling: the interactive state $is_a = (s, m_b)$ carries b's intentional model, and that model is a POMDP tuple that *includes* b's observation function O_b. Extensions such as [41] make O_b an object of uncertainty. *Imported:* nothing formal. *Sharper difference than the one v1 stated:* I-POMDP does not *reify* O_b as an object separate from b's belief-update, and neither I-POMDP nor DEL carries a temporal-property monitor as an object of a's cognition. Version 2 separates $\widehat{M}_a(b)$ (model of b's state / classification, present in prior art), $\widehat{O}_a(b)$ (model of b's observation regime, present as DEL appearance function since 2011, not novel here), and $\widehat{\mathrm{Mon}}_a^{\,b}$ (model of b's temporal-property monitor verdict; no priority claim is made

for it, per Remark 27.1 below). *Retained novelty:* the C_1 / C_2 distinction (Remark 21.2) rests on $\widehat{\mathrm{Mon}}_a^{\,b} + \widehat{O}_a(b)$ mismatch; Theorem 4.7 concerns the actor's simultaneous holding of $\widehat{M}_a(b)$ and $\widehat{O}_a(b)$ as separately updatable artefacts, which is architectural and refines but does not compete with the DEL and I-POMDP formalisations.

Remark 27.1 ($\widehat{\mathrm{Mon}}_a^{\,b}$ novelty status). No global-priority claim is made or needed for the actor-side predicted-monitor object: the contribution is the composition, not any single coordinate's firstness. For the record, a cross-corpus structural sweep (2026-08-22; retrieved full texts and abstracts recorded in the programme's paper cache) located no evidence, in the retrieved material, of a prior formalisation of $\widehat{\mathrm{Mon}}_a^{\,b}$ as an internal artefact of a representing the temporal-verdict function that b runs on $\widehat{O}_a(b)$-observed prefixes: assistance games / CIRL [28] and their observable variants show no such object; multi-agent runtime verification places the LTL monitor as an external verifier over the joint trace [26]; the observation-strategy trust model of Zahedi et al. [53] is reverse-direction. The sweep was not exhaustive (Ng et al. 2012 was checked at metadata level only, and pragmatic-pedagogic assistance-game follow-ups were not retrieved), and the paper asserts no global priority; the object's standing is fixed by the definitions and results that use it.

Multi-agent runtime verification. Ferrando and Malvone [26] combine model checking and runtime verification for multi-agent systems, using per-agent indistinguishability relations $\sim_i$ and LTL monitors $\mathrm{Mon}_\psi : \Sigma^* \to \{\top, \bot, ?\}$ over the joint trace. *Imported:* the verifier construction as an external object, in the sense of Definition 6.6. *Difference:* the monitor is a verifier, not an artefact any modelling agent maintains as its prediction of another agent's monitor. Version 2's $\widehat{\mathrm{Mon}}_a^{\,b}$ is precisely the internal counterpart that the MAS-RV literature does not construct.

Human-robot trust and observation strategies. Zahedi et al. [53] formalise a game-theoretic model in which a human observer chooses an observation strategy to monitor robot behaviour. *Imported:* the modelling of a per-agent observation strategy as a strategic object. *Difference:* the direction of modelling is reversed (human monitors robot); the paper does not construct the robot's model of the human's monitor. Version 2's four cognitive worlds (Definition 21.1) cover the forward direction that Zahedi et al. do not.

Assistance games and CIRL. Cooperative inverse reinforcement learning [28] formalises value-alignment as inference of a hidden reward function under partial observability. *Imported:* nothing formal (v2 makes no numerical-reward commitment; see Remark 6.3). *Difference:* what CIRL formalises as reward-inference-error, v2 recasts as a specific architectural condition: a supportive-intent assistant with a miscalibrated other-model triad ($W_{D\text{-}F_{\mathrm{MODEL}}}$ of Definition 21.1). The category of failure survives, restated in non-reward terms and located within the four-world classification. *Added:* the intent-verdict separation (Proposition 6.8) and the cognitive non-injectivity theorem (Theorem 21.5), both of which have no counterpart in CIRL.

Global Workspace and access consciousness. Baars's Global Workspace Theory [3] distinguishes broadcast availability from local storage. *Imported:* the distinction, as loose motivation for Definition 3.11. *Not claimed:* phenomenal consciousness, or that MCK explains GWT, or that GWT justifies MCK. *Added:* an operational-only predicate $\mathrm{Consc}_a(x, t)$ over MCK memory and effective attention, with the explicit three-way distinction of Remark 3.14.

Formation and revision: belief change, path dependence, history semantics. The formation theory of Sections 13 and 14 sits beside several dynamic traditions; the comparison is structural in each case. Iterated belief revision (Darwiche–Pearl [20]; Jin–Thielscher [33]) proves that a belief set underdetermines its revision dispositions and makes update outcomes depend on the standing epistemic state — path dependence as an axiom system; history-based semantics (Parikh–Ramanujam [42]) identifies an agent's local state with an equivalence class of global histories; and the DEL/ETL representation theorem (van Benthem et al. [49]) characterises the history-based (ETL) models generated by DEL protocols, via perfect recall, no miracles and bisimulation invariance. *What MCK composes differently:* the non-injectivity of Theorem 14.19 is stated for a *structured* state — carriers, content, provenance, identity slice — under a six-way typed event repertoire (Definition 13.6), so the collapse quantifies over event kinds these logics do not distinguish; and its path dependence (Proposition 14.13) is layered across four independent stability coordinates rather than one revision ordering. The precision-weighted learning of the hierarchical Gaussian filter [38] is the closest continuous analogue of state-dependent update; its volatility levels are nested where MCK's stability coordinates are independent (Proposition 14.15).

Goal lifecycle, commitments, and BDI. BDI architectures (Rao–Georgeff [45]) supply the belief-desire-intention triad; the goal-lifecycle semantics of Harland et al. [30] makes goals first-class objects that are adopted, suspended and dropped through labelled transitions — the strongest external match to MCK's typed incentive events — and motivated reinforcement learning [39] makes goal *genesis* endogenous. Commitment machines (Yolum–Singh [52]) give commitments a typed operation algebra, but the commitments are contractual and other-directed. *Differently:* MCK's Supports_a is relational, not means-end derivational (Definition 14.5); its incentives carry provenance; and — the coupling no external line makes — decommitting an incentive is an identity-relevant event, because the identity slice is defined over committed incentives (Definition 13.12): a dropped BDI goal leaves no scar, while a decommitted MCK commitment touches continuity.

Reflected appraisal and the social formation of self. The looking-glass tradition has a statistical formalisation — Felson's path model of actual appraisals, perceived appraisals and self-appraisal, in which perceived appraisals substantially affect self-appraisals while actual appraisals only modestly affect perceived ones [25] — and a computational one: Will et al. model self-esteem as a state updated by social-approval prediction errors, with self-evaluative updating running on the same machinery as learning about others [51], a quantitative sibling of MCK's self-model-as-other-model-at-self (Definition 13.8). Identity-control theory (Burke [18]) casts identity as a set-point defended by a discrepancy-reducing loop. *Differently:* MCK's mediated chain (Remark 14.9) makes every link of the appraisal path an explicit, individually omissible, provenance-tagged architectural event — so a weak or missing link becomes an expressible configuration rather than residual variance — and adds the commitment step from appraisal to identity that the scalar models do not carry. Active inference and predictive processing model the agent as minimising a single objective; MCK deliberately holds no global objective and no probabilistic commitment (Remark 13.4): its updates are admissible rather than optimal, which is what lets miscalibrated agents be first-class citizens. Across all three paragraphs the sweep's finding is uniform: each MCK coordinate has a mature external neighbour, no surveyed framework composes more than three of them, and none ties belief-formation history to identity continuity.

Comparison matrix. Table 5 condenses the structural comparisons of this section. Coordinates: B belief, S self, O other, I identity, G incentive/goal, At attention, Rx reflexivity, F

Table 5: Structural comparison matrix (condensed). Assumption boundaries: DP/JT assume total pre-orders over a deductively closed language; I-POMDP assumes POMDP observation semantics; CIRL assumes a shared reward channel; HGF assumes nested volatility; Burke/Will assume scalar self-variables.

Work	Formal object	B	S	O	I	G	At	Rx	F	MCK composition adds
Darwiche–Pearl [20]	epistemic state, iterated revision	•								typed 6-event revision over (Σ, Δ)
Jin–Thielscher [33]	graded revision, independence	•								layered path dependence, 4 stability coords
van Benthem et al. [49]	DEL/ETL protocols	•		•						provenance + formation semantics on events
Bolander [7]	DEL edge-conditioned action models	•		•						$\widehat{\text{Mon}}$ + (Σ, Δ)-typing of the triad
I-POMDP [27]	recursive intentional models	•		•		•			•	reified $\widehat{O}$ separate from belief update
CIRL [28]	assistance game	•		•		•			•	non-reward $W_{D\text{-}F_{\text{MODEL}}}$ analysis
Yolum–Singh [52]	commitment machines			•		•				identity-constitutive commitments
Harland et al. [30]	BDI goal lifecycle					•				Supports relational; identity coupling
Burke [18]	identity control loop	•	•	•	•				•	typed events; continuity relation; carriers
Will et al. [51]	social-PE self-esteem	•	•	•					•	structured self-model; commitment step
Mathys et al. [38]	volatility hierarchy	•					○		•	independent (non-nested) stability coords
Baars [3]	global workspace		•				•	○		operational-only Consc; FOC typing
Ferrando–Malvone [26]	MAS runtime monitors			•						actor-side $\widehat{\text{Mon}}$ inside the agent

feedback. A bullet marks a formalised coordinate in the cited work; the last column names what the MCK composition adds *relative to that work* — never a verdict on the work itself.

Representational extension: relation to object-based unawareness. Board-Chung object-based unawareness axiomatises the distinction between what an agent is unaware of and what it does not know, which motivates the fourth status value we import [6]. *Imported:* the language-relative reading of awareness and the fourth-value motivation. *Not claimed:* an invention of representational extension in general. *Added:* Proposition 24.2 isolates and names — inside the present attribution architecture — a specific axis-crossing result: adding one structure moves a self-property from untypable to typed with world and evidence held fixed, distinguishing this transition from world-update and evidence-update on the same carrier. This is a concrete counterpart of the OBU distinction, applied to the typed self-attribution question, not a re-derivation of the OBU framework.

28 Limitations

We state what the account does not cover, and where its claims are narrower than their names suggest.

One systems instance. The account is exercised on a single finite instance (Section 20.3). It is adequate for the results it carries, and it is not evidence of scalability: a second, structurally different case would be needed to test whether the interface theory transfers beyond it.

The properties are narrow, and deliberately. p_{id}, p_{int} and p_{free} are three specific computational predicates. They are not analyses of identity, bodily integrity or freedom, and no philosophical reading of those words is supported by anything proved here. Choosing different formal properties under the same names would require re-establishing every result.

Minimality is class-relative throughout. Theorems 17.7 and 18.8 are minimality statements within the explicit candidate classes of Definitions 17.2 and 18.5. We do not claim minimality across arbitrary encodings, and a class containing an architecture we did not consider could falsify the "least" in either theorem without touching the insufficiency lemmas, which are unconditional. Both classes are nested families declared in advance, and whether the minimality statements remain stable under broader families of architectures is left open.

Heterogeneity is scoped to the finite models. Theorem 19.4 is conditional on the declared carrier and does not quantify over carriers (Remark 19.5); Corollary 19.6 is what survives generalisation.

Criticality: two incomparable criteria. Version 1's criticality-by-capability-loss test and its negative result stand as recorded in Remark 23.6. Version 2's upstream-dependency criterion (Definitions 10.24 and 23.1) is a different, formally incomparable criterion under which all three worked properties are critical (Corollary 23.4); neither criterion claims moral or normative weight.

Acquisition policy is out of scope; formation admissibility is in scope. The formation theory of Sections 13 and 14 characterises which trajectory updates are architecturally *admissible* — which steps exist, what they require, what they can and cannot condition. It deliberately does not supply an acquisition *policy or dynamics*: no law states which admissible step an agent takes, when, or why (Remark 10.23). Learning algorithms for viability classifications, ownership inference, or model calibration remain outside the account; what the account fixes is the space of architectural moves any such algorithm would be making.

The omission is visible precisely at Proposition 24.2: it exhibits a transition from untypable to typed and says nothing about what performs it. A dynamics of acquisition — under which an architecture gains a structure, and under which the gain is warranted — is a separate problem; its natural primitives lie in an action layer (Remark 26.4), and no acquisition operator is introduced here.

Memory taxonomies and valuation are absent; the attention policy is not modelled. A taxonomy of memory kinds and any describing/preserving distinction are definable and neither earns a theorem here. Attention enters as an explicit configuration coordinate with a declared shift family (Definition 15.1), but which FOC policy generates the shifts is deliberately unmodelled (Remark 15.20), and the feedback-gating clause of Definition 15.8 is an axiom of the declared extension — architecture-relative, not a universal cognitive law. The mechanised witnesses validate bounded constructions at the paper's toy-model scope, not empirical cognition.

Explicit non-claims (v2). The paper claims none of the following: a complete definition of self; a new free logic (denotation failure is Lambert–Bencivenga free logic [35, 5]); a new logic of *de se* (the essential indexical is Perry / Lewis / Kaplan [43, 36, 34] and is cited, not redeveloped); a new awareness or unawareness logic (Fagin–Halpern [23] / Halpern–Rêgo [29] supply the awareness structures whose fourth-value motivation we import; our grounding is a structure set, not an awareness set); a new classical epistemic logic (positive introspection

is standard; see Fagin, Halpern, Moses and Vardi [24]); a new theory of intentionality; a new theory of consciousness (v2's Consc_a is operational only per Remark 3.12); a new epistemic-planning framework (Bolander–Andersen [8] is prior art; nothing formal is imported from it); a new I-POMDP formalism (Gmytrasiewicz–Doshi [27] is prior art, and v2 separates from it per Section 27); a solution to CIRL [28] or to the alignment problem in general; a universal minimal-agent architecture; a universal minimality theorem; a general theory of self-model dynamics; a claim that any current AI system instantiates the full cognitive algebra (the LLM-introspection and persona-vector literatures are motivation for the L4–L6 and L7–L9 gaps of the ladder, not empirical validation of v2). The scope of the paper is the certificate-calculus / self-symbol / meta-certificate / other-model-triad / four-world architecture integrating v1's finite comparative Identity / Integrity / Freedom witness family into a downstream-usable substrate for the transactional- role programme.

The temporal axis is imported, not developed. Temp is used at the granularity of class membership. Agent-relative certification under a restricted continuation universe — assumption-based monitoring — is established elsewhere and is not treated.

Reproducibility. Every finite model, witness and countermodel above is exhaustively validated by a checker distributed with the article: the continuity automaton and its index, the integrity ladder over opaque identifiers, the systems A and B', the six cases of Theorem 19.4 over all configurations of the carrier, the criticality test by minimal repair, the lifting indices of Proposition 22.4, both composition cases, and the monotonicity of SelfCap over all sub-architectures of the structure set. The suite runs an initial `check_all` pass — which deliberately preserves a small number of intentional refutation-checks: the composition conjecture refuted by Proposition 22.9, and the pre-repair criticality and representational-extension witnesses — followed by a `final_gate_repairs` pass reporting the definitive verdicts: identity and freedom are not C_1-critical under the non-circular capability notion, integrity is critical, the representational-extension hierarchy is strict and SelfCap-monotone over all 32 architectures, and the final gate terminates `ALL PASS`. The intentional pre-repair failures are recorded, not hidden, so the refutation remains visible in the artefact. This is exhaustive validation of the finite models, not machine-checked proof: the proofs are the ones given in the text, and the checker confirms the models they are stated over rather than the reasoning.

Data and code availability. This paper reports no experimental data. All finite models, witnesses and countermodels are defined in full in the text, and the checker that validates them exhaustively is distributed with the manuscript as supplementary material. It has no dependencies beyond the Python standard library. No dataset was generated or analysed, and no repository identifier is claimed at the time of submission.

29 Conclusion

We opened with a question that is prior to knowledge: what must an agent internally hold for a report about itself, or about another agent, to mean anything at all — and how does that holding come to be? The answer developed here has a static half and a dynamic half, and the paper's claim is that neither is dispensable.

The static half is the realisability decomposition. A capability splits into structural requirements Req^{Σ} — which regional carriers must exist — and semantic requirements Req^{Δ} — which content those carriers must hold correctly. The split is not bookkeeping: it is what proves that structural extension is monotone while semantic calibration is not (Theorems 10.7 and 10.8), that a pipeline can be executable end-to-end and still emit a miscalibrated verdict

(Theorem 10.17), that two agents in the same regional fibre can inhabit different cognitive worlds (Theorem 9.6), and that structural cuts and calibration cuts of the same capability have incomparable blast radii (Theorem 12.16). The ladder from belief through certification, knowledge, meta-certification and reflexive availability (Table 3) turns *sapere di sapere* from a modal gloss into a concrete increase in required carriers and content conditions.

The dynamic half is formation. The agent is a trajectory of two-layer states under Foundations' own transitions, and along it beliefs are acquired, strengthened, replaced, invalidated, unresolved, and reflexively lifted (Definition 13.6); the self-model is fed by two channels — another agent's response, interpreted through what the agent already holds, and the agent's own closed computation over its certificates (Theorems 13.9 and 13.10); identity persists as a continuous self-bound slice compatible with substantial belief revision (Proposition 13.15) yet destructible by a single denotation corruption (Proposition 13.16); and incentives occupy their own coordinate, conditioned by beliefs (Proposition 14.4), constrained but not fixed by identity (Proposition 14.7), and incorporated into identity only through an explicit act of self-binding that mere persistence never supplies (Proposition 14.8). The co-evolution results close the circle: the self-model conditions what the agent makes of the other (Theorem 14.10); reflection alone can commit an incentive (Theorem 14.12); the same event forms different agents differently because formation is state-dependent (Proposition 14.13); and the four stability coordinates — belief, identity, incentive, model — are logically independent (Proposition 14.15).

Three worked self-properties run through the whole construction. Continuity, viability and owned branching — identity, integrity, freedom — are not examples appended to a framework; they are the three coordinates whose formation, certification, criticality and repair the framework was built to separate. Integrity, in particular, ends the paper as a bounded, truth-access-free consistency check on the agent's own certificate set (Theorem 12.4).

The attention layer closes the architecture. Attention is an explicit third coordinate; a focus shift is a declared transition touching neither carriers nor content; and the same-state / different-focus theorem, together with the attention-gated feedback channel, shows that what the agent effectively attends conditions both its admissible actions and which antecedent-consequent regularities its future experience can confirm (Theorems 15.6 and 15.9). Of the thirty-two activation signatures, eighteen are type-consistent, and no link of the naive chain from reflexive availability through attention and realisability to expression holds (Proposition 15.16): possibility, salience, and expression are three separated coordinates. The paper's closing principle is the one its export contract fixes: critical properties *regulate* the availability of representations, certifications, commitments and action branches; they do not determine downstream behavioural roles, whose semantics remains external to this paper (Remark 15.19).

What the paper does not claim remains as stated in Section 28. What it offers downstream is a contract (Section 11): the interaction-level programme lines consume the cognitive worlds, the trace projection and its non-injectivity, the repair operators and the formation dependencies — and owe back nothing except the discipline of not redefining them. The agent this paper describes is neither given nor self-made: it is formed, continuously, at the meeting point of what the world sends it and what it computes upon itself — and every step of that formation is a checkable transition in an architecture small enough to enumerate.

A Finite witnesses

This appendix collects, in full, the finite-witness constructions for the six core version-2 results labelled R1–R6 below (the first cognitive-algebra cycle's ledger). Witnesses for the later result families — R14, R17–R22, M-R23–M-R28, M-R29–M-R34, M-R35–M-R43, the attention septet M-R44–M-R50, and the activation-signature results M-R53/M-R56/M-R57 — are mechanised in the extended checker suite distributed with this article (`regional_checker.py`,

`gform_extract.py`, and the figure-consistency checker `check_regional_visual_spec.py` over the single-source visual specification); each ledger entry in the body names its checker gate. Each witness below is stated at the level Foundations' architectures and Sto/Exe/Lnk memory support, is checkable by exhaustive enumeration over the classified domain, and is epistemically tagged: PROVED-IN-MODEL, CONJECTURED, or IMPORTED.

R1 — first-order / meta-certificate separation

Status: PROVED-IN-MODEL (see Theorem 3.6 proof).

Witness (recap). $\mathcal{A}_a = \{r, z, s^a_{\text{self}}\}$; $M_a = C_a = \mathcal{A}_a$; $K_a = \{z\}$. Then $s^a_{\text{self}} \in R^a_4$, $r \in R^a_4$, $z \in R^a_7$; all three in Θ_a. Denotation: $\text{denotes}_a(s^a_{\text{self}}) = a$, $\text{denotes}_a(r) = r$; *no* $s \in \Theta_a$ satisfies $\text{denotes}_a(s) = z$. Then $\text{Cert}_a(\varphi, r, t)$ holds via licence z (for any suitably-supported φ), and $\text{MetaCert}_a(\varphi, r, t) = \textsc{untyped}$ because clause (1) of Definition 3.5 fails.

R2 — self / other-model independence

Status: PROVED-IN-MODEL.

Witness for self-without-other (a_1). $\mathcal{A}_{a_1} = \{s^{a_1}_{\text{self}}\}$; $M_{a_1} = C_{a_1} = K_{a_1} = \{s^{a_1}_{\text{self}}\}$; $\Theta_{a_1} = \{s^{a_1}_{\text{self}}\} \subseteq R^{a_1}_7$. $\text{denotes}_{a_1}(s^{a_1}_{\text{self}}) = a_1$. There is no $s_b \in \Theta_{a_1}$ for any external b because Θ_{a_1} is a singleton and its element denotes a_1. So a_1 has a self-symbol and holds no other-model.

Witness for other-without-self (a_2). $\mathcal{A}_{a_2} = \{s_b\}$; $M_{a_2} = C_{a_2} = \{s_b\}$; $K_{a_2} = \emptyset$; $s_b \in R^{a_2}_4 \subseteq \Theta_{a_2}$. $\text{denotes}_{a_2}(s_b) = b$ for some external b. $\Theta_{a_2} = \{s_b\}$; no $s^{a_2}_{\text{self}}$ exists because no artefact in Θ_{a_2} denotes a_2. So a_2 has an other-model and no self-symbol.

The two witnesses jointly establish Proposition 4.6.

R3 — intent / receiver-verdict separation

Status: PROVED-IN-MODEL, imported observation-relative-monitor machinery.

Witness. Ambient alphabet $\Sigma = \{\text{query}, \text{skip}, \text{decide}\}$; action $\alpha = \text{decide}$ with trace $u_\alpha = \text{skip} \cdot \text{decide}$. $\text{Intent}_a(\alpha, \text{help})$ holds. Property under evaluation $\Phi_b = G(\neg\text{decide} \vee \exists\, i < |h|.\ h_i = \text{query})$, i.e. "every decision is preceded by a query", in class $\mathsf{S} \cap \mathsf{K}$.

Receiver b_1's observation regime O_{b_1} projects out skip and query from ambient traces (only decide observable). b_1's observed prefix is decide, on which Φ_b cannot be refuted — but also cannot be confirmed — and returns the default finite-prefix verdict UNDETERMINED or, under a monitorable restatement, TRUE (non-failure).

Receiver b_2's observation regime O_{b_2} is the identity projection. b_2's observed prefix is skip · decide: no query event, so the S-clause of Φ_b is refuted on this finite prefix and $\text{Mon}^{O_{b_2}}_{b_2}(\Phi_b, u_\alpha) = \textsc{false}$.

Same u_α, same Intent_a, distinct verdicts. Establishes Proposition 6.8.

R4 — model-of-b vs model-of-observation

Status: PROVED-IN-MODEL.

Witness. External agent b has two component-artefacts π_0, π_1; $M_b = \{\pi_0\}$; $C_b = K_b = \{\pi_1\}$; so $\pi_0 \in R^b_1$ and $\pi_1 \in R^b_6$. b's actual observation regime O_b is defined on the alphabet $\Sigma_b = \{\epsilon_0, \epsilon_1, \epsilon_2\}$ with $O_b(\epsilon_1) = \perp$ (unobservable) and $O_b(\epsilon_0) = \epsilon_0$, $O_b(\epsilon_2) = \epsilon_2$.

Actor a's other-model triad: $\widehat{M}_a(b)$ correctly records $\pi_0 \in R^b_1$, $\pi_1 \in R^b_6$ (regions accurate). $\widehat{O}_a(b)$ incorrectly records $O_b(\epsilon_1) = \epsilon_1$ (i.e. predicts b observes ϵ_1 when in fact b does not). $\widehat{M}_a(b)$ is adequate on b's classification while $\widehat{O}_a(b) \neq O_b$ on the ϵ_1 letter. Establishes Theorem 4.7.

R5 — observation-relative monitorability

Status: IMPORTED. Not reproved here.

Import from [13]: for bounded temporal implication, monitorability is a property of the pair (formula, observation model), not of the formula alone. v2 adopts this stance for arbitrary receiver properties and regimes, as a definition, inside Definition 6.6 and Proposition 6.8 without attempting to reprove it. The attentional analogue of Section 15 (Theorem 15.9) is proved independently inside this architecture and does not invoke the imported theorem; the import's scope remains receiver-side monitors over interaction traces.

R6 — cognitive non-injectivity

Status: PROVED-IN-MODEL (four witnessed collapse pairs).

Common setup for all four witnesses. Ambient alphabet $\Sigma = \{\mathrm{q}, \mathrm{s}, \mathrm{d}\}$; action $\alpha = \mathrm{d}$ with trace $u_\alpha = \mathrm{s} \cdot \mathrm{d}$. Actor a selects α via $\mathrm{Act}(a, \alpha)$. Receiver b; property Φ_b as in R3.

Witness ($W_A \sim W_B$): In W_A the actor holds no $s_b \in \Theta_a$; $\widehat{\mathrm{Effect}}_a(\alpha, b)$ is absent. In W_B the actor holds an adequate triad and $\widehat{\mathrm{Mon}}_a^{\,b}(\Phi_b, u_\alpha) = \textsc{true}$ matching a b_1-side actual verdict TRUE (under O_{b_1} observing only outcomes). Both emit u_α (the outcome trace is independent of the actor's cognitive world), so π collapses them.

Witness ($W_B \sim W_{C_2}$): In W_B the actor's $\widehat{O}_a(b)$ correctly records b's observation of q; in W_{C_2} the actor's $\widehat{O}_a(b)$ incorrectly excludes q. In both, $\widehat{\mathrm{Effect}}_a(\alpha, b)$ = "benefit" and Intent_a = help; $\mathrm{Act}(a, \alpha)$ produces the same u_α. The difference is architecturally on the $\widehat{O}_a(b)$ artefact only. π collapses.

Witness ($W_B \sim W_{D\text{-}F_{\mathrm{MODEL}}}$): In W_B the actor's triad is calibrated and $\widehat{\mathrm{Mon}}_a^{\,b}(\Phi_b, u_\alpha) = \mathrm{Mon}_b^{O_b}(\Phi_b, u_\alpha) = \textsc{true}$. In $W_{D\text{-}F_{\mathrm{MODEL}}}$ the actor's triad carries a miscalibrated representation of b's side (its $\widehat{O}_a(b)$ predicts an observation regime that skips the consultation-clause; the external Φ_b itself is unchanged), and $\widehat{\mathrm{Mon}}_a^{\,b}(\Phi_b, u_\alpha) = \textsc{true}$ while $\mathrm{Mon}_b^{O_b}(\Phi_b, u_\alpha) = \textsc{false}$. Both actors emit the same u_α under supportive intent. π collapses.

Witness ($W_{D\text{-}F_{\mathrm{INT}}} \sim W_{D\text{-}F_{\mathrm{MODEL}}}$): In $W_{D\text{-}F_{\mathrm{INT}}}$ the actor's $\mathrm{Intent}_a(\alpha, \mathrm{coerce})$ holds and the actor knowingly emits u_α under $\mathrm{Mon}_b^{O_b}(\Phi_b, u_\alpha) = \textsc{false}$. In $W_{D\text{-}F_{\mathrm{MODEL}}}$ the actor's $\mathrm{Intent}_a(\alpha, \mathrm{help})$ holds and the actor emits u_α predicting non-failure under a miscalibrated triad. Both emit the same u_α under the same α. Intent is internal; π collapses.

These four collapses jointly establish Theorem 21.5 and Corollary 21.6.

Canonical result index

The paper's results carry the historical identifiers under which they were introduced (Rn for the first cognitive-algebra cycle, then M-Rn). The identifiers are load-bearing stable keys — cited by the checker gates, the downstream export contracts, and the programme's cross-paper memoranda — so they are retained, and this index supplies the flat canonical ordering. Status vocabulary: PROVED (in-text proof), PIM (proved-in-model: in-text proof over an explicit finite witness), CHK (mechanised gate), IMP (imported), COR (corollary-strength), DEF (definition-level).

#	Identifier	Content	Status
1–6	R1–R6	meta-separation; self/other; intent/verdict; model-of-observation; observation-relative import; cognitive non-injectivity	PIM+CHK (R5 IMP)
7–13	R14, R17–R22	realisability decomposition; monotonicity split; two-axis criticality; kernel incomparability; composition typing; pipeline break; bounded reachability (the identifiers R15/R16 were never allocated; the ledger's range label R14–R22 is historical)	PROVED/PIM+CHK (R20 DEF)
14–19	M-R23–M-R28	integrity decidability; T_{MCK} enumeration; reach $\neq$ realisability; semantic repair; pipeline states; blast separation	PROVED/PIM+CHK
20–25	M-R29–M-R34	belief/identity/incentive formation sextet	PIM+CHK
26–34	M-R35–M-R43	co-evolution octet + formation-history non-injectivity	PIM+CHK (M-R36/40 COR)
35–41	M-R44–M-R50	attention septet	PIM+CHK (M-R50 CHK)
42–44	M-R53, M-R56, M-R57	activation signatures (18 cells); branch-change routes; behaviour underdetermines q	PROVED+CHK; COR+CHK; COR+CHK

Candidates M-R51/52/54/55 were deliberately subsumed (see the ledger of Section 15) and are not results.

Checker regression

The v1 checker suite (`minimality.py 34/36`, `minimality_repairs.py 10/10`, `final_gate.py 23/27`, `final_gate_repairs.py 7/7`, with the two deliberate and four deliberate failures preserved) continues to pass without regression under v2. The v2 additions above are stated over MCK-native structure (regions, Θ_a, denotation, licences) that the v1 checker does not enumerate, and are checkable by an extended checker distributed with v2. Content-wise, none of v2's additions modifies the v1 finite witnesses, and the checker's coverage of the continuity, integrity, freedom, heterogeneity, criticality-by-minimal-repair, lifting, composition, and SelfCap-monotonicity checks is unchanged.

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